How Does Data Access Shape Science? The Impact of Federal Statistical Research Data Centers on Economics Research *

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Abstract

How does data access shape the rate, quality, and policy relevance of academic research? To study this question, we estimate the impact of access to confidential microdata through the progressive geographic expansion of the U.S. Census Bureau's Federal Statistical Research Data Centers (FSRDCs). FSRDCs boost the use of confidential microdata and help applied researchers publish more policy-relevant articles in top outlets. Besides direct data use, this effect also reflects spillovers to exposed researchers who are more likely to explore new topics and methods related to FSRDC-enabled research. Our findings underscore the importance of data access for both scientific progress and policy.

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1 Introduction

Modern science is largely empirical. In fields as diverse as astronomy, chemistry, and environmental sciences, researchers increasingly rely on large-scale, centralized datasets rather than on data curated for a single question (Hill et al., 2020; York et al., 2000). Economics is no exception (Backhouse and Cherrier, 2017). The share of theoretical papers published in top economics journals decreased from 50.7% in 1963 to 19.1% in 2011 (Hamermesh, 2013), while empirical work has surged during this period (Angrist et al., 2020). A key factor behind this growth has been the availability of microdata on consumers and firms, often from administrative government records (Card, 2022; Heckman, 2001). Around 20% of recent articles in the five most prestigious economics journals use such data (Currie et al., 2020).

Data at the level of individual units are unique in their potential to inform evidence-based policies across a broad range of economic and social phenomena (Cole et al., 2020). Academic economists have therefore urged broad access to large-scale microdata, citing significant scientific benefits (Card et al., 2010). However, the level of granularity afforded by microdata poses significant privacy risks, prompting government agencies to adopt tight confidentiality and security standards (Abowd and Lane, 2004; Foster et al., 2009). Understanding how and to what extent research access to confidential data entails benefits would complement current debates, which are primarily centered on ensuring privacy protection, even at the cost of reduced accessibility (Abowd and Schmutte, 2019; Chetty and Friedman, 2019; Lane, 2021). Despite this urgent need, evidence of the impacts of increasing data access remains thin.

It is not a given that broadening access to confidential data will benefit research and policy-making. First, access may not translate into effective use. Confidential data require stringent security restrictions, which may limit adoption and, therefore, any scientific benefits. Second, greater data availability need not improve research quality. Access to a common pool of data could intensify competition and encourage researchers to prioritize speed over rigor (Hill and Stein, 2025). More broadly, it may redirect attention toward questions that are easier to study rather than those that are most scientifically important. In the words of the late Bob Solow, researchers may

move away from substantive "puzzles that need to be explained" toward more marginal questions simply because an "enormous bunch of data" is readily available (Dizikes, 2019; Hoelzemann et al., 2024). These considerations make it necessary not only to establish whether greater data access generates meaningful benefits, but also to quantify their magnitude and understand how they arise.

We shed light on these questions by studying how access to confidential microdata distributed by the U.S. Census Bureau shapes the quantity, quality, and policy impact of economic research. Our focus is motivated by the fact that they are perhaps *the* pre-eminent source of restricted-access data in the United States. Examples include the Longitudinal Business Database (LBD) and the Longitudinal Employer and Household Dynamics (LEHD) database. Importantly, researchers wishing to analyze these data must be physically present at the Census Bureau's headquarters or in a secure data enclave, known as "Federal Statistical Research Data Center" (FSRDC). A network of 31 FSRDCs was established nationwide between 1994 and 2019. As multiple analyses will show, the timing and location of FSRDCs were partly driven by factors governing geographic equity rather than by pre-existing trends in microdata use. Therefore, the focus on FSRDC data provides quasi-experimental variation to identify the effects of confidential data access on economics.

We create a novel longitudinal dataset measuring the publication output of individual economists using EconLit, which we combine with hand-curated sources to assess the use and impact of FSRDC data. Our central regressions estimate the effects of FSRDC openings on empirical researchers at institutions with an FSRDC relative to researchers without local access. Our preferred specification includes researcher and university-tier $\times$ year fixed effects, controlling for time-invariant researcher quality and time-varying shocks that might correlate with FSRDC openings. We find that local access substantially increases the use of confidential data. Although researchers could access such data through collaborators or by traveling to another city, an FSRDC opening at their institution increases usage by 116%–118% relative to the sample mean. We also find a large increase in citations to prior work based on confidential data, suggesting that local access increases exposure to FSRDC-based research and may, in turn, shape the research trajectories of non-users.

Next, we explore how data access affects the productivity of empirical researchers. Contrary

to views that link data access to more marginal research, we find that treated empiricists produce around 18% more publications in top-ranked outlets. This result is strengthened when considering citation-weighted publications, suggesting that confidential data are particularly useful for boosting the scientific quality of empiricists' output. Moreover, FSRDCs lead empirical economists to publish research that receives more citations from policy documents. This result is not accompanied by an explicit shift of emphasis toward more policy-oriented research topics. Instead, access to microdata yields findings that are both more scientifically impactful and more relevant outside academic circles (Card et al., 2010; Chetty, 2012; Einav and Levin, 2014).

Our findings are robust to several potential concerns. First, event studies and formal tests support our identification strategy (Callaway and Sant'Anna, 2021; Rambachan and Roth, 2023). Qualitative evidence from interviews also suggested alternative designs, which we use to exploit within-institution variation in exposure to data access. Our results remain similar when comparing treated empiricists with colleagues in theoretical fields or applied fields that rely on other types of data. Second, university-level analyses show that FSRDCs are not systematically opened at institutions experiencing rising research intensity or hiring sprees. Third, we consider alternative definitions of FSRDC exposure. The results are robust to excluding host institutions and instead treating researchers at nearby institutions as exposed. The effects grow stronger with proximity to an FSRDC and are largest for researchers with access on their own campus, even among institutions that jointly operate a common FSRDC. Finally, we find no evidence that our results are driven by the sorting of more productive economists toward institutions with an FSRDC.

We investigate the mechanisms behind these findings. Our results are only partly driven by researchers who directly adopt confidential data, suggesting spillovers to non-users (Myers and Lanahan, 2022). These spillovers, however, are concentrated among economists who are more exposed to FSRDC research, either because they work on related topics or have colleagues who use FSRDC data. The effects disappear in departments without data adoption and among authors whose work does not build on FSRDC-based research. These patterns suggest that local data access also affects academic output by increasing exposure to research based on confidential data. In turn, this

exposure appears to raise productivity by shifting research direction. First, empirical researchers become more likely to explore topics commonly associated with FSRDC data. Second, treated researchers increase their use of other administrative datasets and quasi-experimental research designs. Taken together, FSRDC openings improve the research output of non-users by shifting both the topics they study and the data and methods they use.

Despite documenting how FSRDCs increase the diffusion of confidential data, the number of publications using these data remains limited. Accordingly, we explore how our findings can inform the design of the data infrastructure in the United States. Our analysis finds that geographical barriers disproportionately penalize researchers from less prestigious institutions or with fewer resources for accessing confidential data. While the optimal level of data access depends on social preferences around privacy (Abowd and Schmutte, 2019), there might be ways to increase the diffusion of confidential data holding the current regulatory environment constant. We discuss the possibility of increasing the number of data centers with "back-of-the-envelope" calculations. While a complete cost-benefit analysis is beyond the scope of our paper, even our conservative estimates suggest that further expansion of the FSRDC network would benefit science.

Our work contributes to different strands of research. New legislation in the U.S. (such as the Evidence-Based Policy Act and the Federal Data Strategy) places government microdata at the heart of policymaking (Lane, 2021). The implicit assumption is that expanding access to restricted data will generate policy-relevant research whose benefits outweigh the associated privacy risks. Yet, while some have hailed current data access programs as successful, others have criticized them for being costly and cumbersome, with uncertain impact (Atrostic, 2007; Card et al., 2010; Cole et al., 2020). Ironically, most of these debates have themselves been data-free. We provide the first systematic investigation of how access to confidential government microdata affects research, offering key parameters for this discussion (Abowd and Schmutte, 2019; Foster et al., 2009). Further, by examining the policy consumption of research, we show how confidential data can benefit evidence-based policymaking (Hjort et al., 2021; Yin et al., 2022).

Second, we contribute to meta-research on the economics profession. This includes work

documenting the empirical turn in economics (Currie et al., 2020; Hamermesh, 2013). Our results show that increased access to high-quality data is an important factor driving the impact and credibility of empirical scholarship (Angrist et al., 2020; Brodeur et al., 2020). In addition, we contribute to the growing scholarship on labor markets for economists, including research on status dynamics, credit attribution, and editorial roles (Card et al., 2020; Feenberg et al., 2017; Heckman and Moktan, 2020; Sarsons et al., 2021). Our results hint that confidential data pose asymmetric hurdles for researchers depending on their resource endowments. Democratizing data access could help level a field characterized by large inequities in status and resources (Wapman et al., 2022).

Finally, we contribute more broadly to the economics of the scientific process (Wang and Barabási, 2021). A vibrant strand of research has examined how access to research tools shapes knowledge production (Biasi and Moser, 2021; Furman and Stern, 2011; Furman and Teodoridis, 2020; Murray et al., 2016; Waldinger, 2016). While this work has primarily focused on research materials, it has rarely examined access to data (Hill et al., 2020). Relative to prior studies of how data availability affects innovation at the level of technologies or firms (Nagaraj et al., 2020; Nagaraj, 2022; Tranchero, 2026; Williams, 2013), we study how data access shapes the productivity and research trajectories of individual scientists. This allows us to examine not only researchers' output, but also the methods they employ and the topics they pursue (Hoelzemann et al., 2024; Truffa and Wong, 2022). We further document spillovers beyond direct users, showing that the effects of data access extend through exposure to research enabled by those data.

2 Empirical Setting

2.1 Confidential Microdata in Economics Research

Microdata can be defined broadly as unit-level data that provide information about the characteristics of individual people or entities (Card, 2022; Heckman, 2001). This type of data can be collected to answer specific research questions, like in the case of the Panel Study of Income Dynamics (PSID). However, recent years have seen a growing diffusion of microdata not initially collected for research purposes (Goroff et al., 2018; Groves, 2011). Government agencies are major sources

of such information, routinely storing administrative records as part of their normal operations. Typical examples include unemployment insurance claims, Medicare data, or tax records. Other agencies, such as the U.S. Census Bureau, collect individual-level information through statistical surveys and censuses as part of their mandate to assemble timely data on the nation's demographic and economic trends (Foster et al., 2009).

There is little doubt that the increased availability of microdata has played an essential role in the empirical turn of economics (Card, 2022; Backhouse and Cherrier, 2017). This is especially true for government records, which usually comprise large samples that allow tracking of individual units over time with little attrition (Cole et al., 2020). Einav and Levin (2014) provide anecdotal evidence on impactful studies based on government data studying topics ranging from broadband internet to Medicaid expansion. However, the same features that make microdata invaluable for research also put respondents' privacy at risk (Bowen, 2024). Distributing firm- or individual-level information such as wages, health records, or similarly sensitive information poses obvious concerns. Government agencies thus face a trade-off between providing access to microdata for research and their duty to protect the confidentiality of the information entrusted to them (Abowd and Schmutte, 2019; Foster et al., 2009).

Government agencies have experimented with several second-best solutions, ranging from anonymized public-use samples to synthetic data (Abowd and Lane, 2004; Weinberg et al., 2007). Unfortunately, none comes close to the research potential of the universe of individual-level information (Cole et al., 2020; Goroff et al., 2018). Most statistical agencies have eventually resorted to granting vetted researchers direct access to confidential microdata for pre-approved projects, with results released only after careful review. Yet government agencies differ in how they provide this access. Some countries, especially in Northern Europe, trade off control for easier access via virtual private networks (VPNs) or dedicated remote devices with biometric authentication. In others, such as the United States, researchers must access confidential microdata in secure facilities on specialized devices where data use is closely monitored.[1] This data enclave model provides

[1]In 2019, the Census Bureau began to provide remote virtual access to a select number of FSRDC researchers not

greater security, but imposes substantial costs on universities and researchers.

2.2 The FSRDC Network

We focus on the FSRDC program created by the U.S. Census Bureau. The program originated with the establishment of the Center for Economic Studies (CES) in 1982 to combine microdata collected through the Bureau's activities and provide restricted access to researchers (Atrostic, 2007). The objective was to enable research that could improve Census Bureau data programs while complying with its legal mandate to protect confidentiality.[2] However, researchers had to access the data at Census Bureau headquarters near Washington, D.C., making access costly and inconvenient (McGuckin et al., 1993). To reduce these barriers, CES spearheaded the creation of additional secure facilities where researchers could access confidential data. These data enclaves, known as FSRDCs, are hosted by partner institutions, often organized into local consortia to share operating costs (CES, 2017). By 2019, the FSRDC network expanded to 31 data centers thanks to financial support from the National Science Foundation (NSF) (CES, 2019).[3]

Each FSRDC is a research facility that meets several physical and computer requirements to ensure the confidentiality of sensitive data. First, each branch has controlled access doors, security cameras, and a Census employee on-site. No data reside in the FSRDC, since all statistical analyses

working with records originating from the IRS. This development does not affect our analyses, as it applies only to projects completed after our sampling period. Appendix C provides more details on this recent development.

[2]The data collected by the U.S. Census Bureau is tightly regulated. Title 13 of the U.S. Code provides a legal framework for acquiring, using, and protecting confidential data, ensuring it is used only by authorized personnel for statistical purposes. Moreover, the Bureau integrates data from other government sources. These data are governed by similar confidentiality provisions in Title 26 of the U.S. Code (for IRS records) and in the Confidential Information Protection and Statistical Efficiency Act (for other statistical agencies). From a legal standpoint, the Census Bureau is authorized to give access to confidential microdata only insofar as it derives a clear benefit for its programs and as long as users are sworn to observe the limitations imposed by Title 13. A detailed discussion of the Census Bureau's objectives for sharing its data is in Appendix C.1.

[3]The only FSRDCs to close during our sampling period were those at Carnegie Mellon University (CMU), opened in 1996 and closed in 2004, and RTI International, opened in 2010 and closed in 2018. In 2018, the FSRDC network formally became part of the Center for Enterprise Dissemination, with no change to its operations.

are performed on Census Bureau servers via a secure client physically located in the data center. Second, researchers can access an FSRDC only after submitting an application that outlines the research question and the data needed to answer it. Approval requires passing an evaluation of the merits, feasibility, disclosure risks, and benefits for the Census Bureau. Third, pre-approved researchers must receive a Special Sworn Status after thorough security checks. Special Sworn Status individuals take an oath of confidentiality and are subject to the same legal obligations and penalties as Census Bureau employees. Fourth, results produced in an FSRDC undergo a disclosure review before they can be exported.

While CES was originally established to provide access to manufacturing data (McGuckin et al., 1993), FSRDCs now support research on a much broader range of phenomena using datasets that have become household names among economists, such as the LBD or the LEHD (CES, 2017). Two developments drove this expansion in scope. On the one hand, CES staff and FSRDC users have contributed to the creation or improvement of databases as part of their research projects; a recent example is the creation of the Management and Organizational Practices Survey (Bloom et al., 2019). On the other hand, other federal agencies began making their confidential data available through the same infrastructure. The agencies partnering with the FSRDC network include the Agency for Healthcare Research and Quality, Bureau of Economic Analysis, Bureau of Justice Statistics, BLS, NCHS, and National Center for Science and Engineering Statistics.[4]

Several accounts suggest that the FSRDC network enabled path-breaking advances in economics (CES, 2017).[5] However, the strict security requirements imposed by the Census Bureau have limited

[4]We term the data collectively provided through this program as "FSRDC data" and use it to mean all confidential microdata distributed by the FSRDC network, regardless of how the dataset was created or the specific government agency that collected the original records. Different agencies began distributing datasets at different times, but unfortunately we cannot precisely track this expansion in our study. In our estimates, we rely on a tight set of time fixed effects to control for the introduction of new datasets. In Appendix F, we provide additional details on the most widely used FSRDC datasets.

[5]For instance, CES' establishment-level data are credited to have facilitated a generalized shift from "representative firm" thinking toward research that takes into account intra-industry heterogeneity (McGuckin, 1995).

the use of administrative data relative to European countries (Cole et al., 2020). Researchers have argued that such limitations risk undermining U.S. leadership in cutting-edge empirical research (Card et al., 2010). The gradual expansion of the FSRDC network was intended to address this concern, but its effectiveness remains unclear. The program is also costly: by some estimates, maintaining the network cost over $7 million per year as of 2021, not accounting for set-up expenses and the costs of space for the data enclaves themselves.[6] These considerations make it important to assess the effects of the FSRDC program on academic research and policymaking.

3 Research Design

There are two key challenges in empirically assessing the impact of data access on research. First, researchers do not systematically cite data sources and may be influenced by data they do not use directly, making it difficult to track both data use and access. Second, the correlation between the use of specific data and publication quality is likely to provide an upward-biased estimate of the causal effect of data access. Researchers using such data might produce better research not because of the data themselves, but because they have greater resources or are more creative. The empirical challenge is thus to find a research design that provides (a) a measure of access to confidential data independent of their use and (b) credible variation in the availability of the same data.

In this paper, we employ the staggered geographical expansion of the FSRDC network as a source of variation in data availability for academic economists. Even though researchers could, in principle, access confidential FSRDC data through collaborators or by visiting the Census Bureau's headquarters, co-location with an FSRDC decreases the barriers to using them. In particular, our research design leverages the requirement that FSRDCs' confidential microdata can be analyzed only in secure facilities. This allows us to proxy data access at the level of individual researchers as the presence of an FSRDC in their institution. To avoid confounding effects from the generalized growth of empirical research, our main design compares empirical economists co-located with

[6]Estimates are taken from https://deepblue.lib.umich.edu/response_FSRDC_Directors_2021. A cost-benefit assessment is provided in Section 7 and Appendix G.

FSRDCs to empirical economists lacking local access.[7] Ruling out the presence of pre-trends in our estimates would further validate this research design.

A potential concern with our identification strategy is that FSRDC locations are not randomly assigned. If time-invariant university attributes such as specialization or resource endowments drive location choice, we can absorb them using fixed effects. More concerning are two possibilities. First, changes in research resources or other productivity shocks concurrent with an FSRDC opening could threaten the internal validity of our estimates. Second, FSRDCs may be established at institutions expected to benefit more from confidential data. Selection on expected gains does not, by itself, bias a difference-in-differences estimate, but it may limit external validity (Ghanem et al., 2025). To assess these concerns, we combine archival evidence, nine interviews with FSRDC administrators, and quantitative tests of the factors predicting FSRDC openings (see Appendix A).

Opening a new FSRDC requires universities to submit a competitive application to the NSF, jointly evaluated with the Census Bureau (Atrostic, 2007). Archival and interview evidence suggest that this process reflected both local demand and factors that weakened sorting. First, the timing of FSRDC openings often depended on idiosyncratic factors. Interest in establishing a center sometimes followed changes in university leadership, and openings could be delayed for years even after an NSF grant was awarded.[8] These factors make it less likely that the precise timing of openings systematically coincided with concurrent productivity shocks. Second, geographic equity was an explicit consideration in the network's expansion. As a former FSRDC administrator explained, "Many institutions were interested in opening an RDC, but the NSF was interested in

[7] One natural concern is that trends in the growth of empirical research might underlie both the increasing use and impact of confidential data. For instance, changes in editorial preferences at top journals might lead to productivity gains for applied researchers, even in the absence of local FSRDC openings. However, using empirical economists who lack FSRDC access as a control group allows us to control for these trends.

[8] In unreported analyses, we examine whether researchers respond in anticipation of an FSRDC opening. We define the anticipation period as the interval between the announcement of an NSF grant supporting a new FSRDC and the start of its operations. On average, this period lasts 1.42 years, although in some cases it extends to four years. We find no evidence that project approvals increase during this period. Instead, the increase emerges only after the FSRDC becomes operational, and longer announcement-to-opening gaps do not predict larger post-opening effects.

kind of parity across the U.S. so that researchers in one part of the country had the same access as researchers in another part of the country did" (interview T14). As a result, even top-tier universities with substantial resources could wait years for local access when a nearby FSRDC already existed.[9] Consistent with this account, Appendix Table A1 shows that geography and the size of the potential user community predict FSRDC openings, while research productivity trends do not.

Taken together, this evidence suggests that institutional frictions and geographic considerations played an important role in the timing and location of FSRDC openings, reducing concerns that they simply tracked changes in research productivity. We assess the remaining identification concerns directly using alternative sources of variation (Section 5.3), extensive robustness checks (Section 5.4), and selection bias decomposition (Appendix A).

4 Data

Our research question requires us to collect data on a few key dimensions: (a) identifying the set of relevant academic economists and their affiliations, (b) matching academics with the quantity, quality, and policy impact of their publication output, (c) identifying researchers who are empirically oriented, and (d) measuring the diffusion of FSRDCs and the adoption of confidential FSRDC data.

4.1 Building the Universe of Publishing Economists

The main data source we leverage is EconLit, a proprietary database of economic scholarship curated by the American Economic Association (AEA). Compared to other popular databases of scientific publications, EconLit has broader coverage of economics journals and includes *Journal of Economic Literature* (JEL) codes that classify articles by topical focus. However, EconLit lacks unique author and affiliation identifiers, which prevents us from reliably linking researchers with their scientific output. To reconstruct authors' publication records, we need to disambiguate publication metadata, a common but difficult and time-consuming task in bibliometric analyses. We disambiguate our data in several steps outlined below (and detailed in Appendix B). We start

[9] For example, Stanford University had to wait 11 years to open its FSRDC branch due to the presence of the relatively nearby Berkeley FSRDC. See Appendix A for additional institutional detail.

with the complete set of 839,513 scientific articles published in 1,856 journals between 1990 and 2019. While starting with such a heterogeneous body of articles makes the disambiguation task harder, including every paper is essential because it allows us to detect mobility events by tracking changes in academic affiliations across published work.

We then proceed in two steps. First, we standardize the 178,798 affiliations in EconLit to identify researchers' locations and treatment status over time, while also restricting the sample to U.S.-affiliated economists who could be co-located with an FSRDC. We include all U.S. doctorate-granting institutions in the 2018 Carnegie Classification of Institutions of Higher Education, as well as major institutions active in economic research, such as the IMF, RAND Corporation, World Bank, and regional Federal Reserve offices. Using fuzzy matching and manual checks, we standardize the 11,491 distinct spellings associated with the 439 U.S. research institutions in this expanded list.

Second, we disambiguate researchers' names using a graph-theoretic procedure (Önder and Schweitzer, 2017). This approach assumes that the combination of first, middle, and last names uniquely identifies each economist, while conservatively treating ambiguous names without a middle name (Card et al., 2020). To avoid including researchers in unrelated fields who only occasionally publish in economics journals, we match our data to 19 yearly lists of AEA members spanning 1993–2019. We also exclude researchers who are or have been affiliated with the U.S. Census Bureau or any partner agency, since they may enjoy privileged access to confidential data.[10] This procedure yields a starting sample of 14,589 U.S.-based economists who have been AEA members, which we later refine based on their methodological orientation. Appendix B provides additional details on the construction of the researcher panel and validates our author disambiguation against the OpenAlex database.

4.2 Publication-Level Information

The EconLit database provides article metadata, including authors, journal, year of publication, JEL codes, and, for most articles, abstracts. We collect missing abstracts from sources such as

[10] Our results are almost identical if we include non-AEA members or researchers affiliated with the Census Bureau. Results are available from the Authors upon request.

Google Scholar, EconStor, and JSTOR. We then augment EconLit with annual citation counts from SSCI/Web of Science and information from acknowledgment sections obtained from Crossref and Dimensions. We construct individual-level productivity measures using all articles published in journals that are i) indexed in Web of Science, ii) published in English, and iii) listed in SCImago under the subject areas "Economics, Econometrics and Finance," or "Business, Management and Accounting." This results in a final set of 188,181 articles published in 158 journals during 1990–2019. We can match academic citation data for 97.2% of these articles.

In addition to scientific impact, we are also interested in directly assessing the impact of FSRDC access on the policy relevance of economics research. To do so, we leverage novel data from Overton (see Appendix E for details). In particular, Overton collects data on scientific articles cited by policy documents from a wide range of institutions, including government reports and international organizations. Policy sources are collected directly from organizations' websites and merged into the bibliographic records using title, authors, and year of publication. We match these data to our EconLit sample, resulting in a unique database that enables examination of economic research consumption by policy sources (Yin et al., 2022).

We compile a list of all articles that *directly* use restricted-access data available only through an FSRDC. We identify these articles using several complementary strategies. Projects using confidential data are expected to disclose this use in the paper's acknowledgments. We therefore search the Web of Science, Scopus, JSTOR, and Google Scholar for common expressions indicating the use of FSRDC data. We also exploit the CES requirement that researchers submit a working paper for online publication upon project completion. We collect the metadata for all CES working papers and manually match them to published articles.[11] We also capture how FSRDC data affect research *indirectly* by enabling findings that shape subsequent work. We do so in two ways. First,

[11] The requirement to publish FSRDC output in the CES working paper series is difficult to enforce, which is why we also rely on the additional steps detailed in Appendix C.4. Moreover, we cannot separately identify papers stemming from internal Census projects that are not subject to the FSRDC application procedure. However, this should not affect our analyses because we exclude researchers formally affiliated with the U.S. Census Bureau.

we identify articles that cite papers using FSRDC data and thus explicitly build on their findings.[12] Second, we tag papers with JEL codes most representative of FSRDC research. This captures work that is thematically close to research conducted in FSRDCs, regardless of whether it cites that research.

Finally, we characterize the methodological approach of each article. We first use a machine learning classifier that estimates the probability that an article is empirical based on its journal, title, abstract, year of publication, and JEL codes.[13] Following Angrist et al. (2020), we classify a paper as empirical if it uses data to estimate an economically meaningful parameter (see Appendix D for more details). The results of this classification effort are highly reliable, as validated in several manual checks. In turn, we use this publication-level classification to characterize the methodological orientation of each publishing economist in our sample. We also follow the approach of Currie et al. (2020) to code more detailed information about each paper's research design (Brodeur et al., 2020). We tag each paper that explicitly mentions a particular method (e.g., difference-in-differences) or data type (e.g., administrative data) in the title or abstract. The complete list of keywords used is reported in Appendix C.6.

4.3 Researcher-Level Information

Thanks to the host of article-level variables outlined above, we can compute several metrics that capture the research productivity of each researcher. In particular, we compute yearly publication counts weighted in three ways: by the prestige of the outlet, by scholarly citations received, and by policy impact. The first metric sums the number of publications in the top field and top five economics journals, which we collectively refer to as "top publications."[14] The second metric captures scientific impact by weighting publication counts by the citations received in a window up

[12] We exclude papers directly using FSRDC data from this count because they are likely to mechanically cite prior FSRDC research using the same or related datasets. Such citations may reflect the need to reference earlier work that developed the data or documented its construction, rather than broader exposure to FSRDC-based findings.

[13] We are indebted to Ryan Hill for sharing the code and the training data originally used in Angrist et al. (2020).

[14] These journals are especially important for career progression in economics. We rely on the list assembled by Heckman and Moktan (2020) and reported in Appendix B.

to five years after publication. Finally, we use the count of articles that receive mentions in policy documents as a third outcome.

One of our objectives is to measure changes in research trajectory after obtaining access to confidential microdata (Furman and Teodoridis, 2020). This is possible by leveraging the JEL codes, a hierarchical taxonomy of research topics in economics. We rely on 2-digit JEL codes to track the topical focus of research (Card et al., 2020) and explore the likelihood that a researcher writes a paper with JEL codes she has never used in her previous work. While a potential shortcoming of this metric is that JEL codes are author-assigned, all our models will include researcher fixed effects, effectively controlling for researchers' differing styles in assigning such codes (details in Appendix C.6).

We use our article-level methodological classification to characterize each scholar's research orientation. In our primary analyses, we define an empiricist as a researcher for whom a majority of their early publications are empirical.[15] We validate this classification in several ways. First, we use a case-control approach to compare our classification of editorial board members across journals with different methodological orientations, such as the *Journal of Economic Theory* and *AEJ: Economic Policy*. Second, we compile records of Ph.D. students graduating from U.S. universities from the annual JEL listings and compare our classification with their dissertation fields. Both tests confirm the face validity of our approach (Appendix D.2). Finally, in the results section, we repeat our analyses using progressively more stringent cutoffs for defining empirical scholars and obtain consistent results.

We use this classification to define our main sample of 10,060 empirical economists, yielding an unbalanced panel of 155,720 researcher-year observations after imputing missing years between

[15] We classify researchers using up to their first five publications, or all observed publications if fewer than five are available. This measure has the advantage of being available for every publishing researcher in our sample. Results change little if we instead classify empiricists based on all lifetime publications, only publications predating co-location with an FSRDC, the first publication alone, or a more restrictive majority of the first seven publications. The pre-FSRDC measure is not defined for researchers who start their careers at institutions already hosting an FSRDC, which leads us to prefer the definition in the main text. Appendix D provides additional details and validation.

a researcher's first and last observed publication. For years without publications, we also impute institutional affiliation, which may introduce measurement error when affiliation changes are observed across non-consecutive years. We assign the old affiliation to the first third of the missing years and the new affiliation to the remaining two-thirds, but the data are largely unchanged under alternative imputation rules. Appendix Figure B1 summarizes the construction of the author-year panel from the bibliographic records.

4.4 University-Level Information

We collect the details of FSRDC openings from the 15 CES research reports published between 2005 and 2020. Figure 1 synthesizes the expansion of the FSRDC network in time and space. Overall, the figure confirms the oral accounts testifying a conscious effort by the U.S. Census Bureau and the NSF to provide geographic balance in access. We see that the data centers spanned different regions of the United States, starting from some of the major centers of economic research (e.g., Boston, Berkeley, Los Angeles) but also leaving out until much later other prominent universities if relatively close-by alternatives were present (e.g., Stanford, Yale, Princeton). The complete list of 31 FSRDCs established in our sampling period is in Appendix C.

Data on the research intensity of economics departments come from Kalaitzidakis et al. (2003), who rank institutions based on publications weighted by journal prestige. This ranking fits our purposes well because it considers publications from 1995 to 1999, predating the establishment of most FSRDCs. We use it to classify universities into seven tiers based on economics department rank.[16] These data show that less research-intensive institutions gained access at a pace similar to that of more prestigious peers (Appendix Figure H1). This balanced expansion of the FSRDC network (Appendix Figure H2) is consistent with our interview evidence and reduces concerns that high-status universities were explicitly prioritized.

[16]This choice yields a roughly equal number of observations across tiers, but our results are robust to alternative choices. The ranking in Kalaitzidakis et al. (2003) includes only the top 200 research institutions worldwide, of which 88 are in the U.S. We assign institutions not included in the ranking to a residual tier (Appendix B.2).

4.5 Summary Statistics

Table 1 reports summary statistics for the dataset we assembled. Panel A summarizes the cross-section of 10,060 empirical researchers in our main sample, representing roughly 63% of all U.S.-based publishing economists. Nearly 4% of these researchers have directly used FSRDC data in published work, while 28.4% have cited at least one paper using FSRDC data. The average researcher publishes 1.9 articles in top outlets over the period they appear in our sample. However, the number of policy-relevant articles is close to 3.9, suggesting that many policy-relevant findings are not published in the most prestigious outlets. Appendix E provides additional details on our policy citation data.[17]

Our data show that, among the 10,060 researchers in our sample, about 79% never had access to an FSRDC at their employing institution. Appendix Figure C1 further distinguishes researchers who always had access (532), gained access through an FSRDC opening at their institution (928), or moved to an institution with an FSRDC (615). Panel B reports summary statistics for the unbalanced researcher-year panel, which spans 1990–2019. The average researcher publishes about 0.13 papers per year in top economics journals. Across researcher-years, 0.005 papers directly use FSRDC data and 0.041 build on prior research using these data.

5 Results

Our baseline specification is at the empirical researcher i, university j, and year t level and takes the following form to estimate the effect of local data access:

$$y_{i,j,t} = \alpha + \beta PostFSRDC_{j,t} + \mu_i + \delta_t \times \omega_j + \epsilon_{i,j,t}, \tag{1}$$

where the dependent variable $y_{i,j,t}$ is publications by researcher i affiliated to university j in year t. The main independent variable $PostFSRDC_{j,t}$ is a time-varying dummy that takes the value 1 after a researcher gains access to an FSRDC facility at her university and 0 otherwise.

[17] Descriptive patterns for articles using FSRDC data are also reported in Nagaraj et al. (2025), which uses a subset of the data collected for this study to report descriptive metrics aimed at a U.S. policymaking audience.

The specification includes researcher fixed effects (μ_i) that control for time-invariant differences across researchers. Finally, university tier-by-year fixed effects ($\delta_t \times \omega_j$) control for time trends in productivity or resources that might be specific to universities of different research intensity.[18] We present results with the treatment dummy coded as an absorbing state, which means that individuals do not lose the treated status even if they move to a non-treated institution. This approach accounts for potential delays in the publication process and makes it straightforward to interpret $\hat{\beta}$ as an average post-access DID estimate.

5.1 Impact of FSRDC Openings on the Diffusion of Confidential Data

We begin by examining whether FSRDC openings increase the use of confidential data. Table 2 reports that local access increases the use of FSRDC data by about 0.006 papers per year, or 118% relative to the sample mean of 0.005. The estimate is only slightly smaller when we include university tier-by-year fixed effects (Column 2). We obtain similar results using alternative measures of FSRDC use, such as having an FSRDC project approved or submitting a working paper to CES (Appendix Table H1). We also find a meaningful increase in citations to prior work using FSRDC data (Columns 3 and 4). Although the increase relative to the sample mean is smaller, around 26%–42%, this result is striking because lack of direct access should not prevent researchers from learning about published work.[19] This suggests that opening an FSRDC increases local economists' exposure to prior research using confidential data. Both results are robust to specifications that account for the large number of zeros in these outcomes (Appendix Table H2).

The validity of our regressions hinges on the assumption that treated and control units would have followed parallel trends absent treatment. We assess this assumption by estimating an event-study version of the specifications in Table 2. Specifically, we estimate $y_{i,j,t} = \alpha + \Sigma_z \, \beta_z \times 1(z) + \mu_i + \delta_t \times \omega_j + \epsilon_{i,j,t}$, where μ_i and $\delta_t \times \omega_j$ represent researcher and university tier $\times$ time fixed effects,

[18] We also show results with separate university and time fixed effects. In Section 5.3, alternative identification strategies leveraging within-university variation enable the inclusion of tighter university-by-year fixed effects.

[19] Note that we exclude from the count of citing papers those that directly use FSRDC data. This ensures that the increase is not mechanically driven by papers citing prior work that uses the same or related data.

as before. z represents the "lag," or the years elapsed since an empirical researcher first received access to confidential data via a local FSRDC. Figure 2 reports the effects of FSRDC access on data use (panel (i)) and citations to FSRDC-based research (panel (ii)). Both panels show no evidence of pre-trends; rather, the effects emerge gradually and stabilize after three to five years, a pattern consistent with publication lags and the gradual diffusion of data in academic research.

5.2 The Effects of Data Access on Scientific Productivity

Having established that local FSRDC access increases the diffusion of confidential data, we next estimate its effects on researchers' productivity. We use a specification similar to the one above, but broaden the outcomes beyond research directly using FSRDC data. For each researcher, we count the number of papers published yearly in top field outlets and top five generalist journals, which carry outsized weight in determining career trajectories (Heckman and Moktan, 2020). To measure scientific impact, we use a second dependent variable based on the number of citations each paper receives up to five years following its publication. Finally, we use the number of publications cited in policy documents to capture the policy relevance of research.

Table 3 presents the results from this analysis. For each outcome variable, the first model reports results with university and year fixed effects, and the second model presents results that include time trends for university tiers. Across the board, the results suggest that FSRDC access raises the productivity of empirical researchers. Empirical economists produce about 0.023 more publications in the top journals, which, compared to a baseline of 0.126, is an 18.3% increase in output. Columns 3 and 4 show that the increase in productivity does not come at the expense of reduced impact: citation-weighted publications increase by about 1.24–1.71 (25%–35% increase against a baseline of 4.9). This result suggests that access to confidential data leads to significantly higher-impact publications.

Next, we estimate the effect of FSRDC access on the policy impact of economic research. Columns 5 and 6 show that applied researchers are more likely to have their work referenced by policy sources after gaining access. The increase relative to the sample mean is slightly larger for U.S.-based policy sources, confirming the impact of federal data infrastructure on evidence-based

policy discussions in the United States (Appendix Table H3). Interestingly, these effects do not appear to be driven by a shift toward more policy-oriented work. Such a shift should be reflected in greater use of language emphasizing policy implications or in a move toward policy-related JEL codes. We find no evidence of either pattern in article abstracts or author-reported JEL codes (Appendix Table H4). Instead, our results are consistent with the idea that access to better data increases the scientific quality of applied research and, in turn, its policy relevance.

We assess the reliability of our research design by examining pre-trends in the outcome variables. Figure 3 reports event-study estimates analogous to those in Figure 2 for the measures of research productivity discussed above. Across all panels, publication outcomes are flat before treatment, remain low in the first few years after an FSRDC opening, and then gradually increase before stabilizing at significant levels. Appendix Table H5 and Figure H3 repeat the analysis using the doubly robust DID estimator of Callaway and Sant'Anna (2021), confirming that our results are not driven by heterogeneous responses to FSRDC openings at different times. Finally, we formally assess the parallel trends assumption bounding the change in differential trends between treated and control economists required to render our estimates insignificant. Appendix Figure H4 shows that deviations from the pre-treatment trends would need to be implausibly large to account for our findings.[20]

5.3 Alternative Identification Strategies

Our analysis identifies the effect of FSRDCs by comparing empirical researchers at institutions that host them with those lacking local access. The divergence in confidential data use and productivity following the opening of a local FSRDC allows us to estimate the causal effect of increased data access. While the institutional setting and the absence of pre-trends support this research design, our interviews with individuals familiar with the history of the FSRDC network suggested alternative sources of variation that we use to validate our findings.

Academic economists tend to specialize along methodological lines and devote themselves

[20]For instance, for publications in top journals, we can reject a null effect unless we allow the linear extrapolation across consecutive periods to deviate by more than 16% from the linear trend estimated in the pre-period.

primarily to empirical work or theoretical modeling (Backhouse and Cherrier, 2017). Rather than evaluating our research question only among empirical researchers, we can also confirm our baseline result by using theoretical economists at the same university as a control group. Accordingly, we add theoretical economists to our sample and interact the independent variable with $Empiricist_i$, a dummy that takes the value of one for the economists whom we classify as empiricists. This allows us to include university-by-year fixed effects, providing even more fine-grained controls for university-level research dynamics and time trends. In particular, exploiting only within-university variation rules out potential university-wide confounders that might correlate with opening an FSRDC (e.g., a sudden influx of funding in the economics department).[21] The results, reported in Appendix Table H7, confirm our main estimates.

Nevertheless, using theoretical economists as a control group does not rule out productivity shocks specific to empirical researchers that coincide with FSRDC openings. For instance, host institutions might be increasing support for applied work. We address this concern in two ways. First, we exploit within-university variation in "exposure" to FSRDCs across empirical fields. Our interviews suggest that FSRDC access is less relevant in fields such as development or behavioral economics, providing variation among empiricists in the same university. Although less precisely estimated, Appendix Table H8 shows that the benefits are concentrated among researchers in fields most likely to benefit from Census microdata.[22] Second, we find no increase in grant funding or research assistant support acknowledged in empiricists' papers after an FSRDC opens (Appendix Tables H22 and H10). While this test cannot rule out all unobserved confounds, it reduces concerns that our results reflect broader resources for applied research rather than FSRDC access.

During our interviews, we also discovered that when a consortium submitted an FSRDC appli-

[21] This specification differs from a triple-difference estimate because the term "Post" is not defined for never-treated units due to the staggered rollout of FSRDCs. We can sidestep the need for a matching estimator because empiricists are compared with theorists who gain access at the same time by being affiliated with the same institution. Appendix Figures H6 and H7 show the raw publication outcomes for empirical and theoretical economists, respectively.

[22] Relatedly, our main results are robust to controlling for field-specific time trends using JEL field $\times$ year fixed effects (Appendix Table H9).

cation, the decision on which member would host the data center often resulted from a compromise among the interested parties. This suggests that within-consortium variation in distance from the hosting institution could also be used to identify the effect of data access while holding constant unobserved local factors shared by consortium members. For instance, this would account for investments in empirical research made by consortium members around the same time or if they all show a rising demand for confidential microdata. Leveraging this intuition, Appendix Table H11 shows that our results are mostly robust to within-consortium estimates.[23] In sum, the combined evidence from these additional research designs confirms that access to confidential microdata increases the quantity, quality, and policy relevance of empirical research.

5.4 Other Robustness Checks

We conduct several additional tests to assess the robustness of our findings. First, we address concerns about the endogenous placement of FSRDCs. The absence of pre-trends in individual productivity, together with the evidence in Appendix A, already reduces concerns that our results reflect systematic sorting of FSRDC openings. We examine this issue directly using a university-year panel. In addition to finding significant aggregate gains at institutions opening FSRDCs, the university-level estimates show that these effects are concentrated among empiricists (Appendix Table H12). The event-study estimates in Appendix Figure H8 show no evidence that FSRDCs systematically opened at institutions already experiencing rising research intensity.[24]

Second, we test whether our results are driven by the researchers or institutions most directly involved in establishing an FSRDC. We identify the recipients of NSF grants supporting each new center, and Appendix Table H13 shows that our results are robust to their exclusion. We obtain similar estimates when restricting the sample to researchers at universities located in the same city

[23] Appendix Figure H5 illustrates how the effect of FSRDC access rapidly declines with distance, even among consortium members. We thank an anonymous referee for suggesting this alternative identification strategy.

[24] We also estimate the effects separately by FSRDC adoption cohort. The estimates are larger for earlier cohorts and smaller for later cohorts, although later adopters have substantially shorter post-treatment windows over which publication effects can emerge (Appendix Table H6).

as an FSRDC but that do not host the data center themselves (Appendix Table H14). This test is particularly informative because researchers at nearby non-host institutions are unlikely to have been directly involved in establishing the FSRDC but can still benefit from improved local access.

Third, we consider alternative measures of "exposure" to confidential data in both geographic and intellectual space. Appendix Table H15 shows that an economist's likelihood of using FSRDC data declines rapidly with distance from a data center. Consistent with this pattern, our estimated effects grow stronger with proximity and are largest for researchers who enjoy access on their own campus. Similarly, the effects increase monotonically as we use more stringent definitions of empirical researchers (Appendix Figure H9). The placebo regressions in Appendix Table H16 yield precisely estimated zeros for theoretical economists, reassuring us that our findings are not an artifact of the threshold used to classify researchers as empiricists.

One might worry that researcher mobility drives our results, especially since we infer move timing from publication data. Appendix Table H18 shows that our estimates are robust to excluding researchers who gain access through mobility events. A related concern is that FSRDC openings attract more productive economists, making our effects reflect a zero-sum reallocation. We find no evidence that FSRDC openings systematically change department size, researcher quality, or empirical composition (Appendix Table H19), nor that they affect researcher exit rates overall or among empiricists (Appendix Table H20). Our estimates are also nearly unchanged when we control for researcher age (Appendix Table H17), reducing concerns that changes in researcher composition explain our estimates.

6 Mechanisms

A. Spillovers from Data Access: So far, we have shown that confidential data access leads to greater productivity for empirical economists, but note that we did not restrict the sample to those who adopt the data directly in their work. This leaves open the question of the extent to which productivity effects are driven by researchers who directly employ FSRDC data. Co-localization with an FSRDC may affect research quality through two channels: (1) directly, by allowing

researchers to use confidential data to produce more impactful research; and (2) indirectly, through spillovers to researchers who do not adopt the data themselves.

We assess the role of spillovers by examining how much our results are driven by researchers who directly use FSRDC data. Table 4 reports our main estimates after excluding these researchers. The coefficients are 21%–25% smaller but remain large and significant. Thus, while FSRDC openings raise productivity partly by enabling direct data use, they also generate substantial spillovers to researchers who do not adopt them directly (Myers and Lanahan, 2022). This points to an additional mechanism by which FSRDC access shapes economics: exposure to research based on confidential microdata. Consistent with this interpretation, our earlier finding that FSRDC openings increase citations to FSRDC research suggests that local access raises exposure to such work,[25] with potential downstream effects on research trajectories.

Appendix Figure H10 provides two tests of this exposure mechanism. First, we interact FSRDC access with an indicator for researchers who cited work based on FSRDC data. The positive productivity effects are concentrated among those whose work builds on prior FSRDC research. Second, our interviews suggest that researchers often learn about the potential of FSRDC data through colleagues' work (Appendix C.6). We therefore separately estimate effects for treated empiricists in departments with and without colleagues using confidential microdata. We find no positive spillovers for researchers with fewer colleagues directly using FSRDC data. Together, these results suggest that spillovers from co-location operate by increasing researchers' exposure to work based on restricted-access FSRDC data.

B. How Exposure to FSRDCs Increases Research Quality: We explore two specific channels through which awareness of past research carried out with FSRDC data might improve research output. First, researchers might be inspired to formulate new research questions close to the topics investigated with FSRDC data. We examine whether researchers exposed to FSRDCs are more likely to explore topics such as labor, trade, or firm productivity. Table 5 confirms that the likelihood

[25] Appendix Figure H11 shows that an economist's likelihood of using and citing confidential data declines rapidly with distance. The effects are stronger closer to a data center and largest for researchers with access on their own campus.

of working on JEL codes commonly associated with FSRDC data goes up by 9.3% over the sample mean, while there is no statistically significant increase for the remainder of JEL codes.[26] This result is robust to alternative ways to select which JEL codes most represent FSRDC-based research and gradually attenuates as we make the selection less stringent (Appendix Table H21).

Second, exposed researchers might learn from the research design of FSRDC studies, leading them to adopt similar methods or data. Table 5 shows that treated researchers increase mentions of quasi-experimental methods. The absence of a similar effect on laboratory experiments or randomized control trials implies that this result is not a byproduct of a broader "credibility revolution." Learning about FSRDC data might also inspire the researcher to search for new datasets with similar characteristics and lower bureaucratic hurdles. This theme also emerged in our interviews, with respondents noting how administrative data from foreign countries often offer similar advantages and are easier to access (Appendix C.6). Indeed, we find evidence that researchers are more likely to employ microdata from other sources, while a similar increase is absent for traditional research surveys.

C. Distributional Effects and Collaboration: Beyond its average effects on research output, FSRDC access may also reshape the distribution of research opportunities among researchers. We examine whether these gains are concentrated among researchers who were already highly productive, measuring baseline productivity as average annual publications before gaining local FSRDC access. Appendix Table H24 shows that the estimated gains are largest among researchers with lower baseline productivity, consistent with FSRDC access narrowing productivity differences within treated institutions. One possible interpretation is that highly productive researchers are better able to overcome geographic access barriers through existing resources or professional networks, making local access especially valuable for researchers with fewer alternatives.

Access may also reshape research opportunities by changing the set of collaborators researchers can draw on. Appendix Table H23 shows little change in aggregate measures of collaboration.

[26] We also explore whether FSRDC access encourages scholars to explore new topics. Column 1 of Table 5 indicates a higher propensity to explore new JEL codes, but the results are imprecise.

However, the composition of new collaborations changes after access. Appendix Table H25 shows that treated researchers form more new collaborations with junior scholars and researchers with Census-relevant expertise, as well as more collaborations with researchers at distant institutions. New local collaborations decline slightly. Taken together, these results suggest that FSRDC access affects not only who benefits from confidential data, but also how researchers assemble the teams through which they pursue new research opportunities.

7 Increasing the Diffusion of Confidential Microdata

Our analysis shows that FSRDC openings increase scientific productivity and policy impact among applied researchers. These gains raise the broader question of how to expand access to confidential data while preserving existing confidentiality protections. Although the optimal level of access depends on social preferences over privacy (Abowd and Schmutte, 2019), there are two margins for increasing use within the current regulatory framework. One is to reduce frictions that limit the use of existing FSRDCs. The other is to open additional facilities, which requires weighing the benefits of greater access against its costs. While a complete analysis is beyond the scope of our paper, we use our reduced-form estimates to provide guidance on both.

A. Barriers to Using Confidential Data: The use of FSRDCs rapidly declines with geographical distance from the closest FSRDC, more than halving for a distance of 50 miles (Appendix Figure H11). Yet we also noted that some scholars can write several papers using confidential data, even when affiliated with universities farther from an FSRDC. What could be driving this heterogeneity? Researchers' financial constraints are a natural candidate. We learned during our interviews that researchers often add research assistants to their FSRDC project and help them obtain Special Sworn Status. In this way, they can bypass the hurdle of physical presence in an FSRDC by relying on their assistant to access data in the enclave. This practice suggests that access constraints might be more binding for researchers who lack the resources to recruit assistants.

Figure 4 reports estimates of FSRDCs' effect on the use of confidential data at increasing distances from the hosting institution. Each panel separately shows the impact for researchers we

classify as having access to more resources. While we do not have precise data on individual research budgets, we employ several proxies for researchers' resources: affiliation with an FSRDC consortium[27] (Panel (i)), the prestige of researchers' affiliation (Panel (ii)), and the availability of external funding (Panel (iii)). All three panels paint a consistent picture: geographical distance is especially binding for less resource-endowed researchers. Panel (iii) is especially stark in showing how usage outside a radius of five miles from the host institution is mainly confined to researchers who can secure external funding sources.

Documenting the role of resource constraints in accessing confidential data has distributional implications, especially because access to data disproportionately benefits marginalized researchers (Nagaraj et al., 2020). The research impact of FSRDCs could be magnified by expanding how many researchers use confidential data, even without increasing the number of FSRDC facilities. These results help support recent efforts by the Census Bureau to provide virtual access to confidential Title 13 data. While this program seems valuable, the open question remains how to increase accessibility for the data that cannot be shared remotely.

B. Assessing the Cost-Effectiveness of FSRDCs: To assess whether further expansion of the FSRDC network would be justified, we conduct a "back-of-the-envelope" cost-effectiveness analysis. A complete cost-benefit assessment is beyond the scope of our paper: we lack data for disciplines outside economics, and we cannot measure the benefits occurring to the Census Bureau from FSRDC research. However, we can map the productivity effects of FSRDCs into monetary benefits and compare them with operating costs (details in Appendix G). Specifically, our analysis requires an estimate of (i) the cost of operating an FSRDC, (ii) the value of an economics publication, and (iii) the number of additional publications induced by FSRDCs.[28]

A rough assessment of FSRDCs' operational costs can be derived from the official documen-

[27] Researchers affiliated with consortium members usually have their data access fees waived, while non-affiliated users usually pay a fee of around $15,000-20,000 per year.

[28] This evaluation is necessarily speculative, so our preference is to make choices that provide a conservative assessment of FSRDCs' benefits. Depending on the application, some parameters might be preferred over the ones we use. For this reason, we leave the full menu of choices in the Appendix G for interested readers.

tation on the CES website. We use the upper bound yearly estimates of $185,000 per data center, and we estimate that, on average, 77% of that amount pertains to economics research projects.[29] Evaluating the monetary value of a scientific publication is more challenging. In Appendix G.2, we discuss three alternative approaches. Our preferred estimate is based on the implicit value that universities attribute to an additional top publication, as revealed by their faculty salaries. This approach yields an estimated present value of $26,916 per additional publication, the most conservative among our potential estimates (Appendix Table G1).

We then compare the costs of FSRDCs' operations with the value of the additional publications they generate. We present our results as a function of two interrelated parameters: the effect of FSRDC access on researchers' productivity and the number of consortium partners that sustain the expenditures for the data enclave. Appendix Figure G2 shows how much a publication must be valued to justify opening an FSRDC. Using our estimates in Table 3, we find that the value of an FSRDC is evident for medium-sized consortia of four or five institutions despite conservative assumptions about the value of additional publications. When considering institutions with a larger number of applied scholars and treatment effects reflecting the dynamically increasing impact of FSRDCs documented in Figure 3, even individual universities could justify opening an FSRDC.

Strikingly, investments in FSRDCs appear more effective at fostering scientific production than alternative uses of the same resources (Appendix Figure G3). Relative to estimates of the impact of research grants to economists, our estimates imply that resources invested in an FSRDC yield 1.7–6.5 times more publications.[30] Taken together, this analysis suggests that the value added by FSRDCs likely exceeds their operating costs despite relatively limited direct use. The case for operating an FSRDC is even stronger for institutions that place greater value on scientific

[29] We include in the calculation the start-up funding of $300,000 usually offered by the NSF. However, these cost estimates do not include the in-kind goods the hosting institution provides (e.g., physical space). A detailed discussion of these choices is in Appendix G.1.

[30] Estimates from Arora and Gambardella (2005) suggest that allocating $100,000 of NSF funding to an economist yields 0.48 additional top publications, compared with 3.14 implied by our estimates for a mid-sized FSRDC consortium. Section G.4 in Appendix G details our procedure and alternative benchmarks.

publications, experience above-average local use, or participate in larger consortia.

8 Conclusion

We assemble a novel longitudinal dataset of U.S.-based academic economists and exploit the staggered expansion of U.S. Census Bureau FSRDCs to study how increased data access shapes economic science. We first show that researchers co-located with an FSRDC are much more likely to use confidential data or to build on work that does so. We then assess the effects on scientific output. After gaining FSRDC access, researchers are more likely to produce impactful and policy-relevant work. We also document substantial spillovers to applied economists who do not directly use these data. Researchers exposed to work using confidential data are more likely to build on it, exploring new questions and adopting similar research designs. Back-of-the-envelope calculations suggest that FSRDCs are highly cost-effective investments in the productivity of applied economists.

Our results have implications for ongoing policy discussions around the use of confidential microdata in academic research. To the best of our knowledge, we provide the first quasi-experimental evidence on the important role of these data in economics. Our findings suggest that broader access to confidential data can foster scientific progress, but that the current restricted-access regime disadvantages researchers with fewer resources. Fully harnessing the potential of U.S. federal microdata for research and policy may require legislative changes to broaden access. Even within current regulations, however, our results suggest that further expansion of the FSRDC network may be warranted. Further, the spillovers we document imply that evaluations of data access programs should extend beyond direct users. Data access can shape research by changing the topical and methodological direction of non-users as well, generating broader gains in scientific output.

Our work has some limitations. First, we do not aim to provide a policy evaluation of the FSRDC network. Doing so would require extending the analysis to other disciplines that benefit from FSRDC data, such as health policy and demography. Moreover, the Census Bureau created the FSRDC network to generate direct benefits for its data programs (Foster et al., 2009). A full evaluation from the Census Bureau's perspective would therefore need to account for these benefits,

which we cannot capture in our analysis. Still, we show that, at least in economics, FSRDC access has substantial researcher-level effects, increasing the production of higher-quality scientific output that receives more attention in policy documents. Even if our cost-benefit assessment understates the total value generated by FSRDCs, it points to high returns from these data-access institutions.

Second, our analysis faces some data limitations. Our extensive robustness checks rule out the most obvious confounds, but we suffer from the common pitfalls of studies based on bibliographic data. In particular, our within-researcher estimates could not consider dynamic changes in connections to editors, which might directly affect scientific productivity, or in researchers' status, which might affect the impact of their work. Further, we do not observe specific datasets that became available over time through FSRDCs and cannot examine their impact separately. Finally, our results are based on a particular type of confidential data that entails geographical constraints. While this is crucial to our research design, it also limits the generalizability of our results to contexts where researchers face similar barriers to accessing data.

Despite these caveats, our analysis helps inform the debate around confidential data access. Data are the lifeblood of modern economics, which means that the ability to understand and intervene in the economy is fundamentally "constrained by the extent and quality of the available data" (Griliches, 1994). While more research is needed to design a data access infrastructure that maximizes research use and ensures privacy protection, our results indicate that the payoffs of doing so are vast for both research and policy.

References

Abowd, J. M. and J. Lane (2004): "New approaches to confidentiality protection: Synthetic data, remote access and research data centers," in *International Workshop on Privacy in Statistical Databases*, Springer, 282–289.

Abowd, J. M. and I. M. Schmutte (2019): "An economic analysis of privacy protection and statistical accuracy as social choices," *American Economic Review*, 109, 171–202.

Angrist, J., P. Azoulay, G. Ellison, R. Hill, and S. F. Lu (2020): "Inside job or deep impact? Extramural citations and the influence of economic scholarship," *Journal of Economic Literature*, 58, 3–52.

Atrostic, B. (2007): "The Center for Economic Studies 1982-2007: A brief history," CES working paper.

Backhouse, R. E. and B. Cherrier (2017): "The age of the applied economist: The transformation of economics since the 1970s," *History of Political Economy*, 49, 1–33.

Biasi, B. and P. Moser (2021): "Effects of copyrights on science: Evidence from the WWII Book Republication Program," *American Economic Journal: Microeconomics*, 13, 218–60.

Bloom, N., E. Brynjolfsson, L. Foster, R. Jarmin, M. Patnaik, I. Saporta-Eksten, and J. Van Reenen (2019): "What drives differences in management practices?" *American Economic Review*, 109, 1648–1683.

Bowen, C. M. (2024): "Government Data of the People, by the People, for the People: Navigating Citizen Privacy Concerns," *Journal of Economic Perspectives*, 38, 181–200.

Brodeur, A., N. Cook, and A. Heyes (2020): "Methods matter: P-hacking and publication bias in causal analysis in economics," *American Economic Review*, 110, 3634–3660.

Callaway, B. and P. H. Sant'Anna (2021): "Difference-in-differences with multiple time periods," *Journal of Econometrics*, 225, 200–230.

Card, D. (2022): "Design-based research in empirical microeconomics," *American Economic Review*, 112, 1773–1781.

Card, D., R. Chetty, M. S. Feldstein, and E. Saez (2010): "Expanding access to administrative data for research in the United States," in *Ten Years and Beyond: Economists Answer NSF's Call for Long-Term Research Agendas*, American Economic Association.

Card, D., S. DellaVigna, P. Funk, and N. Iriberri (2020): "Are referees and editors in economics gender neutral?" *The Quarterly Journal of Economics*, 135, 269–327.

CES (2017): "Center for Economic Studies and Research Data Centers Research Report 2016," .

——— (2019): "Center for Economic Studies and Research Data Centers Research Report 2018," .

Chetty, R. (2012): "Time trends in the use of administrative data for empirical research," NBER Summer Institute presentation. Available at the author's website.

Chetty, R. and J. N. Friedman (2019): "A practical method to reduce privacy loss when disclosing statistics based on small samples," in *AEA Papers and Proceedings*, vol. 109, 414–20.

Cole, S., I. Dhaliwal, A. Sautmann, and L. Vilhuber (2020): *Handbook on Using Administrative Data for Research and Evidence-Based Policy*, JPAL and MIT Press.

Currie, J., H. Kleven, and E. Zwiers (2020): "Technology and big data are changing economics: Mining text to track methods," *AEA Papers and Proceedings*, 110, 42–48.

Dizikes, P. (2019): "The productive career of Robert Solow," *MIT Technology Review*.

Einav, L. and J. Levin (2014): "Economics in the age of big data," *Science*, 346, 1243089.

Feenberg, D., I. Ganguli, P. Gaule, and J. Gruber (2017): "It's good to be first: Order bias in reading and citing NBER working papers," *Review of Economics and Statistics*, 99, 32–39.

Foster, L., R. Jarmin, and L. Riggs (2009): "Resolving the tension between access and confidentiality: Past experience and future plans at the U.S. Census Bureau," *Statistical Journal of the IAOS*, 26, 113–122.

Furman, J. L. and S. Stern (2011): "Climbing atop the shoulders of giants: The impact of institutions on cumulative research," *American Economic Review*, 101, 1933–1963.

Furman, J. L. and F. Teodoridis (2020): "Automation, research technology, and researchers' trajectories: Evidence from computer science and electrical engineering," *Organization Science*, 31, 330–354.

Ghanem, D., P. H. Sant'Anna, and K. Wüthrich (2025): "Selection and parallel trends," *arXiv preprint arXiv:2203.09001*.

Goroff, D., J. Polonetsky, and O. Tene (2018): "Privacy protective research: Facilitating ethically responsible access to administrative data," *The ANNALS of the American Academy of Political and Social Science*, 675, 46–66.

Griliches, Z. (1994): "Productivity, R&D, and the Data Constraint," *American Economic Review*, 84, 1–23.

Groves, R. M. (2011): "Three eras of survey research," *Public Opinion Quarterly*, 75, 861–871.

Hamermesh, D. S. (2013): "Six decades of top economics publishing: Who and how?" *Journal of Economic Literature*, 51, 162–172.

Heckman, J. J. (2001): "Micro data, heterogeneity, and the evaluation of public policy: Nobel lecture," *Journal of Political Economy*, 109, 673–748.

Heckman, J. J. and S. Moktan (2020): "Publishing and promotion in economics: The tyranny of the top five," *Journal of Economic Literature*, 58, 419–470.

Hill, R. and C. Stein (2025): "Race to the bottom: Competition and quality in science," .

Hill, R., C. Stein, and H. Williams (2020): "Internalizing externalities: Designing effective data policies," *AEA Papers and Proceedings*, 110, 49–54.

Hjort, J., D. Moreira, G. Rao, and J. F. Santini (2021): "How research affects policy: Experimental evidence from 2,150 Brazilian municipalities," *American Economic Review*, 111, 1442–1480.

Hoelzemann, J., G. Manso, A. Nagaraj, and M. Tranchero (2024): "The streetlight effect in data-driven exploration," UC Berkeley, University of Vienna, and University of Pennsylvania.

Kalaitzidakis, P., T. P. Mamuneas, and T. Stengos (2003): "Rankings of academic journals and institutions in economics," *Journal of the European Economic Association*, 1, 1346–1366.

Lane, J. (2021): *Democratizing Our Data: A Manifesto*, MIT Press.

McGuckin, R. H. (1995): "Establishment microdata for economic research and policy analysis: Looking beyond the aggregates," *Journal of Business & Economic Statistics*, 13, 121–126.

McGuckin, R. H., R. H. McGukin, and A. P. Reznek (1993): "Research with economic microdata: The Census Bureau's Center for Economic Studies," *Business Economics*, 52–58.

Murray, F., P. Aghion, M. Dewatripont, J. Kolev, and S. Stern (2016): "Of mice and academics: Examining the effect of openness on innovation," *American Economic Journal: Economic Policy*, 8, 212–52.

Myers, K. R. and L. Lanahan (2022): "Estimating spillovers from publicly funded R&D: Evidence from the U.S. Department of Energy," *American Economic Review*, 112, 2393–2423.

Nagaraj, A. (2022): "The private impact of public data: Landsat satellite maps increased gold discoveries and encouraged entry," *Management Science*, 68, 564–582.

Nagaraj, A., E. Shears, and M. de Vaan (2020): "Improving data access democratizes and diversifies science," *Proceedings of the National Academy of Sciences*, 117, 23490–23498.

Nagaraj, A., F. Stipanicic, and M. Tranchero (2025): "Restricted-Access Data in Economics: Adoption, Diffusion, and Impact of US Census Bureau's Microdata," *Harvard Data Science Review*.

Önder, A. S. and S. Schweitzer (2017): "Catching up or falling behind? Promising changes and persistent patterns across cohorts of economics PhDs in German-speaking countries from 1991 to 2008," *Scientometrics*, 110, 1297–1331.

Rambachan, A. and J. Roth (2023): "A more credible approach to parallel trends," *Review of Economic Studies*, 90, 2555–2591.

Sarsons, H., K. Gërxhani, E. Reuben, and A. Schram (2021): "Gender differences in recognition for group work," *Journal of Political Economy*, 129, 101–147.

Tranchero, M. (2026): "Data-driven search and the birth of theory: Evidence from genome-wide association studies," *The Wharton School*.

Truffa, F. and A. Wong (2022): "Undergraduate gender diversity and direction of scientific research," *American Economic Review*.

Waldinger, F. (2016): "Bombs, brains, and science: The role of human and physical capital for the creation of scientific knowledge," *Review of Economics and Statistics*, 98, 811–831.

Wang, D. and A.-L. Barabási (2021): *The Science of Science*, Cambridge University Press.

Wapman, K. H., S. Zhang, A. Clauset, and D. B. Larremore (2022): "Quantifying hierarchy and dynamics in US faculty hiring and retention," *Nature*, 610, 120–127.

Weinberg, D. H., J. M. Abowd, P. M. Steel, L. Zayatz, and S. K. Rowland (2007): "Access methods for United States microdata," CES working paper.

Williams, H. L. (2013): "Intellectual property rights and innovation: Evidence from the human genome," *Journal of Political Economy*, 121, 1–27.

Yin, Y., Y. Dong, K. Wang, D. Wang, and B. F. Jones (2022): "Public use and public funding of science," *Nature Human Behaviour*, 6, 1344–1350.

York, D. G., J. Adelman, J. E. Anderson Jr, S. F. Anderson, J. Annis, et al. (2000): "The Sloan Digital Sky Survey: Technical summary," *The Astronomical Journal*, 120, 1579.

9 Tables and Figures

Table 1: Summary Statistics

Panel A: Researcher-Level

	N	Mean	Std. Dev.	Median	Min	Max
Ever Had Access (0/1)	10060	0.206	0.40	0	0	1
Year of Access	2075	2009.988	5.94	2011	1994	2019
Ever Used FSRDC (0/1)	10060	0.038	0.19	0	0	1
Ever Cited FSRDC (0/1)	10060	0.284	0.45	0	0	1
Lifetime Top Publications	10060	1.944	3.62	1	0	57
Lifetime Cite-weighted Papers	10060	75.952	210.09	8	0	3776
Lifetime Policy-relevant Papers	10060	3.898	5.70	2	0	79

Panel B: Researcher-Year-Level

	N	Mean	Std. Dev.	Median	Min	Max
Post-FSRDC (0/1)	155720	0.108	0.31	0	0	1
Papers Using FSRDC Data	155720	0.005	0.08	0	0	3
Papers Citing FSRDC Papers	155720	0.041	0.22	0	0	6
Top Publications	155720	0.126	0.40	0	0	7
Cite-weighted Papers	155720	4.907	23.24	0	0	1285
Policy-relevant Papers	155720	0.252	0.58	0	0	11
New JEL Codes (0/1)	155720	0.303	0.46	0	0	1
Papers with FSRDC JEL (0/1)	155720	0.149	0.36	0	0	1
Papers without FSRDC JEL (0/1)	155720	0.201	0.40	0	0	1
Papers Mentioning Admin Data	155720	0.006	0.08	0	0	4
Papers Mentioning Survey Data	155720	0.018	0.14	0	0	6
Quasi-experimental Papers	155720	0.023	0.16	0	0	4
Experimental Papers	155720	0.007	0.09	0	0	5
Year	155720	2006.229	7.69	2007	1990	2019

Note: This table lists summary statistics at the researcher level for 10,060 publishing empirical economists (Panel A) and at the researcher-year level for an unbalanced panel of 155,720 observations (Panel B). See text for details.

Table 2: Effect of FSRDC Access on the Diffusion of Confidential Data

	Papers Using FSRDC Data		Papers Citing FSRDC Papers	
	(1)	(2)	(3)	(4)
Post-FSRDC	0.00590***	0.00582***	0.0175***	0.0107**
	(0.00128)	(0.00141)	(0.00411)	(0.00372)
Researcher FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
University FE	Yes	No	Yes	No
University Tier × Year FE	No	Yes	No	Yes
N	155615	155622	155615	155622

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data diffusion among empirical economists. *Papers Using FSRDC Data* is the count of articles that directly use confidential data accessible only in an FSRDC. *Papers Citing FSRDC Papers*is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). *Post-FSRDC* equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. The number of observations changes slightly due to the inclusion of different fixed effects. Standard errors are clustered at the university level. *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively.

Table 3: Effect of FSRDC Access on Research Output

	Top Publications		Cite-weighted Publications		Policy-relevant Publications	
	(1)	(2)	(3)	(4)	(5)	(6)
Post-FSRDC	0.0240***	0.0230**	1.706***	1.239**	0.0361**	0.0362**
	(0.00665)	(0.00839)	(0.43773)	(0.42415)	(0.01374)	(0.01139)
Researcher FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No
University FE	Yes	No	Yes	No	Yes	No
University Tier × Year FE	No	Yes	No	Yes	No	Yes
N	155615	155622	155615	155622	155615	155622

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of research output for empirical economists. *Top Publications* is the count of articles published in the most prestigious economics journals (the list of top five and top field journals from Heckman and Moktan, 2020). *Cite-weighted Publications* is the count of articles weighted by the number of citations received up to five years following their publication. *Policy-relevant Publications* is the count of articles with at least one citation from a policy document. *Post-FSRDC* equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. The number of observations changes slightly due to the inclusion of different fixed effects. Standard errors are clustered at the university level. *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively.

Table 4: Effect of FSRDC Access on Research Output for Non-FSRDC Users

	Top Publications		Cite-weighted Publications		Policy-relevant Publications	
	Baseline	Indirect	Baseline	Indirect	Baseline	Indirect
	(1)	(2)	(3)	(4)	(5)	(6)
Post-FSRDC	0.0230**	0.0181*	1.239**	0.930*	0.0362**	0.0280*
	(0.008)	(0.008)	(0.424)	(0.374)	(0.011)	(0.011)
Researcher FE	Yes	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	155622	149113	155622	149113	155622	149113

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of research output for empirical economists after removing from the sample researchers using FSRDC data. Columns (1), (3), and (5) are our baseline models with the full sample of researchers, while Columns (2), (4), and (6) remove researchers using FSRDC data. *Top Publications* is the count of articles published in the most prestigious economics journals (the list of top five and top field journals from Heckman and Moktan, 2020). *Cite-weighted Publications* is the count of articles weighted by the number of citations received up to five years following their publication. *Policy-relevant Publications* is the count of articles with at least one citation from a policy document. *Post-FSRDC* equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. The number of observations changes slightly due to the inclusion of different fixed effects. Standard errors are clustered at the university level. *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively.

Table 5: Effect of FSRDC Access on the Direction and Design of Empirical Research

Panel A: Research Direction

	New JEL Codes (0/1)	FSRDC JEL Paper (0/1)	Non-FSRDC JEL Paper (0/1)
	(1)	(2)	(3)
Post-FSRDC	0.0141	0.0139*	0.0123
	(0.008)	(0.006)	(0.008)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
N	155622	155622	155622

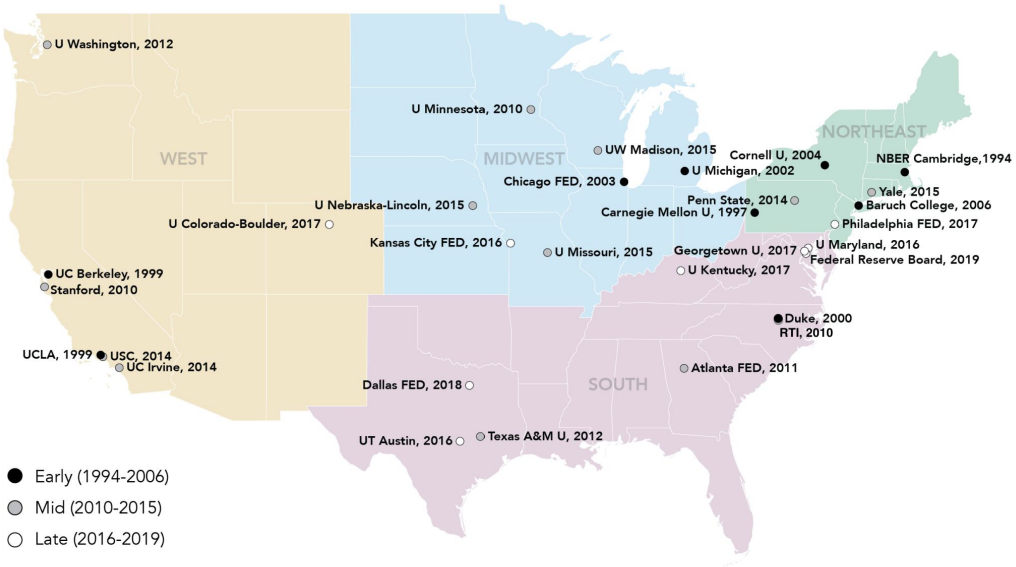
Panel B: Research Design

	Data		Research Methods	
	Admin.	Survey	Quasi-Exp.	Experiment
	(1)	(2)	(3)	(4)
Post-FSRDC	0.00333*	0.00315	0.00632*	-0.00115
	(0.002)	(0.003)	(0.003)	(0.002)
Researcher FE	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes
N	155622	155622	155622	155622

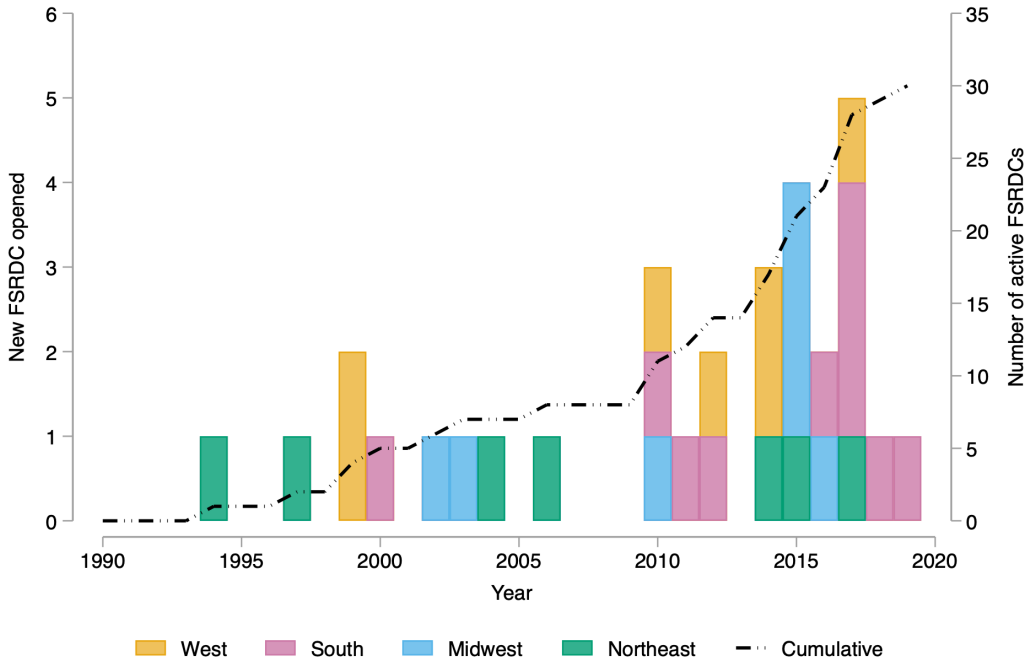
Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on the direction and design of empirical research, excluding papers directly using FSRDC data. *New JEL Codes (0/1)* is an indicator equal to one if the researcher used at least one JEL code never used before. *FSRDC JEL Paper (0/1)* is an indicator equal to one if the researcher used at least one FSRDC JEL code during the year. *Non-FSRDC JEL Paper (0/1)* is an indicator that equals one if the researcher used at least one non-FSRDC JEL code during the year. *Data: Admin* and *Data: Survey* are the number of published articles that mention in their title or abstract keywords related to the use of administrative and survey data, respectively. *Research Methods: Quasi-Exp.* and *Research Methods: Experiment* are the number of published articles that mention in their title or abstract keywords related to the use of quasi-experimental or experimental methods, respectively. *Post-FSRDC* equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. Standard errors are clustered at the university level. *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively.

Figure 1: Expansion of the FSRDC Program over Time and Space

Panel A: Geographic Expansion of FSRDCs



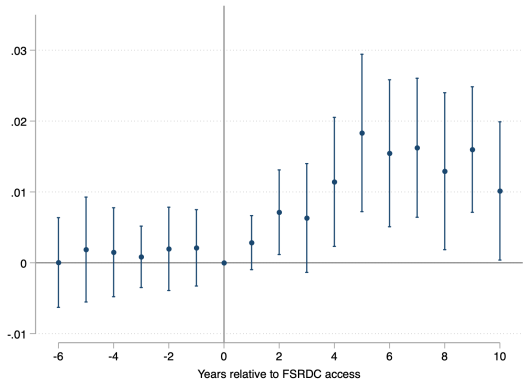
Panel B: Time Expansion of FSRDCs



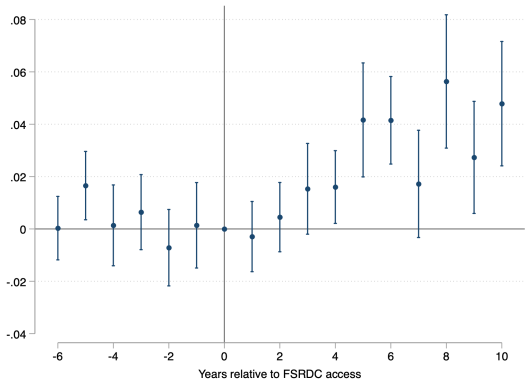
Note: This figure illustrates the timing and geographic scope of the expansion of the FSRDC network. See text for more details.

Figure 2: Time-Varying Estimates of the Impact of FSRDCs on the Diffusion of Confidential Data

(i) Papers Using FSRDC Data



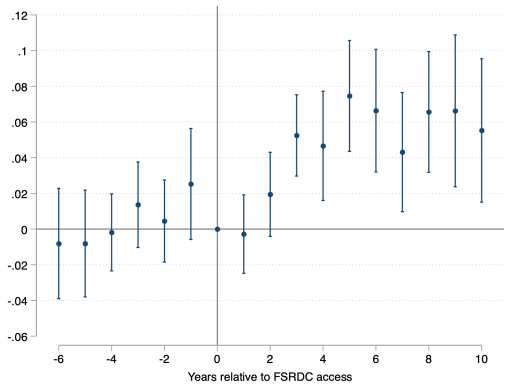
(ii) Papers Citing FSRDC Papers



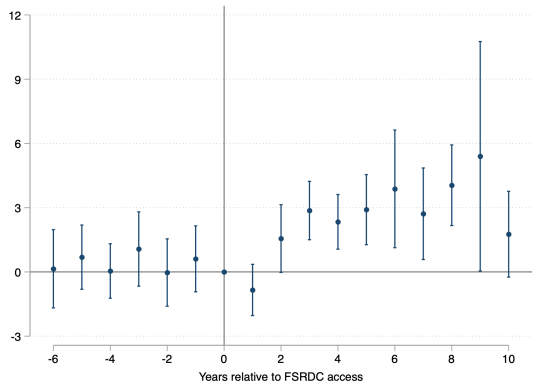
Note: This figure provides visual illustrations of the event study version of the main regression on confidential data adoption among empirical researchers. The main dependent variables are the number of papers written using FSRDC data (panel (i)) or the number of papers that cite papers using FSRDC data (panel (ii)). The chart plots values of β for different lags before and after obtaining access to an FSRDC with 95% confidence intervals. Regressions include researcher and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure 3: Time-Varying Estimates of the Impact of FSRDCs on Research Output

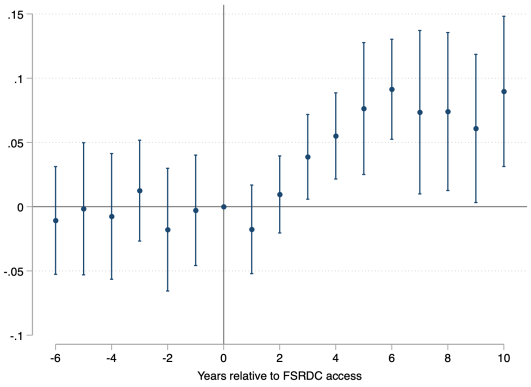
(i) Top Publications



(ii) Cite-Weighted Publications



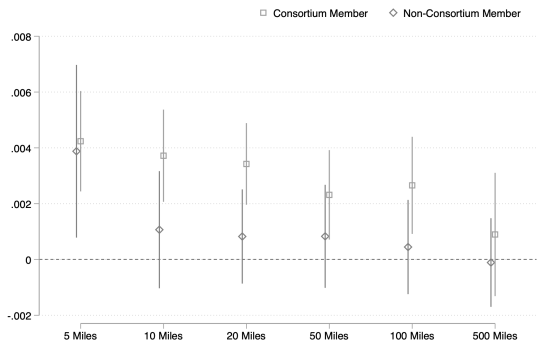
(iii) Policy-Relevant Publications



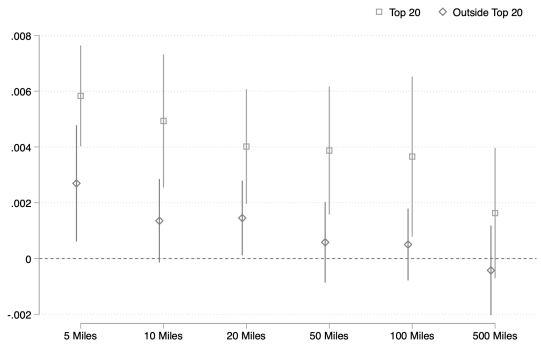
Note: This figure provides visual illustrations of the event study version of the main regressions evaluating the impacts of FSRDC access on measures of research output for empirical researchers. The main dependent variables are the count of papers published in the main economics journals (panel (i)), the count of papers weighted by the number of citations received up to five years following publication (panel (ii)), and the count of papers cited in policy documents (panel (iii)). The charts plot values of β for different lags before and after obtaining access to an FSRDC with 95% confidence intervals. Regressions include researcher and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure 4: Determinants of Confidential Data Use by Distance to the Closest FSRDC

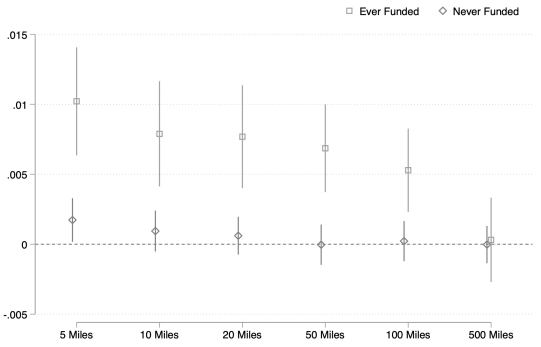
(i) Consortium Membership of Researcher



(ii) University Ranking of Researcher



(iii) External Funding Status of Researcher



Note: This figure explores factors hindering FSRDC usage. We estimate OLS models to evaluate the effect of distance on the count of papers using confidential data. We consider researchers treated if the geometric distance between their institution and the nearest FSRDC is within a certain threshold (ranging from 5 to 500 miles). The regressions are estimated separately per distance threshold. At each distance, we add an interaction for empirical economists of differing consortium membership status (panel (i)), university rank (panel (ii)), and funding status (panel (iii)). Consortium Member: 0/1 = 1 if the researcher is affiliated with an FSRDC consortium partner institution at the time of treatment. Top 20: 0/1 = 1 if the researcher is affiliated with a top 20 institution at the time of treatment, per Kalaitzidakis et al. (2003). Ever Funded: 0/1 = 1 if the researcher acknowledges external funding sources. Regressions include researcher and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

How Does Data Access Shape Science? The Impact of Federal Statistical Research Data Centers on Economics Research

Online Appendix

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A Selection and Timing of FSRDC Openings

This appendix provides additional information on the selection of institutions that host an FSRDC. We combine archival documentation from the U.S. Census Bureau and the National Science Foundation (NSF) with nine semi-structured interviews with FSRDC administrators and users to reconstruct the process governing the expansion of the network.[1]

A.1 Institutional Process Governing FSRDC Openings

In our interviews, we learned that the choice of the earliest locations reflected a combination of host feasibility and local demand. For example, Boston was selected for the first FSRDC in no small part because a Census Regional Office was available to host the center before it was eventually transferred to NBER premises (interview Y79). Other early centers, such as the Carnegie Mellon University (CMU) one, emerged from specific collaborations between university administrators and the Census Bureau (interview S94). In 1998, as the network began to rely on NSF support, the process became more formalized and evolved into the system in place today. Prospective host institutions first consult with the Census Bureau about the feasibility of establishing a new center and subsequently submit a proposal to the NSF, with the review coordinated between the NSF and the Census Bureau (U.S. Bureau of the Census, 1998).

Archival documents identify the criteria used to evaluate these proposals. These include: 1) the cross-disciplinary potential of the proposed FSRDC, 2) whether its location would equitably expand access relative to the existing network, 3) demonstrated demand from local and nearby researchers, 4) the qualifications and staffing of the proposed leadership team, and 5) the adequacy

[1]Before conducting the interviews, we consulted the UC Berkeley Institutional Review Board. Because the interviews focused on factual questions about the FSRDC program and its institutional history, the IRB determined that formal review was not required.

of the institutional resources and secure facilities needed to operate the center (National Science Foundation, 2015). Census Bureau guidelines similarly emphasize the need for a sufficiently large community of prospective users and the institutional capacity to support the center (U.S. Census Bureau, 2019). Thus, institutions initiated applications, but the decision to support a new location ultimately reflected the joint approval of the NSF and the Census Bureau.

Geographic access was particularly important. NSF explicitly asked applicants to discuss the "geographic proximity of the proposed RDC to other RDCs" and whether ease-of-access considerations made the proposed location particularly valuable (National Science Foundation, 2015). Our interviews closely match this archival evidence. One administrator explained that "if you're close to an already existing location, that probably hurts your application" (interview S12). Another described the NSF as seeking "kind of parity across the U.S. so that researchers in one part of the country had the same access as researchers in another part of the country did" (interview T14). As a result, substantial local demand did not necessarily imply that an institution would receive an FSRDC quickly, if ever. Stanford, for example, had substantial demand for local access but waited approximately 11 years for an FSRDC branch because of the Berkeley FSRDC, located only about 40 miles away (interview T14).

Local demand was an important component of the selection process, especially after the CMU FSRDC, the second center to open, closed because of limited usage. NSF solicitations ask applicants to document demand at the proposed site and the presence of researchers at surrounding institutions who would use the FSRDC (National Science Foundation, 2015). Our interviewees similarly emphasized that a successful application generally required a sufficiently large potential user community to sustain the center's operations. Obtaining NSF support was competitive, and "lots of places that have tried to get the NSF money haven't been able to" (interview D88). Yet demand alone was not sufficient. Submitting an application also typically required one or more individuals willing to advocate for the center within the university. As one respondent put it, "for an RDC to get established, you're going to need at least one and preferably a few sort of well-connected and very enthusiastic people who want to push for having an RDC" (interview D88). Another administrator described how a faculty advocate "persuaded the Dean that this would be a really good thing for the School. Well, it would be a really good thing for himself, which would be a really good thing for the School and the University" (interview Y79).

These local institutional factors also generated variation in the timing and precise location of FSRDCs. Changes in university leadership could accelerate or delay applications, while in consortia involving several institutions, the choice of which member would physically host the data center often resulted from a compromise among the participating universities (interviews S12, S94). The pace of network expansion was further constrained by the resources available to the NSF and the Census Bureau (National Science Foundation, 2015). Finally, implementation delays also varied substantially across locations. Among centers for which we can identify the NSF award or public announcement date, the time between announcement and the start of operations averaged 1.42 years but reached as long as four years in some cases.

Taken together, the archival and qualitative evidence points to three broad determinants of FSRDC openings: the presence of a sufficiently large potential user community, the objective of ensuring geographic equity in access, and idiosyncratic institutional factors.

A.2 Predictors of FSRDC Openings

The institutional evidence in the previous section suggests that FSRDC openings reflected both demand for local access and the geographic expansion of the existing network. We next examine whether these factors predict the timing of FSRDC adoption and, more importantly for our research design, whether centers systematically opened at universities already experiencing improvements in research productivity.

We construct a university-year panel and estimate Cox proportional hazards models predicting the opening of an FSRDC. Universities enter the risk set before receiving an FSRDC and exit upon adoption. We consider three time-varying predictors. First, we measure the distance from university j to the nearest operating FSRDC. If geographic parity shaped the expansion of the network, universities located farther from an existing facility should be more likely to receive a center. Second, we measure the number of empirical researchers affiliated with the university as a proxy for the size of the potential local user community. Finally, we construct a measure of recent research productivity trends using the three-year rolling slope in publication output. This variable allows us to test directly whether FSRDCs tended to open at institutions where scientific productivity was already improving. All covariates are lagged by one year, and we stratify the baseline hazard by university tier to allow adoption rates to vary across institutions with different levels of research intensity.

Table A1: Cox Hazard Model Predicting FSRDC Opening

	(1)	(2)	(3)
Distance (100 mi)	1.0670*	1.0778*	1.0742*
	(0.03151)	(0.03193)	(0.03197)
N Empiricists	–	1.0241***	1.0232***
	–	(0.00586)	(0.00611)
Productivity trend	–	–	0.905
	–	–	(0.09164)
University Tier Strata	Yes	Yes	Yes
N	7,703	7,703	6,978

Notes: This table reports hazard ratios from Cox proportional hazard models predicting the timing of FSRDC adoption at the university level. The unit of observation is the university-year, and universities exit the risk set upon adoption. *Distance* is measured in 100-mile increments from university j to the nearest operating RDC in year $t-1$. *N Empiricists* is the count of empiricist researchers at university j in year $t-1$. *Productivity trend* is the three-year rolling slope of productivity. All covariates are lagged one year. The models are stratified by university tier, allowing the baseline hazard to vary across tiers. Robust standard errors are reported in parentheses. *, **, and *** denote significance at the 5%, 1%, and 0.1% levels, respectively.

Table A1 reports the results. Distance from the nearest operating FSRDC significantly predicts subsequent adoption, consistent with the emphasis on geographic coverage in the archival and interview evidence. The number of empirical researchers also predicts adoption, confirming that the size of the potential user community mattered for the establishment of a new center. By contrast, the recent productivity trend is not statistically significant. Coupled with the absence of university-level pre-trends before an opening (Appendix Figure H8), these results provide no

evidence that FSRDCs systematically opened earlier at universities already experiencing more rapid increases in scientific productivity. Taken together, these results closely mirror the institutional process described in Section A.1: geographic and institutional constraints generated variation in access that was not directly tied to changes in research productivity, strengthening the basis for our identification strategy.

A.3 Assessing Bias from Nonrandom FSRDC Placement

The evidence above makes clear that FSRDC locations were not randomly assigned. In particular, institutions with larger communities of potential users were more likely to obtain a center. However, selection based on expected treatment benefits does not, by itself, violate the identifying assumption underlying difference-in-differences. Following Ghanem et al. (2025), the relevant issue is whether the information used to select institutions into treatment also predicts changes in untreated potential outcomes.

To see this distinction, let W_j denote the information available to the NSF and the Census Bureau when deciding whether university j should receive an FSRDC. Treatment assignment may depend on the expected gain from access, $E[Y_{j,\text{post}}(1) - Y_{j,\text{post}}(0) \mid W_j]$. For example, universities with more empirical researchers may be expected to benefit more from access to confidential microdata. Such selection may limit the external validity of our estimates if treated institutions benefit more from FSRDC access than the average university. However, it does not mechanically violate the identifying assumption underlying difference-in-differences. Bias arises if the information used in treatment assignment also predicts changes in untreated potential outcomes, $Y_{j,\text{post}}(0) - Y_{j,\text{pre}}(0)$ (Ghanem et al., 2025).

We assess this concern using the bias decomposition proposed by Ghanem et al. (2025). We conduct the analysis at the university-year level, where FSRDC treatment is assigned, using never-treated universities as the comparison group. The outcome is the aggregate number of top-field publications produced at university j in year t. Their framework highlights two potential sources of bias that are particularly relevant in our setting. First, if university research output is mean-reverting rather than approximating a martingale, pre-existing differences in productivity levels between treated and control universities may mechanically generate different subsequent changes even in the absence of treatment. Second, universities receiving an FSRDC may have experienced transitory productivity shocks before treatment that persist into the post-treatment period.

Table A2 reports the resulting bias benchmarks. The estimated persistence parameters are close to one, with $\hat{\rho}_1 = 0.980$ based on pre-treatment persistence and $\hat{\rho}_2^0 = 0.987$ based on persistence among control universities. Accordingly, departures from the martingale benchmark imply only modest bias. We also allow the difference in estimated productivity shocks between treated and control universities observed before treatment to persist into the post-treatment period. Under this assumption, the bias-adjusted treatment effect ranges from 3.35 to 3.39 additional publications, compared with a baseline difference-in-differences estimate of 3.43. Under the alternative assumption that the pre-treatment shock differential does not persist after treatment, the corresponding adjusted estimates range from 3.46 to 3.47. Taken together, under these benchmark values, the implied bias is small relative to the baseline DiD estimate. Combined with the institutional evidence above, this analysis reduces concerns that the estimated post-FSRDC increase in research output

reflects differential counterfactual productivity growth at universities selected to receive a center.

Table A2: Ghanem, Sant'Anna, and Wüthrich (2025) Bias Benchmarking

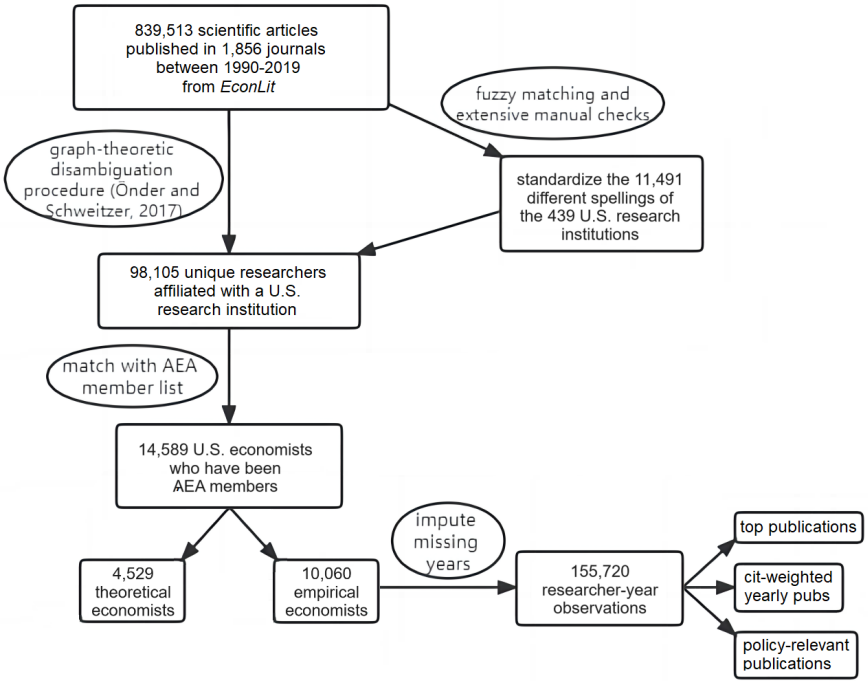
Scenario	Assumption	$\hat{\rho}_2$ Assumption	Total Bias (Δ_{post})	Bias-Adjusted ATT
Panel A: $\Delta_{sel,post} = 0$ (No selection on time-varying shocks)				
1a	Martingale holds	$\rho_2 = 1$	0.0000	3.4309
2a	Pre-treatment persistence	$\rho_2 = \hat{\rho}_1 = 0.980$	-0.0424	3.4733
3a	Control group persistence	$\rho_2 = \hat{\rho}_2^0 = 0.987$	-0.0267	3.4576
Panel B: $\Delta_{sel,post} = \Delta_{sel,pre}$ (Selection pattern continues)				
1b	Martingale holds	$\rho_2 = 1$	0.0799	3.3510
2b	Pre-treatment persistence	$\rho_2 = \hat{\rho}_1 = 0.980$	0.0375	3.3934
3b	Control group persistence	$\rho_2 = \hat{\rho}_2^0 = 0.987$	0.0532	3.3777
Key Estimates:				
Pre-treatment gap (Δ_{pre})			2.1164	
Selection on shocks ($\Delta_{\text{sel,pre}}$)			0.0799	
Baseline DiD estimate			3.4309	

Notes: This table reports bias decomposition results following the GSW (2025) framework. The analysis is conducted at the university-year level, using never-treated universities as the control group. The dependent variable is the sum of top-field publications at university j in year t. The baseline ATT estimate and bias-adjusted ATT estimates are reported under alternative assumptions about the persistence of selection patterns in the post-treatment period.

B Additional Data Details

The primary data source we leverage is *EconLit*, a proprietary database of economic scholarship curated by the American Economic Association. Our version of EconLit includes 839,513 scientific articles published in 1,856 journals between 1990 and 2019 inclusive. Unfortunately, EconLit lacks unique author identifiers, preventing us from easily aggregating the publication data into an author-year-level panel. To reconstruct authors' publication records, we need to disambiguate the author names and affiliations, a complex and time-consuming task that we implement in several steps summarized in Figure B1. Finally, we externally validate our author disambiguation against researcher identifiers from OpenAlex. We describe this exercise in Appendix Section B.3.

Figure B1: Steps Followed for Panel Construction



Note: This figure illustrates the steps we took to construct our panel of U.S.-based publishing economists. See text for more details.

B.1 Data Disambiguation

We disambiguate the names of researchers appearing in EconLit following three steps. First, we replace all non-English characters (e.g., "ó" is replaced by "o") and transform the most common name abbreviations into standardized names (e.g., "Ted" is replaced by "Edward"). Second, we apply the disambiguation procedure developed by Önder and Schweitzer (2017). The method employs a graph-theoretic approach that follows a hierarchical process. After identifying the set of name entries with identical surnames, the algorithm constructs a graph of the relationships of all the corresponding first names to each other. Names associated with a particular surname can be identical, different, or subsets of each other. For example, the first name "Michael J." is identical to "Michael J.", it is different from "Tom", and it is a subset of "Michael". The algorithm classifies "Michael J." and "Michael" as the same person if no other "Michael *x.*", with *x.* different from J., appears in the data. This approach is equivalent to assuming that the combination of first, middle, and last names uniquely identify each economist (Card et al., 2022), while at the same time being conservative in assigning ambiguous names that lack a clear middle name. Third, we performed extensive manual checks and corrected several misclassifications due to either misspellings in EconLit (e.g., "Tabellini, Gudio" instead of "Tabellini, Guido") or ambiguous names (e.g., "David Levine", that could refer to David I. Levine or David K. Levine). This three-step procedure results in a database of 434,938 unique researchers from the 552,570 names originally appearing in EconLit.

Next, we standardize the names of the 178,798 affiliations appearing in EconLit. This step is necessary to pin down researchers' treatment status through their affiliation and restrict the sample

to U.S.-affiliated economists who are ever at risk of being co-located to an FSRDC. We begin with a list of all research universities in the United States taken from the 2018 Carnegie Classification of Institutions of Higher Education.[2] In particular, we consider all doctoral universities (corresponding to the codes 15, 16, and 17 in the Carnegie Classification), to which we add the main institutions active in economic research (such as the IMF, RAND Corporation, World Bank, RTI, and all the regional FED offices). The result is a list of 439 universities and research centers, which we merge with the EconLit record via fuzzy string matching. This is done by employing string partial ratio similarity to consider the information in the affiliation word ordering (e.g., to distinguish "University of Washington" from "Washington University"). We retain all matches achieving a partial ratio similarity score equal to or greater than 90 over 100, and we manually check them.[3] The result is a list of 11,491 different spellings of the 439 U.S. research institutions appearing in the Carnegie Classification.

B.2 Panel Construction

After disambiguating author names and affiliations, we can use bibliographic data to construct an author-year-level database with an annual record for each economist who has published at least one paper in journals indexed in EconLit (Moed et al., 2013). Out of the 434,938 unique researchers in our disambiguated data, we retain 98,105 scholars affiliated with a U.S. research institution for at least one year in our sample period. To further restrict our sample to academic economists, we match these names to nineteen yearly lists of members of the American Economic Association (AEA). This step allows us to avoid confounding effects arising from the inclusion of researchers working in unrelated disciplines but occasionally publishing in economics outlets. This leaves us with 14,589 publishing economists. Finally, since our main specification considers the effect of access to confidential FSRDC data on the research of empirically-minded economists, we use the machine learning classification procedure developed by Angrist et al. (2020) to classify each researcher in our sample based on up to her first five observed publications (see Appendix D for more details). Researchers with fewer than five publications are classified using all publications we observe, without imposing a minimum publication count for inclusion in the sample. We prefer this approach over a classification based on lifetime count since the latter is less robust to the (anecdotally unlikely) possibility that gaining access to an FSRDC shifts a researcher's profile from theorist to empiricist. Furthermore, this allows us to still classify researchers who have only worked at an institution with an FSRDC, which would not be possible if the classification was based only on pre-FSRDC research. The final result is an unbalanced panel of 10,060 empirical economists.

We use this set of authors to derive a panel of 155,720 researcher-year observations by imputing missing years between the first and last years in which we observe a researcher publishing. In practice, we interpolate missing researcher-year pairs by assigning a zero in the count of scientific output. As noted by Moretti (2021), this interpolation choice leads to partially conflating the extensive margin (i.e., the probability of publishing at least one paper) with the intensive margin

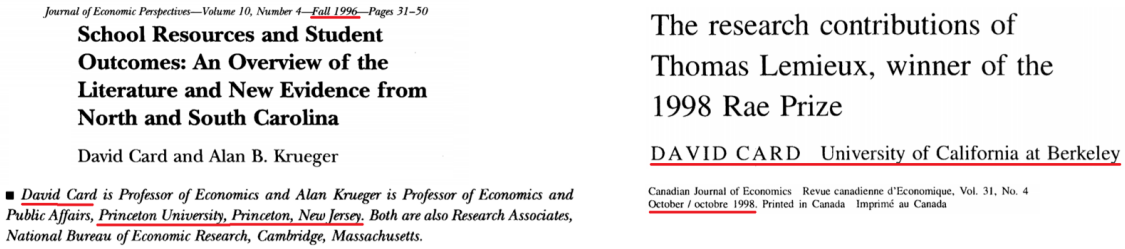
[2]The Carnegie Classification of Institutions of Higher Education is available online at https://carnegieclassifications.iu.edu/.

[3]Manual validation indicates that approximately 86% of affiliation matches with a partial ratio similarity score of at least 90/100 are correct. Because match quality deteriorates below this threshold, we drop lower-scoring affiliations and treat them as non-U.S. institutions.

(i.e., the number of papers published in a given year, given a positive number of publications). This issue could meaningfully change the interpretation of estimates in contexts where observable outputs are relatively rare, such as when inventors obtain a patent (Moretti, 2021). However, this is less of a concern in our case because publishing a paper is more common than patenting, thereby increasing our panel's granularity and reducing the need for imputations.

Our research design takes advantage of the requirement that confidential microdata can be analyzed only at FSRDC facilities. In practice, we can approximate the cost of data access based on the researcher's location relative to the closest data center. We use the information on university affiliations in scientific articles to pinpoint scholars' location and mobility over time (see Figure B2 for an example). In cases where the affiliation changes in non-consecutive years with gaps in between, we assign the old affiliation to the first third of the missing years and the new affiliation to the remaining two-thirds. Fortunately, EconLit has a broad coverage of economics journals, allowing us to record scientific mobility with high precision, even for less prolific authors.

Figure B2: Example of Researcher Mobility from Bibliographic Records



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School Resources and Student Outcomes: An Overview of the Literature and New Evidence from North and South Carolina

David Card and Alan B. Krueger

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The research contributions of Thomas Lemieux, winner of the 1998 Rae Prize

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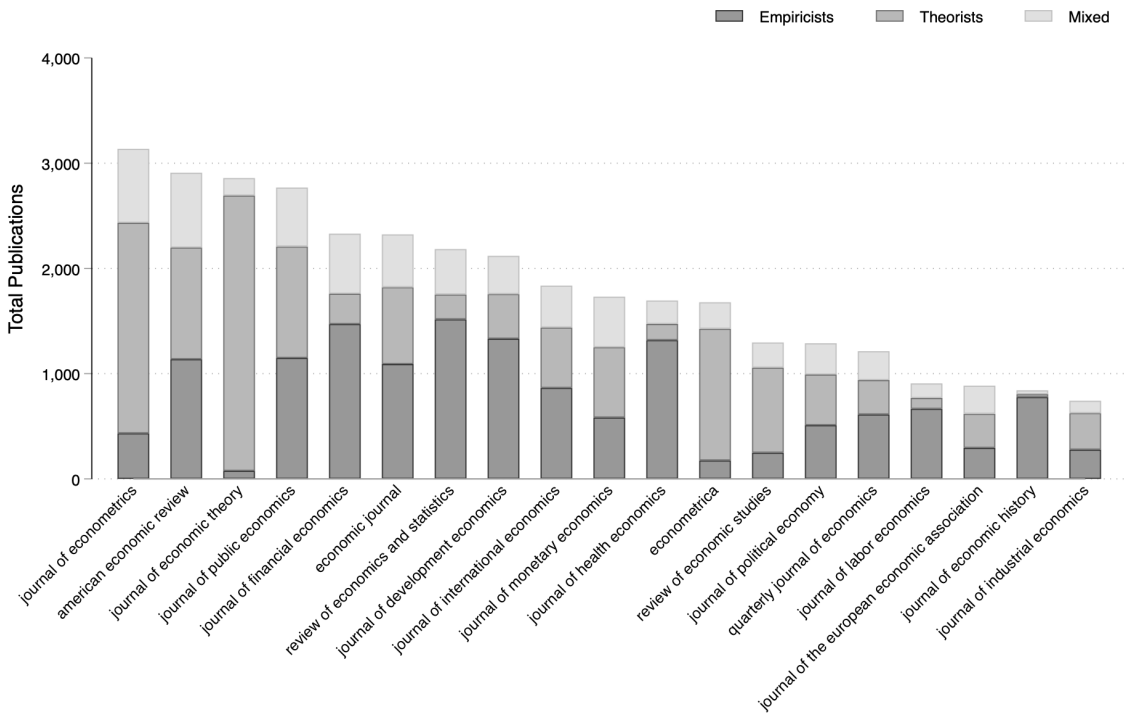
Note: This figure shows how bibliographic records can be leveraged to construct a panel of publishing economists. In particular, the affiliation details permit the detection of mobility events over time and co-location to an FSRDC. In this case, we would be imputing the new affiliation for 1997.

We construct productivity metrics at the researcher-year level by counting the number of articles published each year. We consider "top publications" all papers appearing in the top five and top field journals as defined by Heckman and Moktan (2020). The list includes top field journals (the *Journal of Development Economics*, the *Journal of Econometrics*, the *Journal of Financial Economics*, the *Journal of Economic Theory*, the *Journal of Health Economics*, the *Journal of Industrial Economics*, the *Journal of Labor Economics*, the *Journal of Monetary Economics*, the *Journal of Public Economics*, the *Journal of International Economics*, and the *Journal of Economic History*), high-profile generalist journals (the *Review of Economics and Statistics*, the *Journal of the European Economic Association*, the *Economic Journal*), and the so-called top five journals (the *American Economic Review*, the *Quarterly Journal of Economics*, the *Journal of Political Economy*, *Econometrica*, and the *Review of Economic Studies*). Figure B3 shows the number of articles recorded in EconLit for each main outlet. Next, we augment EconLit by merging each article with its yearly citation count extracted from SSCI/Web of Science. We are able to match 97.2% of our data to the corresponding citation records, thereby constructing citation-weighted metrics of academic productivity. We use this information to adjust yearly paper counts by the number of citations each article receives up to five years following publication. We merge the articles in our data with the list of mentions in policy documents assembled by Overton. Detailed

description of this data source and summary statistics are presented in Appendix E.

Information on the ranking of economics departments is taken from Kalaitzidakis et al. (2003). We use this information to code seven university tiers when estimating models that include university-tier × year fixed effects. We divide the North American institutions reported by Kalaitzidakis et al. (2003) into five groups, each with roughly the same number of observations in our sample. The top tier includes the first five U.S. universities (Harvard, Chicago, MIT, Northwestern, Penn), the second tier the following five (Yale, Princeton, Stanford, UC Berkeley, NYU), the third tier the following ten (from Columbia to Boston U), the fourth tier the following eighteen (from Brown to U Virginia), while the fifth tier includes the remaining ones (from Wash U to Stony Brook). The two residual tiers are composed, respectively, of researchers affiliated with foreign universities or U.S.-based research institutions not included in the ranking.

Figure B3: Count of Articles in the Main Economics Journals (1990-2019)



Note: This figure shows the count of papers in the most prestigious economics journals as recorded by EconLit. We decompose the total yearly publication count into publications authored exclusively by empiricists, publications authored exclusively by theorists, and publications with a mixed authorship team. The list of top five and top field journals is from Heckman and Moktan (2020). See text for more details.

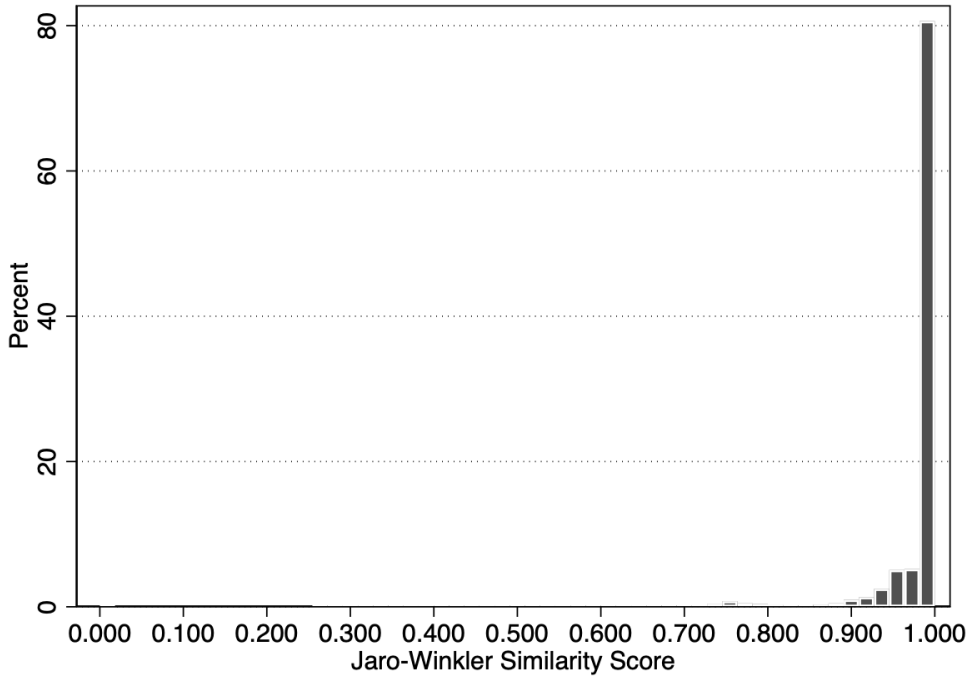
B.3 Data Checks

As an external validation of our author disambiguation, we crosswalk our researcher identifiers with those provided by OpenAlex. We perform the crosswalk at the author-paper level for articles

with a DOI. For each article, we retrieve the full author list from OpenAlex, standardize names using the same procedure as in our data, and sort authors alphabetically by surname and then first name. We then assign each author a sequence number within the article and match our researcher identifiers to OpenAlex identifiers using the DOI and author sequence number.

We assess the quality of these matches by calculating the Jaro-Winkler similarity between each author name in our data and the corresponding name in OpenAlex. This measure ranges from 0 to 1 and places greater weight on common prefixes, making it useful for identifying differences in spelling and transliteration across bibliographic sources. Appendix Figure B4 plots the resulting distribution, which is highly concentrated near one. Of the 381,305 author-paper observations in our master file, 381,254 can be linked to OpenAlex. Among these, 380,921 (99.90%) map to the same researcher in our data and OpenAlex, while only 333 (0.09%) are classified as potential mismatches. The remaining 51 observations (0.01%) cannot be found in OpenAlex. Overall, the close correspondence between the two sets of researcher identifiers provides strong external validation of our author disambiguation procedure.

Figure B4: Author Name Similarity in the OpenAlex Crosswalk



Notes: This figure plots the distribution of Jaro-Winkler similarity scores between author names in our data and the corresponding names in OpenAlex at the author-paper level. The measure ranges from zero to one, with higher values indicating greater name similarity. The sample includes 376,024 author-paper observations for which a similarity score could be computed.

A possible concern with our data construction is that incomplete bibliographic records may introduce measurement error in our dependent variables. To assess EconLit's coverage, we randomly selected 15 researchers from our panel and manually reconstructed their publication records. For

14 of them, we located a CV listing their scholarly output, which we use as the "ground truth" for assessing EconLit coverage. These researchers collectively published eight articles in the top five journals and 21 articles in the top field journals during our sample period. With the exception of one article in the *Journal of Economic History*, all of these publications were correctly recorded in our data, suggesting that EconLit provides reliable coverage of published research.

We also rely on publication records to infer researchers' career histories and mobility. This may introduce measurement error in our treatment variable when researchers move to a university with a local FSRDC but do not immediately publish under their new affiliation. This type of error should, if anything, attenuate our estimates, as some researcher-year observations would be incorrectly classified as untreated despite having local access.

This concern may be more consequential for junior economists, who can take several years to publish their first paper after completing their studies. To assess its importance, we merge our panel with data on researchers' careers from Sarsons et al. (2021). This yields a subset of 349 economists for whom we observe the precise dates of PhD conferral and tenure decisions, along with their publication records. The average time from PhD conferral to first publication is 1.42 years (SD = 2.88), with the vast majority of researchers publishing their first paper within 3 years of graduation. The average time from PhD conferral to tenure is 7.13 years (SD = 2.65), similar to the estimate by Heckman and Moktan (2020). Taken together, these patterns suggest that junior scholars may enter our panel with some delay, but that the resulting attenuation bias is likely to be limited.

C Measuring Diffusion and Impact of FSRDCs

This Appendix provides details on the inception and diffusion of FSRDCs, as well as the measurement of their research impact.

C.1 The Objectives of the FSRDCs Program

The U.S. Census Bureau continually collects individual and business information through statistical surveys and censuses as part of its mandate to assemble timely data on the nation's demographic and economic trends. A crucial challenge in achieving its mission involves earning the trust of households and businesses to elicit complete and truthful answers (Foster et al., 2009; Goroff et al., 2018). For this reason, the data collected by the U.S. Census Bureau is tightly regulated by the U.S. Code. The legal framework regulating the acquisition, use, and protection of confidential data is outlined in Title 13.[4] Among its stipulations is a requirement that Census Bureau employees take an oath to protect data confidentiality and ensure their protection.

The U.S. Census Bureau effectively faces a dual mandate to publish accurate statistics based on its microdata while preserving their confidentiality (Abowd and Schmutte, 2019). From a legal

[4]Data from other Federal Agencies are often integrated and merged with Census records. For instance, the LEHD database was built by linking Census data with IRS records. Such data are governed by the confidentiality provisions outlined in Title 26 of the U.S. Code. For most other statistical data, the legal framework is given by the Confidential Information Protection and Statistical Efficiency Act.

standpoint, Title 13 authorizes the Census Bureau to give access to its confidential information only as it derives a clear benefit for its data programs (McGuckin et al., 1993). The objective of the FSRDC network is thus to maximize the benefit to Census' programs subject to its confidentiality constraints, and not necessarily maximizing confidential data diffusion (Foster et al., 2009). Title 13 also requires that anyone accessing Census data be sworn to observe their confidentiality for life or face severe penalties (including fines of up to $250,000 and up to five years of federal prison), precisely as if they were Census employees. This is why external researchers are required to undergo background checks and face an application procedure that necessarily involves a certain degree of bureaucratic hurdles (and several months).

Therefore, it must be noted that the intention of this study is not to provide a complete policy evaluation of the FSRDC network. First, we would need to expand our focus to other disciplines that benefit from FSRDC data, including demography and especially health policy, which are outside the scope of our dataset. Second, we would need to consider all benefits occurring to the Census' data programs beyond those encapsulated in peer-reviewed publications. Significant examples include improvements to existing datasets, the refinement of survey questionnaires, or even the design of new surveys (Bloom et al., 2019; Chow et al., 2021). Both of these considerations suggest that our analysis can be seen as a likely lower bound on the actual value generated by the FSRDC program for the Census Bureau and society.

Nonetheless, our analysis is important in providing the first data-driven assessment of FSRDCs' success in at least one of their objectives: fostering the diffusion and use of their confidential microdata for research purposes. While the optimal level of data access is ultimately a matter of social preferences (Abowd and Schmutte, 2019), gauging the risks of privacy losses is usually easier than assessing the benefits from data access (Lane, 2021). We contribute to this important debate by quantitatively documenting the significant benefits to research and policy-making that confidential data access entails, beyond the mostly anecdotal evidence existing to date (Bowen, 2024; Card et al., 2010; Einav and Levin, 2014; Lane, 2007).

C.2 Access to FSRDCs Over Time

Table C1 shows the list of the 31 Federal Statistical Research Data Centers opened until 2019.[5] The first was opened in the Census Bureau's Boston Regional Office (Atrostic, 2007). After that, CMU and the CES pioneered a new institutional model where the data center would have been located and operated by a research institution, with the Census Bureau keeping an oversight role. This institutional arrangement became the standard model followed by all subsequent FSRDCs. Interestingly enough, the FSRDC at CMU is also the only one that later closed because of low usage.[6] The closure of the CMU FSRDC prompted a change in the model in which the network is operated: most FSRDCs are now sponsored by consortia of local institutions partnering up to

[5] In 2019, the Census Bureau began to provide remote virtual access to a select number of FSRDC researchers working only with Title 13 data. The pilot has been scaled up during the COVID-19 pandemic, resulting in 83 approved projects by mid-2021. However, the possibility of remote access is not extended to data originating from the IRS – which includes several of the most popular databases such as LBD and LEHD. More details on these developments are available at: https://www.census.gov/about/adrm/fsrdc/about/secure-remote-access.html.

[6] The other exception is the RTI International FSRDC, which was closed as part of a reorganization of the Triangle Research Data Center with Duke University and the University of North Carolina at Chapel Hill.

split the costs of running the data center. Such institutions include universities, regional branches of the Federal Reserve System, and other research centers active in the social sciences, such as the RAND Corporation or the Russell Sage Foundation.

Table C1: Federal Statistical Research Data Centers Opened during our Sample Period

Year opened	Institution of location	City	State
1994	Census Regional Office (then NBER)	Boston (then Cambridge)	Massachusetts
1996*	Carnegie Mellon U	Pittsburgh	Pennsylvania
1999	UC Berkeley	Berkeley	California
1999	UCLA	Los Angeles	California
2000	Duke	Durham	North Carolina
2002	U Michigan	Ann Arbor	Michigan
2003	Chicago FED	Chicago	Illinois
2004	Cornell	Ithaca	New York
2006	Baruch College-CUNY	New York	New York
2010*	RTI International	Research Triangle Park	North Carolina
2010	U Minnesota	Minneapolis	Minnesota
2010	Stanford	Stanford	California
2011	Atlanta FED	Atlanta	Georgia
2012	Texas A&M U	College Station	Texas
2012	U Washington	Seattle	Washington
2014	UC Irvine	Irvine	California
2014	USC	Los Angeles	California
2014	Penn State	University Park	Pennsylvania
2015	U Missouri	Columbia	Missouri
2015	U Nebraska-Lincoln	Lincoln	Nebraska
2015	UW Madison	Madison	Wisconsin
2015	Yale	New Haven	Connecticut
2016	U Maryland	College Park	Maryland
2016	FED Kansas City	Kansas City	Missouri
2017	UT Austin	Austin	Texas
2017	U Colorado Boulder	Boulder	Colorado
2017	U Kentucky	Lexington	Kentucky
2017	Philadelphia FED	Philadelphia	Pennsylvania
2017	Georgetown U	Washington	DC
2018	Dallas FED	Dallas	Texas
2019	Federal Reserve Board	Washington	DC

Note: The FSRDC at CMU closed in 2004, while the FSRDC at RTI International closed in 2018. Additional FSRDCs opened at the University of Utah (2020), University of Illinois Urbana-Champaign (2020), University of Florida (2022), and UNC-Chapel Hill (2022) are outside of our sampling period. To the best of our knowledge, the reported year of opening refers to the date the FSRDC became operative, as can be inferred from CES Yearly Reports and university press releases.

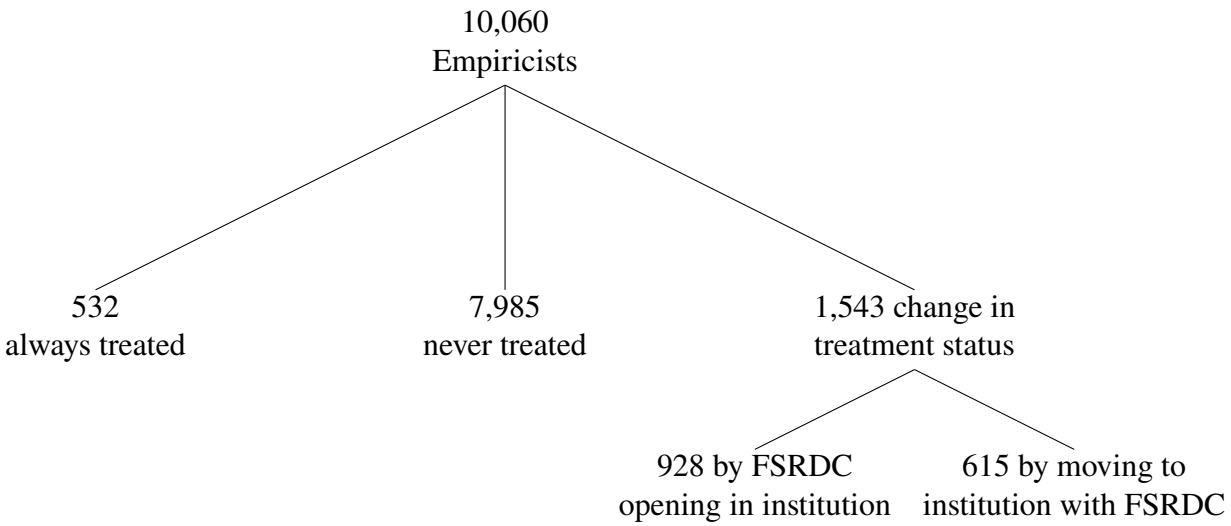
Our primary analysis explores the dynamics of scientific productivity after gaining access to an FSRDC. We define all empirical researchers working at a university with an active FSRDC as "treated".[7] This is consistent with evidence that confidential data diffuse only in the vicinity of the data enclave (Appendix Figure H11). Our interviews further support the importance of

[7]Importantly, this treatment captures local access to an FSRDC rather than approval of a specific research project. We describe the project approval process and its implications for interpreting our treatment in Appendix Section C.3.

geographic proximity in determining effective access. Several FSRDC administrators indicated that, in their experience, even a one-hour commute can be sufficient to discourage a researcher from applying to use confidential data (interviews P39, S94, T52). Nevertheless, we also document that our results are robust when we exclude institutions hosting the data center on their premises and instead consider researchers in non-hosting institutions that are very close-by as treated (Appendix Table H14). Appendix Table H15 further shows the robustness of our main results with alternative definitions of the treatment, such as being affiliated with an institution in the same city as an FSRDC or with an FSRDC consortium member.

Importantly, according to our definition, researchers can gain access to FSRDC data in two ways: when a new data center opens at their current affiliation or when they change employers and move to a new institution with an active FSRDC. Figure C1 provides a breakdown of the researchers in our sample by the modality in which they received access (see also Appendix Figure H2). One might worry that the individuals who advocated opening a local FSRDC might be interested in using it, potentially inflating our estimates. Appendix Table H13 shows that our results are robust to excluding the researchers who appeared in the NSF grants that led to establishing the data centers. An alternative concern is that researchers who gain access by moving to an institution with an active data center might do so precisely because they want to work on confidential data. This endogenous geographical sorting could bias our estimates upward. To assess this concern, we repeated the primary analyses, excluding all scholars who had become co-located with a data center due to changes in affiliation. All the main results are robust to this test (Appendix Table H18). In addition, qualitative evidence from our interviews suggests that this concern is unlikely to hold in practice. While a local FSRDC could, in principle, convince a scholar to accept a job offer, this is usually true only for previous users whose research agenda crucially hinges on having an FSRDC nearby (interview S94).

Figure C1: U.S.-Based Economists by Modality of FSRDC Access



Note: This figure summarizes the treatment status of all empirical economists included in our Panel. See text for more details.

C.3 Project Approval and Access to Confidential Data

Local access to an FSRDC does not automatically authorize researchers to use confidential data. Researchers must first submit a project proposal to the Census Bureau and obtain approval before accessing restricted microdata. An important distinction for our research design is that our treatment captures local access to an FSRDC rather than approval of a specific research project. When an FSRDC opens, we treat all empirical researchers at the institution as gaining access, regardless of whether they subsequently apply to use confidential data. Project approval is thus not a criterion determining treatment status in our analysis.

U.S. Census Bureau guidelines specify five requirements for project approval (U.S. Census Bureau, 2019). First, the project must demonstrate a benefit to Census Bureau programs conducted under Title 13. Second, it must meet a scientific merit requirement. Third, researchers must demonstrate a clear need for non-public data. Fourth, the proposed research must be feasible using the requested data. Finally, the project must satisfy confidentiality and disclosure requirements. Importantly, this process does not rank projects based on expected publication quality, novelty, or policy impact. Furthermore, scientific merit operates primarily as a threshold requirement. For example, prior approval by a federal funding agency such as the NSF or NIH through peer review is sufficient to satisfy this criterion. Our interviews similarly suggested that scientific merit was rarely the binding constraint, with much of the review process focusing instead on feasibility, privacy requirements, and demonstrating benefits under Title 13.

The application process nevertheless imposes meaningful costs on researchers. Our interviewees emphasized both the length and the uncertainty involved in obtaining access. One researcher described the process as requiring scholars to be "a little bit careful and a little bit lucky" (interview S12). The same respondent noted the trade-off involved in deciding whether to use FSRDC data: "the trade-off goes the other way where you say, well, I could be a little bit better with the RDC data, but do I really wanna wait nine months?" (interview S12). Local FSRDC access may reduce some of these administrative costs in addition to the geographic costs of using the data. Census Bureau guidelines assign on-site FSRDC administrators a role in helping researchers develop project proposals (U.S. Census Bureau, 2019). At the same time, final project approval remains centralized at the Census Bureau and is governed by the same review standards regardless of the researcher's location.

Unfortunately, we cannot directly test whether local access affects project approval rates or turnaround times. The Census Bureau does not release information on rejected applications and did not provide these records in response to our FOIA request (FOIA ID No. DOC-CEN-2020-001640). We can observe only approved projects, not the full set of applications or their processing times. This limitation also shapes the interpretation of our estimates. Our treatment captures the effect of reducing barriers to confidential data access under the existing FSRDC screening regime and need not generalize to a setting in which project screening is removed. Importantly, however, project approval does not determine treatment status in our design: all empirical researchers at an institution are considered treated once a local FSRDC becomes operational, regardless of whether they subsequently submit or receive project approval.

C.4 Articles using FSRDC Data

We assembled a novel dataset of all articles that *directly* employ restricted-access microdata accessible only in U.S. Census' secure facilities. When we started our project, there was no official bibliographic record of research using FSRDC data, so we constructed one by carefully searching the bibliome using several complementary strategies.

As a starting point, we exploited the fact that papers using U.S. Census confidential data are expected to indicate it clearly in the acknowledgment of the published version of the paper. We started by collecting all the most commonly used sentences appearing in the acknowledgments of a sample of FSRDC papers (such as "Census Research Data Center", "do not reflect the views of the Census Bureau", and "Special Sworn Status researchers"). Then we searched for them in the main databases of published research, Web of Science and Scopus, which recently started collecting the acknowledgment sections of journal articles. However, we found that many papers do not report the standard disclaimers required by the Census Bureau. We tried to overcome this limitation with additional searches in databases that allow full-text searches, such as JSTOR,[8] Google Scholar and the NBER working paper repository.

We further expanded our search by exploiting the fact that projects approved by the FSRDC are expected to submit a final working paper to the CES for online publication.[9] We collected the metadata of 1,081 working papers and matched them to EconLit through a combination of fuzzy title matching and extensive manual checks. Overall, just about half of these papers were ever published, and only 455 papers could be linked to the corresponding EconLit record.[10] The high share of CES working papers that never get published suggests that output-based assessments of FSRDCs will understate the actual use of the network by academics (see Section C.5 for some alternative approaches).

Finally, we requested access to all approved FSRDC projects via the Freedom of Information Act (FOIA) request to the Census Bureau (FOIA ID No. DOC-CEN-2020-001640). Notably, the Census Bureau subsequently made these records publicly available online.[11] We manually searched the publication records of each researcher whose projects were approved to be carried out in an FSRDC. We supplement these sources with records from FSRDC annual reports, which list research conducted using confidential data. Finally, we manually verify all candidate articles by reading their acknowledgment and data sections and remove false positives. These steps are important because our objective is to identify research using confidential microdata available through FSRDCs rather than research relying exclusively on publicly available Census products.

In total, we found 861 papers published in peer-reviewed journals that could be matched with EconLit. Once we restrict our sample to U.S. researchers affiliated with the AEA, the final sample of papers employing FSRDC data consists of 587 articles written by 509 economics researchers.[12]

[8]The main limitations of JSTOR are that its coverage is unreliable for more recent years and that it does not encompass journals published by Elsevier.

[9]The papers are available online at the following link: https://ideas.repec.org/s/cen/wpaper.html.

[10]Published papers that do not appear in EconLit have either appeared in journals not covered (e.g., the *Strategic Management Journal*) or as a book chapter (mostly in NBER-edited books).

[11]`https://www.census.gov/about/adrm/fsrdc/about/ongoing-projects.html`

[12]During our data construction, we also found several articles from other disciplines, such as sociology, demography,

A more detailed exploration of the characteristics of these papers is provided in Nagaraj et al. (2025).

As an additional validation, we collect data on papers using IPUMS from its official bibliography and test whether FSRDC openings coincide with increased use of this publicly available microdata. Appendix Table C2 shows no significant increase in IPUMS use following local FSRDC access. This placebo result provides further evidence that our measure captures increased use of confidential FSRDC microdata rather than a broader shift toward microdata use at treated institutions.

Table C2: Placebo Test: Effect of FSRDC Access on IPUMS Data Usage

	IPUMS Papers			
	(1)	(2)	(3)	(4)
Post-FSRDC	0.00364	-0.000451	0.000442	-0.00112
	(0.00185)	(0.00179)	(0.00151)	(0.00171)
Year FE	Yes	Yes	Yes	No
Researcher FE	No	Yes	Yes	Yes
University FE	No	No	Yes	No
Tier $\times$ Year FE	No	No	No	Yes
N	155720	155622	155615	155622

Notes: This table presents estimates from OLS models where the dependent variable is the count of papers using IPUMS microdata by researcher i in year t. *Post-FSRDC* equals one in all years after a researcher has been affiliated with a university with an operating FSRDC. Column (1) includes year fixed effects; column (2) includes year and researcher fixed effects; column (3) includes year, researcher, and university fixed effects; column (4) includes researcher and university-tier $\times$ year fixed effects. Standard errors are clustered at the university level. *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively.

C.5 Alternative Measures of FSRDC Use

One potential shortcoming of our measures of FSRDC impact is that they are based on research in peer-reviewed economics journals. However, we might be understating the actual diffusion of FSRDC data if many projects are not published. This would be a concern if papers using FSRDC have a lower propensity to be published, perhaps because they constitute riskier research with a wider variance in scientific quality. To sidestep this concern, we use project approval as an alternative, upstream measure of engagement with FSRDC data that does not depend on whether the resulting research is eventually published. We obtained this information from the list of approved projects published by the U.S. Census Bureau and described in C.4. We supplement these data with similar records from the other agencies participating in the FSRDC program, all obtained with additional FOIA requests.[13] These data give us an upstream measure of data usage that is not dependent on the project's outcomes.

and especially health policy (Foster et al., 2009). We excluded them because they are outside the scope of this analysis.

[13] We obtained data from the AHRQ (FOIA ID No. 2021-00311-FOIA-PHS), the BEA (FOIA ID No. DOC-BEA-2021-000222), the BLS (FOIA ID No. 2021-F-00826) and the NCHS (FOIA ID No. 21-00102-FOIA).

Moreover, projects using Census confidential data must be submitted as working papers to the CES once completed. We record the count of CES working papers up to 2019 inclusive as an additional outcome variable to measure confidential data usage encompassing articles not published in peer-reviewed journals. Interestingly, only 48.4% of CES working papers were published as of June 2020. This figure is close to recent estimates that around 50-74% of NBER working papers are eventually published (Baumann and Wohlrabe, 2020; Lusher et al., 2021). Most CES working papers appeared in a peer-reviewed journal included in EconLit, with just 13% of them appearing as book chapters or in journals of other disciplines. Again, this number is very close to the finding that 12% of published NBER working papers do not appear in economics journals (Lusher et al., 2021).

C.6 Spillovers from Local Access to FSRDCs

A. Articles citing FSRDC papers: Besides using confidential data available only in an FSRDC, we aim to capture how access to confidential microdata might also affect research that does not directly use FSRDC data. Local access to FSRDCs might be shaping the topics or questions researchers decide to work on, for instance, by increasing awareness of past research done with such data or being exposed to the findings of colleagues working in the FSRDC. We measure this type of influence by recording which articles build on the findings of the papers written using FSRDC data. This is done by taking the list of papers using FSRDC data (identified with the procedure of Appendix Section C.4) and downloading their references from the Social Sciences Citation Index (SSCI) curated by Web of Science. We exclude papers directly using confidential microdata from the count since they are likely to mechanically cite other FSRDC papers (for instance, in their data section). The result is a set of papers that build directly on the results made possible by the FSRDC network, even if they do not use its confidential microdata.

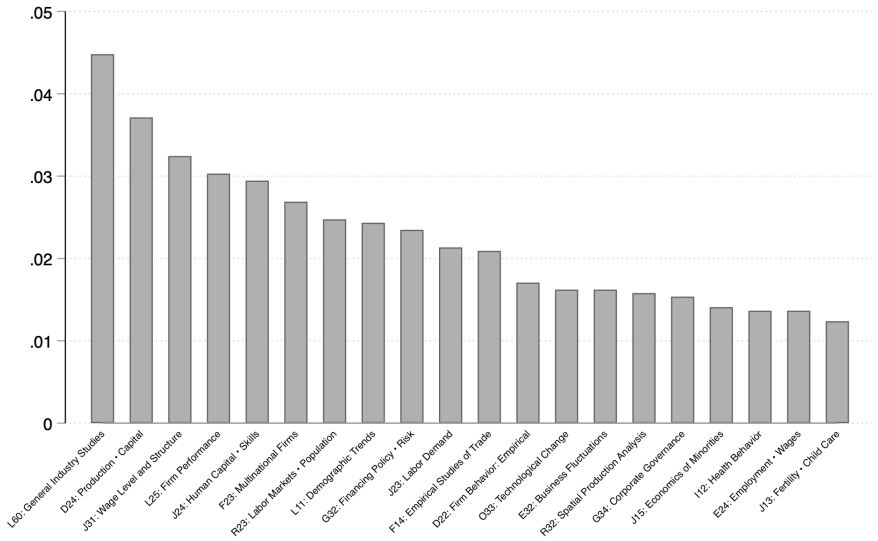
Our interviews provide additional evidence for this awareness mechanism. FSRDC administrators commonly engage in outreach to inform prospective users that "there's all this great data" available through the center (interview S12). More importantly, several interviewees emphasized exposure to colleagues' research as a key channel through which confidential data diffuse locally. One researcher told us that several colleagues in her department began using FSRDC data after being exposed to her work (interview S94). Similarly, one FSRDC director emphasized the importance of word of mouth in spreading the use of confidential data, noting that sheer co-location with a data center is often not sufficient to induce researchers to use it (interview D88).

B. JEL codes: Another approach to measuring spillovers from FSRDC access is to consider its impact on publications that pertain to topics commonly associated with using FSRDC data. Researchers working in areas such as labor, productivity studies, trade, and environmental economics should benefit more from research exposure based on FSRDC data. We capture similarity in research topics using the paper-level JEL codes recorded in EconLit. This is conceptually similar to the approach of Azoulay et al. (2019), who leverage the keyword assigned to all life-science publications by the National Library of Medicine to define proximate research in topical space.

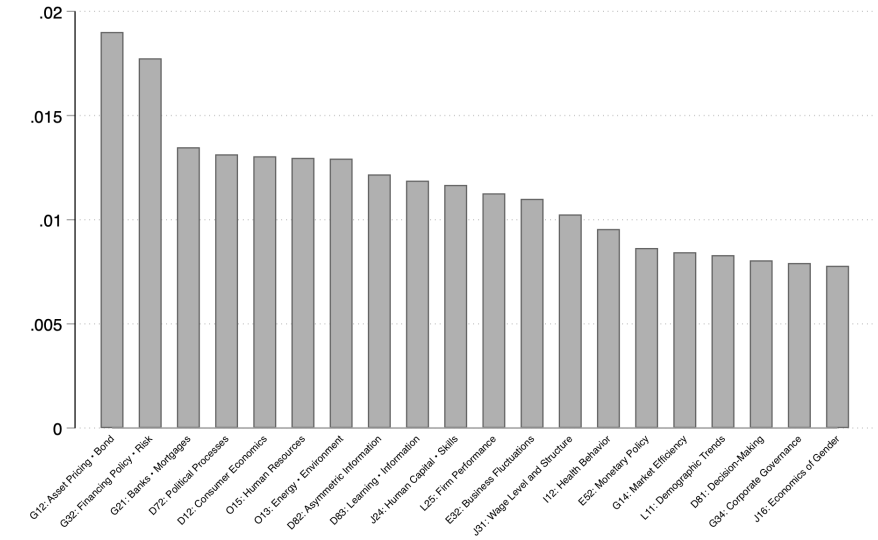
We use the list of FSRDC papers to find the JEL codes most likely to be associated with FSRDC data. Figure C2 displays in Panel (i) the 20 JEL codes with the greatest frequency in papers written in an

Figure C2: List of JEL Codes Associated to Papers using FSRDC Data

(i) JELs most representative of FSRDC research



(ii) JELs most representative of non-FSRDC research



Note: This figure shows the JEL codes most distinctive of research carried out with and without confidential microdata available only in FSRDCs. Each bar represents the share of papers that include that JEL code for each of the two groups of papers. Panel (i) reports the 20 JEL codes with the greatest frequency in papers written in an FSRDC. Panel (ii) reports the 20 JEL codes with the greatest frequency in papers not written in an FSRDC. JEL codes are sorted from largest to smallest difference in frequency. See text for more details.

Table C3: Keywords Used to Tag Articles' Research Design and Data

Category	Keywords (inspired by Currie et al., 2020)
Administrative data	"administrative data" "administrative record" "administrative evidence" "administrative earning" "administrative sample" "administrative report" "administrative panel" "administrative microdata" "administrative regist" "registry data" "register data" "archival" "archives" "tax record" "payroll record" "social security record" "wealth record" "voting record" "earnings record" "personnel record"
Survey data	"survey data" "survey panel data" "survey result" "survey response" "longitudinal survey" "preference survey" "contingent valuation survey" "labor force survey" "values survey" "consumption survey" "population survey" "happiness survey" "national survey" "phone survey" "survey administered" "household survey" "survey evidence"
Quasi-experimental methods	"difference in diff" "differenceindiff" "differences in diff" "differencesindiff" "d-in-d" "diff in diff" "diffindiff" "event stud" "eventstud" "staggered adoption" "regression discontinuit" "regressiondiscontinuit" "regression kink" "regressionkink" "rd resign" "rdresign" "rd estimat" "rdestimat" "rd model" "rdmodel" "rd regression" "rdregression" "rd coefficient" "rdcoefficient" "rk design" "rkdesign" "rdd" "rkd" "instrumental variable" "instrumentalvariable" "two stage least squares" "twostage least squares" "2sls" "tsls" "valid instrument" "exogenous instrument" "iv estimat" "ivestimat" "iv regression" "ivregression" "iv strateg" "ivstrateg" "we instrument" "i instrument" "exclusion restriction" "paper instruments" "weak first stage" "simulated instrument"
Experimental methods	"randomized controlled trial" "randomized field experiment" "randomized controlled experiment" "randomised controlled trial" "randomised control trial" "randomised field experiment" "randomised controlled experiment" "social experiment" "rct" "laboratory experiment" "lab experiment" "public good game" "public goods game" "ztree" "orsee" "showup fee" "laboratory participant" "lab participant" "pre analysis plan" "preanalysis plan" "pre registered" "preregistered" "preregistration" "dictator game" "ultimatum game" "trust game"

Note: This table presents the keywords used to tag each paper as belonging to one of the following non-mutually exclusive categories: administrative data, survey data, quasi-experimental methods, and experimental methods. The list of keywords is built on Currie et al. (2020).

FSRDC. The figure confirms anecdotal evidence that underscores the considerable potential of these data for firm-level analyses. Other common topics include studying labor markets, demographic trends, and international trade. Similarly, Panel (ii) of Figure C2 shows the 20 JEL codes that appear with the greatest frequency in research *not* conducted in FSRDCs. These are more likely to encompass topics like financial markets, development, monetary economics, and theoretical investigations of market efficiency and uncertainty. Appendix Table H21 shows the robustness of our main results to alternative definitions of which JEL codes are the most representative of research enabled by access to FSRDCs.

C. Text-based measures of research design: Finally, we compile data to assess potential spillovers on the adoption of empirical methods and administrative data from other sources. Following Currie et al. (2020), we use a series of regular expression searches to find keywords in the titles and abstracts that identify the use of a certain method (e.g., difference-in-differences) or type of data (e.g., survey data). The list of n-grams we use is presented in Table C3 and builds on the list in Appendix A of Currie et al. (2020). These keywords allow us to tag each paper as belonging to one of the following non-mutually exclusive categories: administrative data, survey data, quasi-experimental methods, and experimental methods.

Table C4 explores what keywords are more likely to be associated with papers using FSRDC data. Confirming our priors, these papers are five times more likely to mention the use of administrative records, while the same is not true for words denoting the use of traditional surveys compiled for research purposes. Further, Table C4 matches our intuition that research carried out in an FSRDC does not employ experimental research design (such as RCTs or behavioral laboratory experiments).

Table C4: Characteristics of Papers Using FSRDC Data

	Administrative Data	Survey Data	Quasi-Exp. Methods	Experimental Methods
	(1)	(2)	(3)	(4)
FSRDC Paper	0.0364***	0.00589	0.0139	-0.0131***
	(0.00811)	(0.00351)	(0.00850)	(0.00041)
Year FE	Yes	Yes	Yes	Yes
N	188181	188181	188181	188181
Mean of DV	0.00644	0.00208	0.0310	0.0112

Note: This table presents estimates from OLS models evaluating the likelihood that papers using FSRDC data mention the keywords listed in Table C3. *, **,*** denote significance at 5%, 1% and 0.1% level respectively. Robust standard errors are reported in parentheses.

Including year fixed effects helps alleviate the concern that these findings are just the reflection of a wider "credibility revolution" in the years of FSRDC activity.

The interviews also suggest that exposure to FSRDC research may lead researchers toward administrative data from other sources. Several respondents noted that comparable microdata from foreign countries can offer many of the same research advantages while imposing substantially lower barriers to access (interviews F72, D88). One respondent suggested that these differences may have contributed to the growing use of foreign administrative data among U.S.-based economists.[14] This qualitative evidence motivates our test of whether local FSRDC access increases researchers' use of administrative data beyond the confidential microdata available through the FSRDC network.

C.7 Illustrative Case: Hsieh and Klenow (2009)

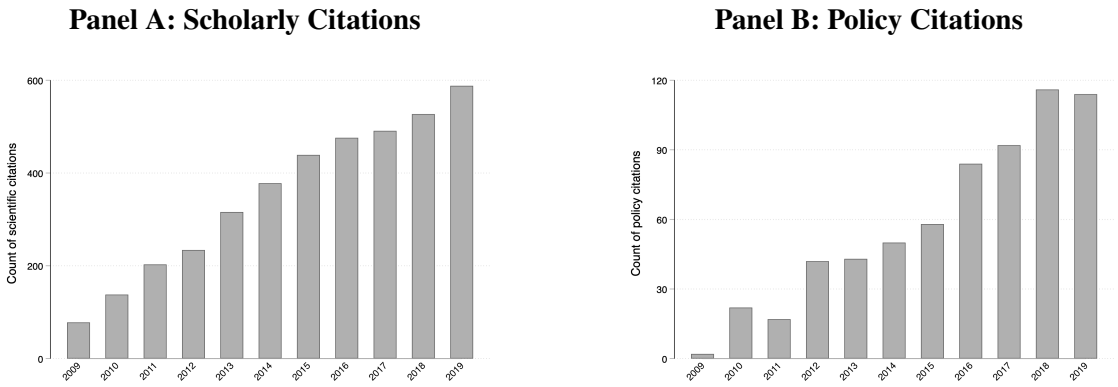
The development of Hsieh and Klenow's work on resource misallocation (Hsieh and Klenow, 2009) illustrates how local access can affect research through both direct access to data and exposure to colleagues who use confidential data. We reconstruct the origins of this project using Census Bureau records and correspondence with both authors.[15]

The paper by Hsieh and Klenow (2009) was carried out using confidential microdata available only at the U.S. Census Bureau (Figure C4). We gathered additional information about this paper from U.S. Census Bureau's records. The misallocation project started in 2006 at the Berkeley FSRDC, where Hsieh was affiliated at the time. In the abstract of the project submitted to the CES, the authors delineated their objective of developing a new methodology to "help shed light on the underlying sources of productivity differences". To do so, they asked for access to

[14] As one respondent explained, "I think at the moment there is this issue which people in the RDC community are sort of talking about, which is that the U.S. data is a lot harder to work with than German data or Dutch data or whatever. [...] And there's a sense in the U.S. perhaps that [...] the data is very good and those who are working with the data can do good stuff. But there's a lot of hurdles to get to it and it's not clear that the data is really even necessarily as good or better than some of the European sorts of data. [...] If you have to look at the U.S., then it is the best data available. But if you just want to test the theory and you don't really care whether it's for the U.S. or Germany or France, the U.S. data may not be quite as good" (interview D88).

[15] At the end of 2019, the paper "Misallocation and manufacturing TFP in China and India" (hereafter, "Misallocation paper") already had over 3,850 citations on Google Scholar, corresponding to over 352 citations every year. Moreover, the paper proved extremely influential in the policy debate (Figure C3). According to data collected by Overton, over 640 policy documents had cited this paper as of 2019.

Figure C3: Impact of the Misallocation Paper by Hsieh and Klenow (2009)



Note: This figure shows the impact of the Misallocation paper by Hsieh and Klenow (2009). Panel A reports the number of scholarly citations received by the Misallocation paper using data from Google Scholar. Panel B reports the number of policy citations received by the Misallocation paper using data from Overton. See text for more details.

confidential U.S. company microdata from the Census of Manufactures (from 1963, 1967, 1972, 1977, 1982, 1987, 1992, 1997, and 2002) and the Annual Survey of Manufacturers (1973-2001). The granularity of the establishment-level data collected by the U.S. Census Bureau was crucial to studying the misallocation of production factors across firms. Indeed, Hsieh told us it would not have been possible to do the same paper without FSRDC data (Hsieh, personal communication, 21st December 2020).

Figure C4: Footnote of the Misallocation Paper by Hsieh and Klenow (2009)

> *We are indebted to Ryoji Hiraguchi and Romans Pancs for phenomenal research assistance, and to seminar participants, referees, and the editors for comments. We gratefully acknowledge the financial support of the Kauffman Foundation. Hsieh thanks the Alfred P. Sloan Foundation and Klenow thanks SIEPR for financial support. The research in this paper on U.S. manufacturing was conducted while the authors were Special Sworn Status researchers of the U.S. Census Bureau at the California Census Research Data Center at UC Berkeley. Research results and conclusions expressed are those of the authors and do not necessarily reflect the views of the Census Bureau. This paper has been screened to ensure that no confidential data are revealed. *chsieh@chicagobooth.edu, pete@klenow.net*.

Note: The figure reports the footnote of the paper by Hsieh and Klenow (2009), highlighting their usage of the Berkeley FSRDC.

We asked both Hsieh and Klenow about the origins of their collaboration and how much the need to physically access FSRDC data shaped their work. Before starting the project, the two authors had been considering the idea but needed data to develop the methodology and conduct empirical validation. Then, one day, Hsieh learned about the data available at the Berkeley FSRDC from a graduate student who was using it for her dissertation. This detail highlights how local FSRDCs can result in knowledge spillovers that substantially alter research trajectories. Interestingly, even though the Berkeley FSRDC had been operating since 1999, Hsieh became aware of the potential of FSRDC data for his research only after seeing the same data used by someone else.

The analysis for the Misallocation paper was supervised mainly by Hsieh, who had easier access

to the Berkeley FSRDC. Indeed, the physical distance between Berkeley and Stanford was a formidable barrier to the use of confidential FSRDC data: as noted by Klenow, "if I hadn't had a co-author at Berkeley, I don't think I would have started the Misallocation paper" (Klenow, personal communication, 8th December 2020). This is notable given that the campuses of UC Berkeley and Stanford are relatively close by, just about 40 miles apart, which prevented Stanford from obtaining its own FSRDC for many years. In 2010, Stanford was eventually allowed to establish a branch of the California FSRDC on its campus. The new FSRDC branch drastically reduced the geographical barriers faced by Stanford researchers. In the official press release of the opening, Nick Bloom was quoted saying that "having to go to Berkeley was an immense waste of time"[16].

D Classification of Research by Methodological Orientation

In this Appendix, we explain the machine learning procedure used to classify the methodological orientation of each EconLit record. We then discuss how we infer each researcher's specialization from their published work and explore the robustness of our results to this classification.

D.1 Article Classification Algorithm

We employ the machine learning classification procedure developed by Angrist et al. (2020) to classify each economics paper as either empirical or theoretical. The algorithm aims to separate research that produces data-based estimates of economically meaningful parameters from purely theoretical or methodological papers. We also classify papers that develop new methods or models but apply them to produce substantively meaningful estimates as empirical. The classifier developed by Angrist et al. (2020) is a logistic ridge regression that fits a dummy variable indicating whether an article is an empirical paper, using article titles, journals, JEL codes, publication years, and abstracts as inputs. The algorithm is trained on a set of 5,469 hand-classified papers and achieves 87% accuracy among articles with an abstract (Angrist et al., 2020). Because abstracts are important inputs to the classification, we supplement missing EconLit abstracts using online bibliographic sources like Google Scholar.

This procedure yields a paper-level score (between 0 and 1) that reflects the probability that an article is empirical. Figure D1 illustrates the classifier for one empirical and one theoretical article, including a theoretical paper whose title or abstract contains terms such as "data" or "estimates" that could otherwise suggest empirical content. We classify all papers with a predicted score greater than 0.5 as empirical.[17] Our data show an increasing share of empirical publications over time, from 52.3% in 1990 to 71.3% in 2019 (Figure D2). Relative to the findings of Angrist et al. (2020), we document a slightly larger share of empirical research. Possible reasons for this discrepancy are the better coverage of abstracts in our data (which allows for a more precise classification) and the fact that we consider a broader set of journals in our analyses.

[16] Source: https://news.stanford.edu/news/2010/february/census-data-center-020210.html

[17] Unlike Angrist et al. (2020), we do not separately classify econometrics articles into an ad-hoc category.

Figure D1: Example of Article Classification using Machine Learning

Panel A: Empirical Article

DO ENERGY EFFICIENCY INVESTMENTS DELIVER? EVIDENCE FROM THE WEATHERIZATION ASSISTANCE PROGRAM*

MEREDITH FOWLIE
MICHAEL GREENSTONE
CATHERINE WOLFRAM

A growing number of policies and programs aim to increase investment in energy efficiency, because conventional wisdom suggests that people fail to take up these investments even though they have positive private returns and generate environmental benefits. Many explanations for this energy efficiency gap have been put forward, but there has been surprisingly little field testing of whether the conventional wisdom is correct. This article reports on the results of an experimental evaluation of the nation's largest residential energy efficiency program—the Weatherization Assistance Program—conducted on a sample of approximately 30,000 households in Michigan. The findings suggest that the upfront investment costs are about twice the actual energy savings. Furthermore, the model-projected savings are more than three times the actual savings. Although this might be attributed to the "rebound" effect—when demand for energy end uses increases as a result of greater efficiency—the article fails to find evidence of significantly higher indoor temperatures at weatherized homes. Even when accounting for the broader societal benefits derived from emissions reductions, the costs still substantially outweigh the benefits; the average rate of return is approximately −7.8% annually. *JEL Codes:* Q4, Q48, Q5.

P(Empirical) = 96.9%

Panel B: Theoretical Article

Econometric Theory, **13**, 1997, 467–505. Printed in the United States of America.

GAUSSIAN ESTIMATION OF MIXED-ORDER CONTINUOUS-TIME DYNAMIC MODELS WITH UNOBSERVABLE STOCHASTIC TRENDS FROM MIXED STOCK AND FLOW DATA

A.R. BERGSTROM
University of Essex

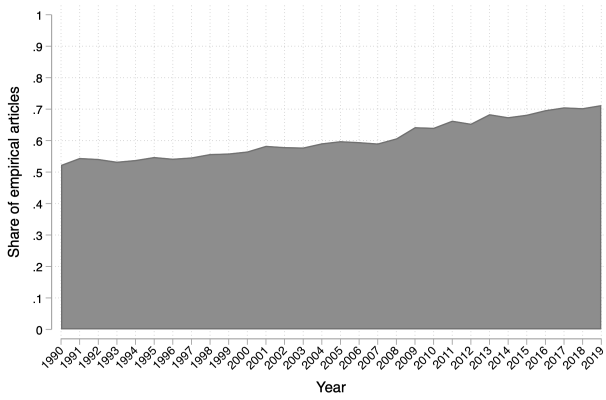
This paper develops an algorithm for the exact Gaussian estimation of a mixed-order continuous-time dynamic model, with unobservable stochastic trends, from a sample of mixed stock and flow data. Its application yields exact maximum likelihood estimates when the innovations are Brownian motion and either the model is closed or the exogenous variables are polynomials in time of degree not exceeding two, and it can be expected to yield very good estimates under much more general circumstances. The paper includes detailed formulae for the implementation of the algorithm, when the model comprises a mixture of first- and second-order differential equations and both the endogenous and exogenous variables are a mixture of stocks and flows.

P(Empirical) = 16.4%

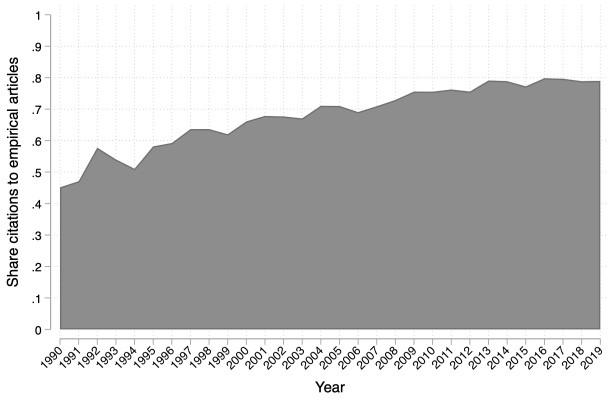
Note: This figure shows the results from the machine learning classification algorithm of Angrist et al. (2020) for two papers in our sample. See text for more details.

Figure D2: Publications and Citations to Economics Papers by Methodological Orientation

Panel A: Empirical Articles



Panel B: Citations to Empirical Articles

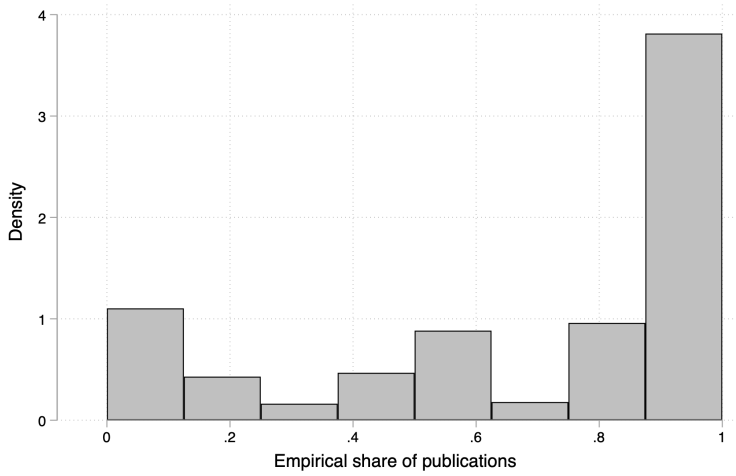


Note: This figure shows the share of empirical articles (Panel A) and citations received by empirical articles (Panel B) over time. See text for more details.

D.2 Researcher Classification and Validation

We use the paper-level classification described above to characterize the methodological orientation of individual researchers. In our baseline definition, we classify a researcher as an empiricist if more than half of up to her first five observed publications are classified as empirical. Focusing on researchers' earliest publications limits the extent to which the classification reflects changes in research orientation later in their careers. Figure D3 reports the distribution of the empirical share

Figure D3: Empirical Articles across Researchers in our Sample



Note: This figure reports the distribution of the share of a researcher's first five publications classified as empirical. For researchers with fewer than five publications, the share is calculated using all publications we observe. In our baseline specification, we classify a researcher as an empiricist if more than half of these publications are empirical.

among the first five publications of researchers in our sample.

We assess the validity of this researcher-level classification using two complementary exercises. First, we compare our classification with the composition of the Editorial Boards of five prominent economics journals spanning the empirical-theoretical spectrum. We collect the information reported on each journal's website as of November 2022 and match its members to the economists in our data. Table D1 reports the share of matched editors whom our procedure classifies as empiricists. The resulting pattern closely tracks the methodological orientation of the journals. Only 4.3% of editors at the *Journal of Economic Theory* are classified as empiricists, compared with 94.3% at *AEJ: Economic Policy* and 97.0% at *AEJ: Applied Economics*. The corresponding shares for *AEJ: Microeconomics* and *AEJ: Macroeconomics* lie between these extremes.

Second, we compare our classification with researchers' dissertation fields, which provide an independent measure of methodological orientation early in their careers. We digitize the annual lists of North American Ph.D. graduates published in the *Journal of Economic Literature* and match these records to researchers in our data. Figure D4 reports dissertation fields separately for researchers whom we classify as empiricists and theorists. The distributions are consistent with the distinction our classifier captures. Researchers classified as theorists are disproportionately concentrated in Microeconomics, Macroeconomics, and Mathematical Methods, whereas Labor, Health, and Development Economics are among the most common dissertation fields among researchers classified as empiricists.

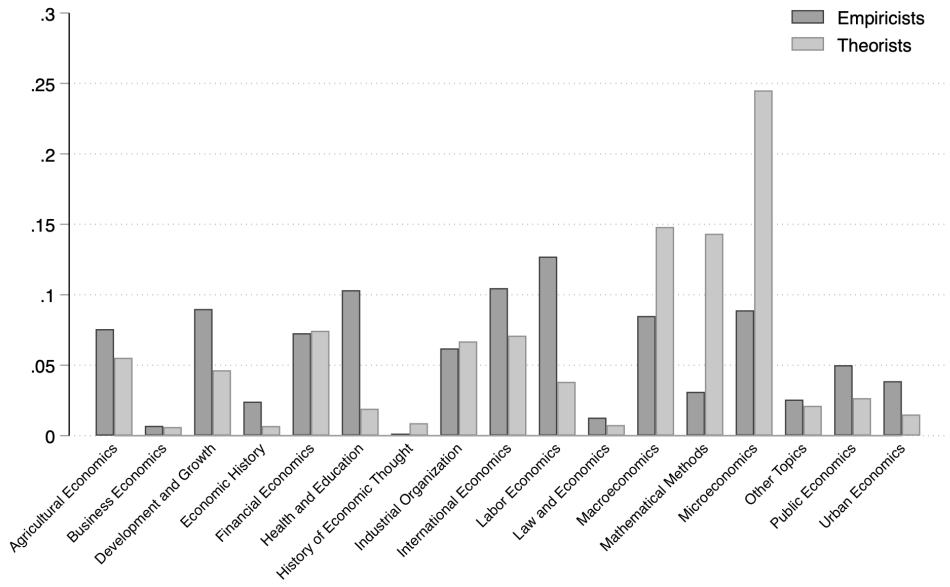
Finally, we examine whether our results depend on the particular publication history used to classify researchers. This is important for two reasons. Although our baseline definition does not require five publications, researchers with shorter publication records are classified using fewer observations.

Table D1: Validation of Researcher Classification Using Editorial Boards

Journal	Number of Editors	Ever U.S.-affiliated	Share Empirical
Journal of Economic Theory	62	47	4.26%
AEJ: Microeconomics	21	19	26.32%
AEJ: Macroeconomics	19	18	61.11%
AEJ: Economic Policy	37	35	94.29%
AEJ: Applied Economics	36	33	96.97%

Note: This table compares our researcher-level methodological classification with the editorial boards of five economics journals as of November 2022. *Ever U.S.-affiliated* reports the number of editors whom we match to a researcher in our sample. *Share Empirical* reports the share of matched editors whom our procedure classifies as empiricists based on the methodological orientation of up to their first five observed publications.

Figure D4: Dissertation Fields of Researchers Classified as Empiricists and Theorists



Note: The figure reports the distribution of dissertation topics for each researcher in our sample that we could match to the *Journal of Economic Literature* (JEL) yearly lists of graduates. We matched 4,903 economists, of which 3,438 were classified as empiricists and 1,465 as theorists. See text for more details.

Furthermore, some of the first five publications may appear after a researcher gains local FSRDC access, raising the possibility that the treatment itself affects the classification. Appendix Table D2 addresses both concerns. Panel A classifies researchers based solely on whether their first publication is empirical, while Panel B instead adopts a more stringent definition based on the first seven publications of a researcher. Panel C constructs the classification using only pre-treatment

information (when available). Researchers whose publication record begins only after an FSRDC is already operating at their institution cannot be classified using pre-treatment publications; for these researchers, we instead use their first publication. Across all definitions, our results remain very similar to our baseline estimates.

Table D2: Robustness of Main Results to Alternative Definitions of Empirical Researchers

Panel A: Classification Based on First Publication

	(1) Top Pubs	(2) Cite-Weighted Pubs	(3) Policy Pubs
Post-FSRDC	0.0260**	1.888***	0.0424***
	(0.00837)	(0.42558)	(0.01136)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
N	146,397	146,397	146,397

Panel B: Classification Based on First Seven Publications

	(1) Top Pubs	(2) Cite-Weighted Pubs	(3) Policy Pubs
Post-FSRDC	0.0246**	1.386**	0.0422***
	(0.00823)	(0.43161)	(0.01143)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
N	157,046	157,046	157,046

Panel C: Classification Based on Pre-Treatment Publications

	(1) Top Pubs	(2) Cite-Weighted Pubs	(3) Policy Pubs
Post-FSRDC	0.0202*	0.983**	0.0271*
	(0.00797)	(0.36849)	(0.01152)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
N	159,416	159,416	159,416

Note: This table presents estimates from OLS models using alternative definitions of empirical researchers. In Panel A, a researcher is classified as an empiricist if their first publication is empirical. In Panel B, a researcher is classified as an empiricist if a majority of their first seven publications are empirical. Panel C uses pre-treatment publication histories whenever available: researchers with publications predating local FSRDC access are classified based on whether a majority of their pre-FSRDC publications are empirical, while researchers whose publication record begins only after an FSRDC is already operating at their institution are classified based on their first publication. *Post-FSRDC* equals one in all years after a researcher has been affiliated with a university with an operating FSRDC. All models include researcher and university-tier × year fixed effects. Standard errors are clustered at the university level. *, **,*** denote significance at the 5%, 1%, and 0.1% level, respectively.

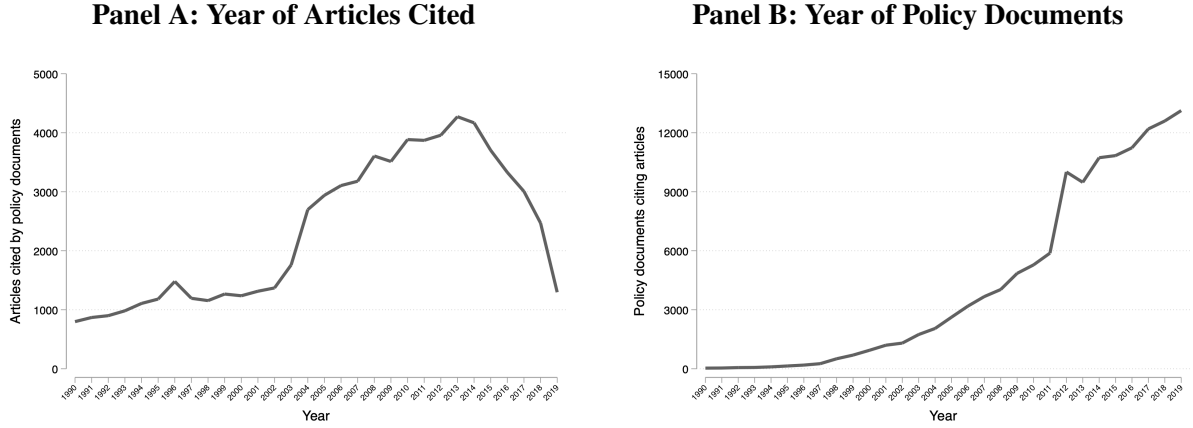
As an additional robustness check, Appendix Figure H9 reports estimates across progressively more stringent definitions of empirical orientation based on the share of a researcher's first five publications classified as empirical. The estimated effects generally increase with researchers' empirical orientation. Appendix Table H16 further shows precisely estimated null effects among researchers classified as theorists. Taken together, these exercises provide evidence that our results are not sensitive to the particular threshold used to distinguish empirical from theoretical researchers.

E Measuring the Policy Impact of Academic Research

E.1 Overton Data

Citations from policy documents (or "policy citations" for brevity) provide a measure of the use of academic research outside academia. Until recently, their use in bibliometric research was limited by the lack of systematic databases that link references in unstructured policy documents to scientific publications (Yin et al., 2022). The growing availability of bibliographic metadata and digitized policy documents has enabled the construction of such measures at scale. We measure the policy impact of economics research using data from Overton, which maintains a large database of policy documents and links their references to the academic literature. The database covers documents produced by government agencies, intergovernmental organizations, think tanks, and other policy institutions across more than 180 countries.[18] Overton also includes historical policy documents dating back several decades, allowing us to capture policy use that may occur well after an article is published.

Figure E1: Year of Publication of Articles Receiving Policy Citations and of the Citing Policy Documents in the Overton Data

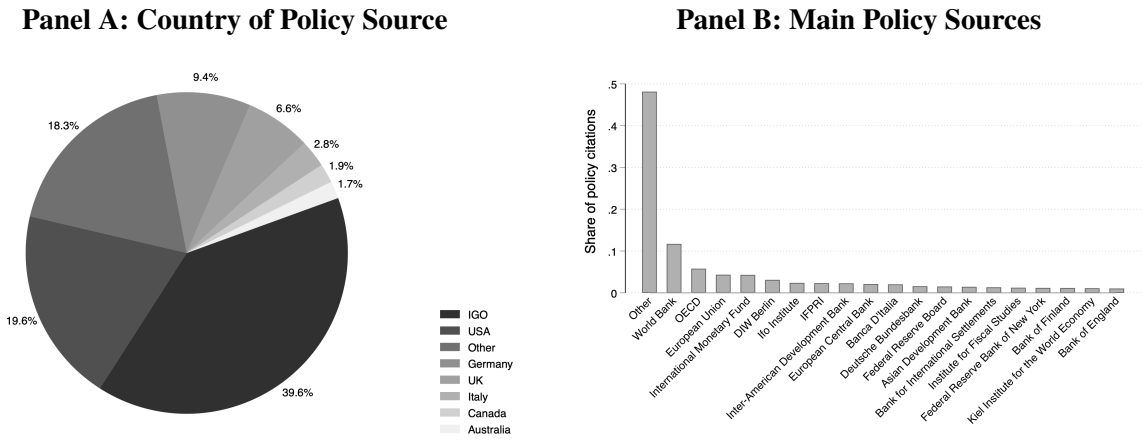


Note: This figure shows the publication year of the economics articles that received at least one policy citation (Panel A) and the publication year of the policy sources that cite them (Panel B). See text for more details.

We merge the EconLit publication data with Overton using article DOIs. In total, we identify 750,842 citations from 129,042 policy documents to 70,895 economics articles, corresponding to

[18] Source: https://help.overton.io/article/what-is-overtons-coverage-and-how-does-it-compare-to-other-systems/

Figure E2: Origin of Policy Citations to Economic Scholarship in the Overton Data



Note: This figure shows the countries of origin of the policy documents citing economics research (Panel A) and their main sources (Panel B). See text for more details.

approximately 38% of the articles in our sample. In this count, we excluded citations from NBER, CEPR, and IZA documents because these sources primarily contain working-paper versions of academic articles rather than policy documents. Figure E1 reports the publication years of cited articles and citing policy documents. Overton coverage is substantially greater for recent policy documents. Consistent with this pattern, articles published after the mid-2000s are more likely to have an observed policy citation in our data. Our empirical specifications include publication-year fixed effects, which absorb common differences in citation coverage across article cohorts.

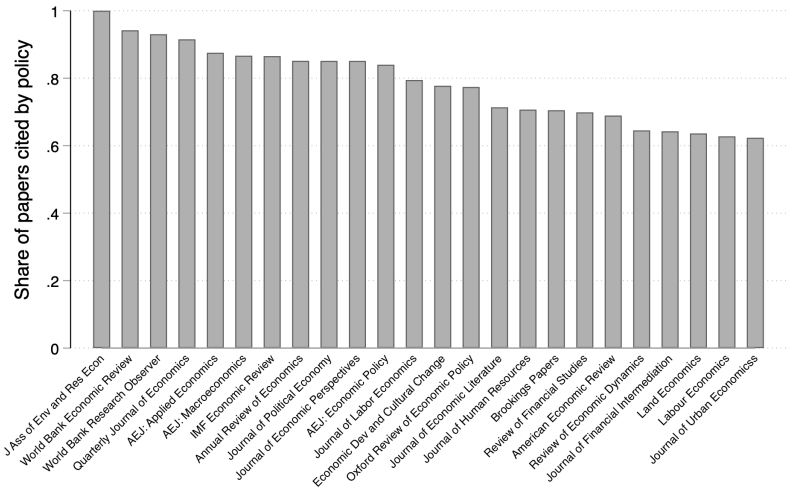
Figure E2 reports the geographic and institutional composition of the policy sources in our data. The United States is the largest individual country of origin, accounting for approximately 20% of policy citations. Intergovernmental organizations collectively account for an even larger share, close to 40%. Figure E3 shows substantial variation across publication outlets: articles published in highly ranked general-interest journals (e.g., the *Quarterly Journal of Economics*) and explicitly policy-oriented outlets (e.g., the *World Bank Research Observer* or the *IMF Economic Review*) are particularly likely to receive policy citations.

Policy citations are positively correlated with conventional measures of academic impact, but capture a distinct dimension of research use. At the article level, the Pearson correlation between policy citations and academic citations is 0.53, consistent with prior evidence that policy and scientific impact are related but distinct (Yin et al., 2022). In unreported analyses, the descriptive relationships reported below are similar when we additionally control for academic citations.

E.2 Policy Relevance of FSRDC-Based Research

We next examine how policy citations vary with the empirical and data characteristics of economics research. Table E1 reports article-level regressions relating policy citations to whether a paper is empirical or uses administrative data. All specifications include publication-year and journal fixed effects. Empirical papers receive substantially more citations from policy documents than theoretical papers. We find that papers mentioning the use of administrative data in their abstract receive,

Figure E3: Share of Articles Receiving at Least One Citation from Policy Documents by Journal



Note: The figure reports the share of published articles that received at least one policy citation for each journal. Only the journals with the highest shares are shown in the graph.

on average, twice as many policy citations (Columns 3 and 4). The association is particularly large for papers using confidential FSRDC data: these articles receive approximately 4.1 additional citations from U.S. policy documents and 5.5 additional citations from non-U.S. policy sources. The fact that the increase of citations from outside the United States is proportionately much smaller fits well the intuition that evidence deriving from FSRDC data should be particularly salient to inform policy-making in the U.S. (Einav and Levin, 2014). These descriptive patterns suggest that research using confidential government microdata is particularly likely to enter policy discussions.

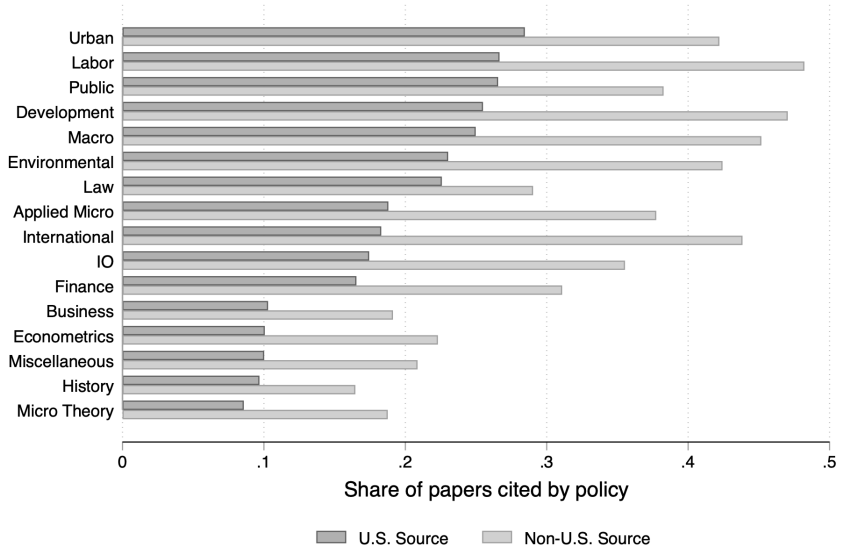
The policy relevance of economics research also varies across fields. Figure E4 shows that papers

Table E1: Policy Citations and Characteristics of Economics Research

	US Cites	Non-US Cites	US Cites	Non-US Cites	US Cites	Non-US Cites
	(1)	(2)	(3)	(4)	(5)	(6)
Empirical (0/1)	0.577***	2.182***				
	(0.04772)	(0.21677)				
Administrative (0/1)			0.697*	0.423		
			(0.28719)	(0.47368)		
FSRDC Use (0/1)					4.137***	5.473**
					(0.47620)	(1.98045)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Journal FE	Yes	Yes	Yes	Yes	Yes	Yes
N	188181	188181	188181	188181	188181	188181
Mean of DV	0.784	3.215	0.784	3.215	0.784	3.215

Note: This table presents estimates from OLS models evaluating the average increase in policy citations for papers that are empirical (columns 1 and 2), mention administrative data in their abstract (columns 3 and 4), and directly use FSRDC data (columns 5 and 6). Papers in columns 3 and 4 are tagged using the keywords listed in Table C3. *, **,*** denote significance at 5%, 1% and 0.1% level respectively. Standard errors are clustered by year of publication.

Figure E4: Share of Articles Receiving at Least One Citation from Policy Documents by Field

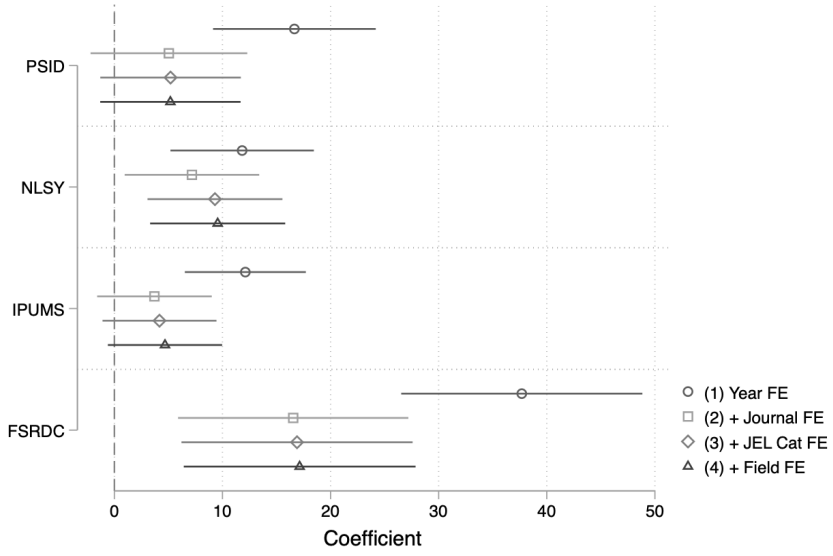


Note: This table reports article-level OLS regressions relating policy citations to characteristics of economics research. Columns (1) and (2) compare empirical and theoretical papers, columns (3) and (4) compare papers mentioning administrative data with other papers, and columns (5) and (6) compare papers directly using confidential FSRDC data with other papers. The dependent variables are the number of citations received from U.S. and non-U.S. policy documents. All specifications include publication-year and journal fixed effects. Papers using administrative data are identified using the keywords reported in Appendix Table C3. Standard errors are clustered by publication year. *, **,*** denote significance at the 5%, 1%, and 0.1% levels, respectively.

in development, labor, and urban economics tend to be very influential in the policy debate, unlike fields such as history or microeconomic theory. This heterogeneity raises a concern: the policy citation premium associated with FSRDC research could reflect differences in the topics or types of research that use these data, rather than the inherent value of confidential government microdata.

To assess whether the policy citation premium is specific to FSRDC-based research or instead reflects a broader increase in policy attention to microdata, we compare articles using FSRDC data with articles using other widely used microdata sources. These include the Panel Study of Income Dynamics (PSID), the National Longitudinal Survey of Youth (NLSY), and IPUMS (the open-access data produced by the U.S. Census Bureau). We estimate article-level regressions in which the dependent variable is the number of policy citations and progressively absorb differences across publication years, journals, JEL codes, and research fields. Figure E5 shows that papers using confidential FSRDC data receive substantially more policy citations than papers using other microdata sources. In the most restrictive specification, FSRDC papers receive approximately 9.5 additional policy citations relative to papers that do not use these microdata sources; the corresponding estimates for PSID, NLSY, and IPUMS are considerably smaller. These differences suggest that the greater policy relevance of FSRDC-based research is not simply a feature of microdata-based research.

Figure E5: Policy Citations to Research Using FSRDC and Other Microdata Sources



Note: This figure presents coefficient estimates from paper-level OLS regressions where the dependent variable is the number of citations received from policy documents. The regressions include separate indicators for papers using FSRDC data, the Panel Study of Income Dynamics (PSID), the National Longitudinal Survey of Youth (NLSY), and IPUMS. The specifications progressively include publication year, journal, JEL code, and field fixed effects. Bars report 95% confidence intervals based on robust standard errors.

E.3 Example: Greenstone et al. (2010)

To illustrate the type of policy use captured by the Overton data, consider the paper "Identifying Agglomeration Spillovers: Evidence from Winners and Losers of Large Plant Openings" published in 2010 by Michael Greenstone, Richard Hornbeck, and Enrico Moretti (henceforth, "Million Dollar Plant paper"). The paper estimates localized productivity spillovers from the opening of large manufacturing plants by comparing incumbent plants in counties that narrowly won or lost competitions for new plants. The analysis relies on several confidential datasets available through the FSRDC network, including the Standard Statistical Establishment List (SSEL), the Annual Survey of Manufactures (ASM), and the Census of Manufactures (CM). Using these data, Greenstone et al. (2010) find that incumbent plants in winning counties experience substantially greater productivity growth following the plant opening.

Such findings have obvious implications for policies aimed at local economic development, as the authors themselves note in the paper's introduction. Indeed, the paper received 101 citations from policy documents through 2019. These citations span a wide range of institutions and countries, including U.S. policy organizations, intergovernmental organizations, and government agencies outside the United States. The first policy citation in our data appears in 2010, the year the article was published, in a Hamilton Project discussion paper from the Brookings Institution.[19]. That report cites Greenstone et al. (2010) as evidence on the productivity effects of industrial clustering

[19]Source: https://www.brookings.edu/wp-content/uploads/2016/06/10_job_creation_bartik.pdf

and considers its implications for place-based economic policy.

Subsequent policy documents use the paper in similar ways. A report prepared for the U.K. Department for Business, Innovation and Skills cites the study as evidence on productivity spillovers from investment,[20], while the Australian Productivity Commission references it in its review of agglomeration economies.[21] Other policy sources, including the Inter-American Development Bank, Bruegel, and the OECD, draw on the paper when assessing the potential effects of place-based policies and investment-promotion programs.

F Evidence on Research using FSRDC Datasets

This Appendix provides additional descriptive evidence on research using confidential FSRDC data. We first describe the development and expansion of the data resources available through the FSRDC network. Using manually coded information on the specific confidential datasets used in each paper, we provide suggestive evidence on the lifecycle of research data sources.

F.1 Development of FSRDC Data Resources

Cognizant of the research potential of its microdata, in 1982, the U.S. Census Bureau established the CES to make these resources accessible to researchers in economics (CES, 2017). The initial focus of CES was the development of research data on the manufacturing sector (McGuckin, 1995). In 1984, CES constructed the Longitudinal Establishment Database (LED) by linking manufacturing records across years. A central contribution of this effort was the creation of Permanent Plant Numbers, which allowed establishments to be linked across survey waves (Atrostic, 2007). CES subsequently expanded these longitudinal records using the Economic Censuses and Annual Survey of Manufactures, eventually creating the Longitudinal Research Database (LRD) (McGuckin et al., 1993).

In the late 1990s, CES began developing a new database, later called LBD (Jarmin and Miranda, 2002). According to Chow et al. (2021), the creation of the LBD was prompted by a desire to test whether results based on the LRD's manufacturing data also applied to other sectors. The LBD was created by merging the Standard Statistical Establishment List with the Economic Census, thus creating a source that contains basic information on the universe of all U.S. business establishments. This data effort was joined by the concurrent development of the LEHD data (Abowd et al., 2004). The LEHD merges worker and employer records from Census Bureau surveys with state unemployment insurance claims to create matched employer-employee panel data. Together, the LBD and LEHD constitute unique research tools to investigate the dynamics of the U.S. economy.

The FSRDC network has also expanded beyond Census Bureau datasets. Other federal statistical agencies, including the National Center for Health Statistics, Bureau of Labor Statistics, and Agency for Healthcare Research and Quality, subsequently began making confidential microdata available through the same infrastructure. FSRDC projects can also contribute to the construction and

[20] Source: https://gov.uk/government/uploads/12-793-investment-intangible-assets-on-productivity-spillovers.pdf

[21] Source: https://apo.org.au/sites/2017-10/apo-nid115951_1.pdf

improvement of research datasets. For example, the recent development of the Management and Organizational Practices Survey (MOPS) has been spearheaded by academic researchers, who have collaborated with the U.S. Census Bureau to design a novel survey instrument (Bloom et al., 2019). As a result, survey and administrative microdata available in FSRDC are increasingly powerful in investigating a broad swath of economic, social, health, and policy questions.

Our interviews similarly emphasized how the development of linked administrative datasets expanded the set of questions economists could study. One respondent described FSRDC data as having "opened up fields of research that weren't possible before" (interview T14). Respondents frequently pointed to work using longitudinal employer and establishment data as an example. In discussing John Haltiwanger's research, one former FSRDC administrator noted that its influence extended across the traditional boundaries of labor and macroeconomics: "I think if you look at Haltiwanger's work, I think his work has transformed, I don't know whether you want to call it labor or macro or, you know, kind of the intersection of labor and macro. I think he's had a big impact" (interview Y79).

F.2 Characteristics of FSRDC Papers

We describe the characteristics of papers that employ confidential Census microdata (see also Nagaraj et al. 2025). Table F1 shows the strength of the statistical association between the use of FSRDC data and several indicators of research quality after controlling for the year of publication. Papers using FSRDC data are more likely to appear in highly ranked journals and receive more academic and policy citations. They are also 36.5 percentage points more likely to receive at least one policy citation. These patterns are consistent with FSRDC data being used in consequential research, although they should not be interpreted as causal effects of data access.

Table F1: Statistical Association Between Proxies of Research Quality and Use of FSRDC Data

	Top Field (0/1)	Five-Year Cites	Policy Relevant (0/1)
	(1)	(2)	(3)
FSRDC Use (0/1)	0.209***	10.57***	0.365***
	(0.02069)	(1.34000)	(0.02199)
Year FE	Yes	Yes	Yes
N	188181	188181	188181
Mean of DV	0.196	12.31	0.377

Note: This table presents estimates from OLS models at the article level evaluating the association between proxies of research quality and the use of FSRDC data. *, **,*** denote significance at 5%, 1% and 0.1% level respectively. Standard errors are clustered by year of publication.

Figure F1 shows the growth in the number of published articles using FSRDC data over time. The growing diffusion of FSRDC data in economics closely mimics the expansion of the FSRDC network exploited in the primary analyses of this paper. However, one possibility is that broader access could expand the number of FSRDC projects while reducing the average quality of the resulting publications. We do not observe such a decline in publication outlets: the share of FSRDC articles appearing in top-field or top-five journals remains relatively stable as the number of FSRDC papers

increases. This pattern is descriptive, but suggests that the expansion in FSRDC-based research was not accompanied by a deterioration in publication outcomes.

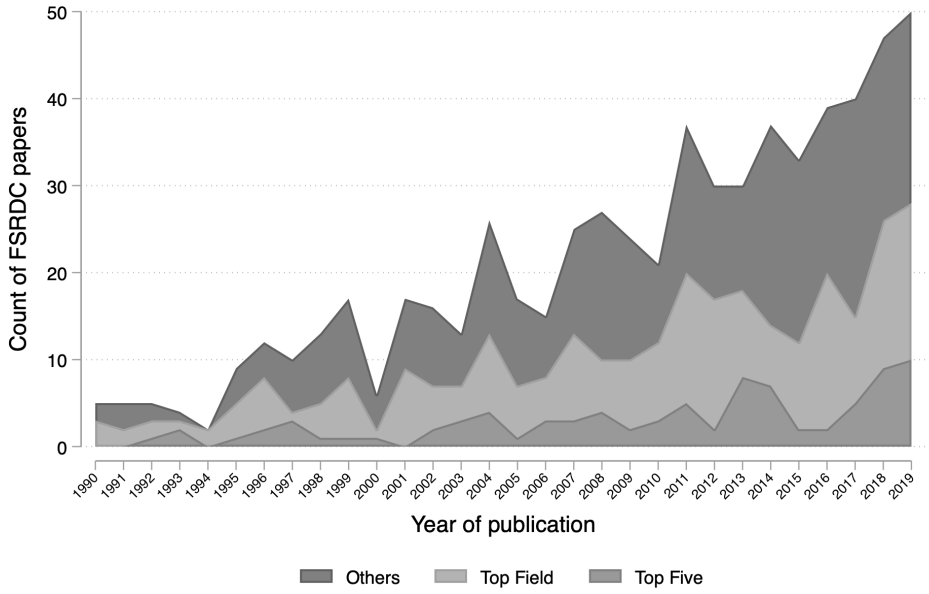
F.3 Dataset-Level Evidence

Our primary analyses exploit the staggered rollout of FSRDCs across the United States to estimate their causal impact on the rate and direction of academic economics research. However, one might wonder how different the effect of an FSRDC that opened in the 1990s could be from that of one that opened in the 2010s. On the one hand, researchers who get access earlier can easily exploit the data to its full potential, unlike those who get access later. This logic would imply diminishing returns to additional FSRDCs because many research lines that could be explored with confidential microdata will be exhausted over time. On the other hand, researchers who gain access later might be able to build off a larger pool of metadata and tacit knowledge about FSRDC datasets. Moreover, one of the objectives of the FSRDC program is to enable the creation of new research data, expanding the menu of available datasets over time. Therefore, it is not clear that the value of FSRDC access should be decreasing over time.

We manually coded the specific confidential datasets used in each FSRDC article.[22] This information allows us to document which data sources account for FSRDC-based research and how their

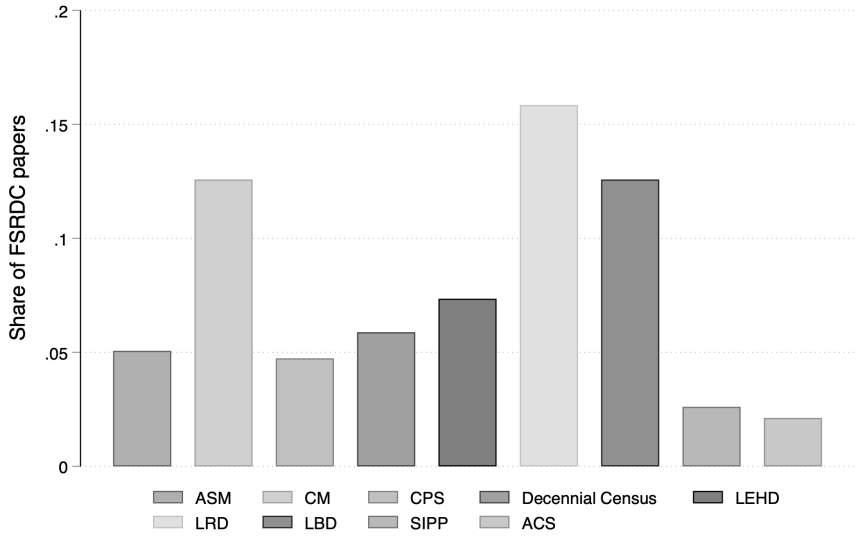
[22]We thank Buyi Geng and Jiamei (Jasmine) Xu for their assistance in coding the confidential datasets used by FSRDC papers.

Figure F1: Count of Yearly FSRDC Articles over Time



Note: The figure reports the yearly count of economics articles using FSRDC data. The area of the plot is colored in different shades of grey to indicate articles published in a top field journal or one of the top five journals.

Figure F2: Confidential FSRDC Datasets Most Frequently Used in Economics Research



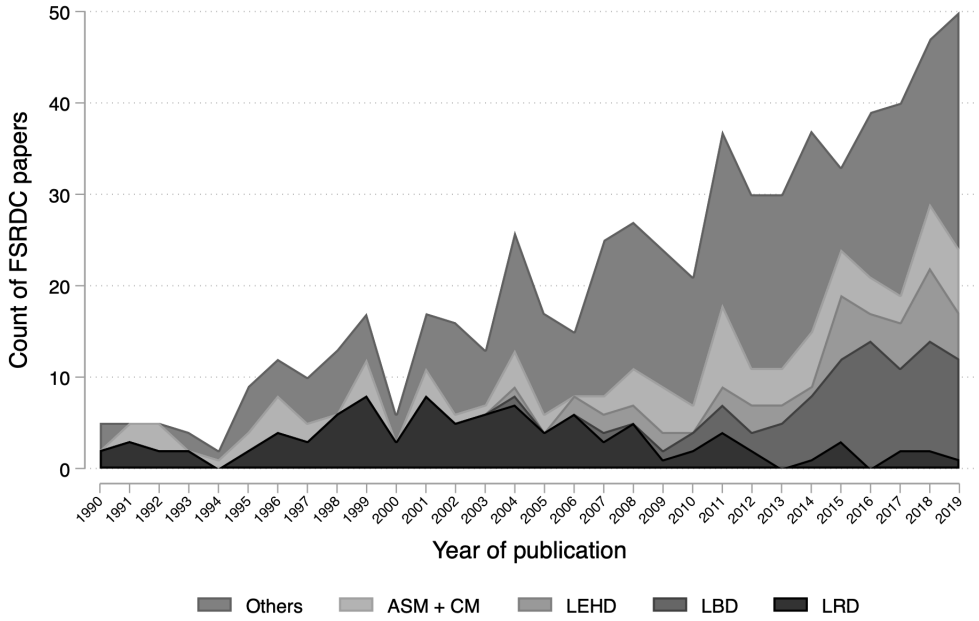
Note: The figure reports the share of articles using FSRDC data that use each of the most common datasets. The datasets shown in the figure include the Annual Survey of Manufactures (ASM), the Census of Manufactures (CM), the Current Population Survey (CPS), the Decennial Census, the Longitudinal Employer-Household Dynamics dataset (LEHD), the Longitudinal Research Database (LRD), Longitudinal Business Database (LBD), the Survey of Income and Program Participation (SIPP), and the American Community Survey (ACS). For this figure, we code the article as using only the LRD or LBD when the CM or ASM data are used in conjunction with the LRD or the LBD, respectively.

use has changed over time. Figure F2 reports the datasets most frequently used in FSRDC articles. The LBD and LRD together account for nearly 30% of the papers in our sample, confirming the continuing importance of datasets on the U.S. manufacturing sector, while datasets such as the Decennial Census and Current Population Survey account for smaller shares.

Figure F3 shows how the use of major FSRDC datasets has changed over time. The LRD was the dominant longitudinal business dataset during the early part of our sample, consistent with Atrostic (2007). Its use subsequently declined as the LBD became increasingly prominent. Use of the LEHD also increased during the later part of the sample, while the Census of Manufactures and Annual Survey of Manufactures are frequently used together with the LRD or LBD. These patterns show that the composition of FSRDC-based research has changed substantially as new data resources have become available.

Finally, we directly assess whether there are decreasing returns to using FSRDC data over time with suggestive evidence on the evolution of research returns to individual datasets. Panel A of Figure F4 plots the probability that an FSRDC paper appears in a top five journal over time. Despite year-to-year variation, there is no clear downward trend. Panel B reports the corresponding relationship for papers using the Panel Study of Income Dynamics (PSID), for which the probability of a top five publication declines over time. Figure F5 suggests one reason why the pattern for FSRDC research may differ. Papers using the LRD exhibit a decline similar to the PSID pattern.

Figure F3: Usage of Specific Confidential Datasets over Time

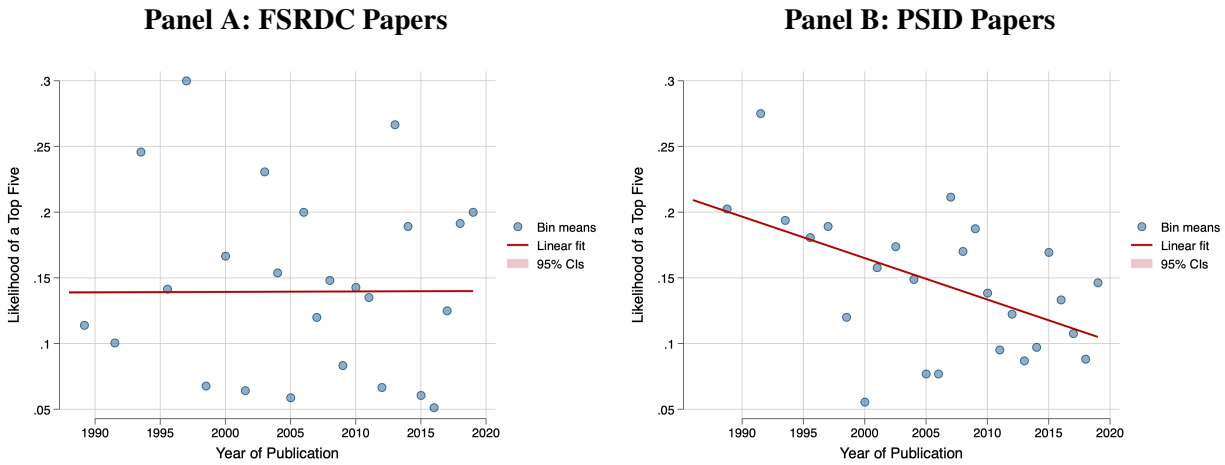


Note: The figure reports the yearly count of economics articles using FSRDC data, divided by the specific confidential dataset used. The figure shows the most common datasets on the manufacturing sector, namely the Longitudinal Research Database (LRD), Longitudinal Business Database (LBD), Longitudinal Employer-Household Dynamics dataset (LEHD), the Annual Survey of Manufactures (ASM) and the Census of Manufactures (CM). For this figure, we code the article as using only the LRD or LBD when the CM or ASM data are used in conjunction with the LRD or the LBD, respectively.

The subsequent introduction and diffusion of the LBD, however, provided researchers with a new longitudinal business dataset. Thus, even if the research value of an individual dataset declines as it matures, the continued addition of new confidential data sources can sustain the value of access to the broader FSRDC infrastructure.

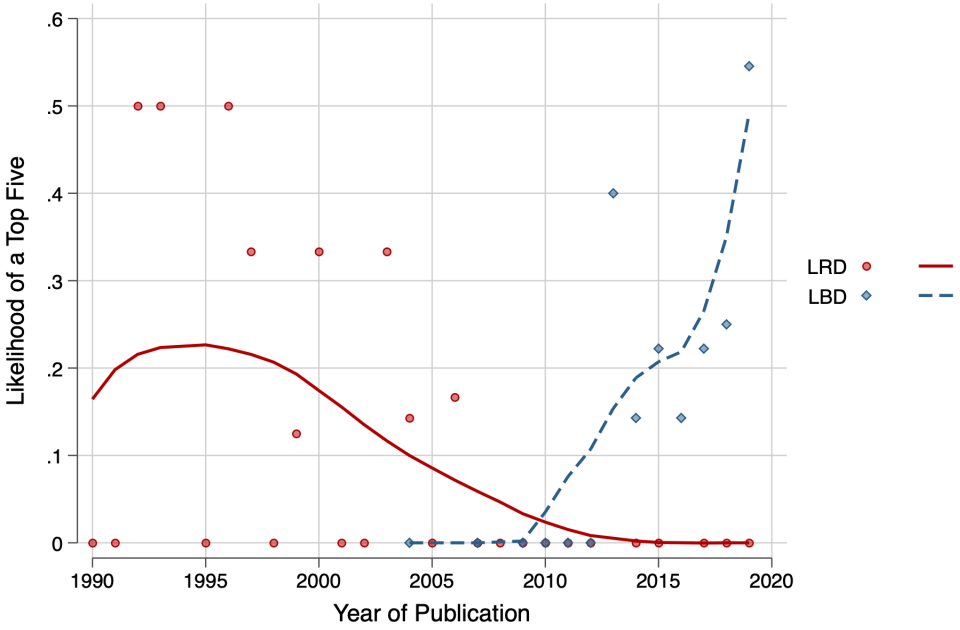
This dataset-level evidence complements the cohort analysis in Appendix Table HH6. The cohort estimates are smaller for later FSRDC openings, although those estimates are difficult to compare because late adopters have substantially shorter post-treatment windows. The evidence in this section highlights a countervailing force: the set of datasets available through the FSRDC network has itself expanded over time.

Figure F4: Likelihood of Being Published on a Top Five Journal, FSRDC Papers vs. PSID Papers



Note: This figure shows a binned scatterplot of the likelihood that an article appears in a top five economics journal over time, separately for papers using confidential FSRDC data (Panel A) and the Panel Study of Income Dynamics (PSID) data (Panel B). See text for more details.

Figure F5: Likelihood of Being Published on a Top Five Journal Conditional on Using FSRDC Data (LRD vs. LBD)



Note: This figure shows a non-parametric binned scatterplot of the likelihood that an article using FSRDC data appears in a top five economics journal over time, separately for papers using the Longitudinal Research Database (LRD) and the Longitudinal Business Database (LBD). See text for more details.

G Cost and Benefit Estimates of FSRDCs

This Appendix presents back-of-the-envelope calculations using our estimated effects to assess the costs and research benefits associated with FSRDC access. We take the perspective of a university or consortium considering whether to open an FSRDC. We first estimate the costs of opening and operating a center, then provide several monetary benchmarks for the value of an additional top publication. We use these inputs to calculate the publication value required for an FSRDC investment to break even and compare the implied research output per dollar with estimates for other forms of research funding. Finally, we use the geographic structure of existing FSRDC consortia to identify potential locations for future expansion.

G.1 Costs of Opening an FSRDC

We evaluate the decision to open a new FSRDC over a twenty-year horizon. We leverage internal estimates from the Census Bureau on the costs for an institution to open an FSRDC location.[23] We use the upper bound estimates, $185,000 annually, which corresponds closely to universities' reported payments.[24] According to Figure G1, the primary cost consists of the salary and overhead of FSRDC administrators, which are Census Bureau employees who oversee the functioning of the FSRDC. The hosting institution also provides the physical space for the data center, with an estimated $150,000 in infrastructure costs to comply with Census and IRS security specifications.[25] Finally, the hosting institution must cover a portion of the ongoing costs of disclosure avoidance reviews, a process that screens research results for sensitive information.

Figure G1: Estimated Costs of Opening and Operating an FSRDC

	Year 1	Year 2	Year 3	Year 4	Year 5	Annual Average Cost
Disclosure services	$ 6,000	$ 12,000	$ 18,000	$ 24,000	$ 24,000	$ 16,800
Infrastructure costs	$ 50,000	$ 50,000	$ 50,000	$ -	$ -	$ 30,000
Administrator Salary & Overhead	$ 129,891	$ 134,220	$ 138,550	$ 142,881	$ 142,881	$ 137,684
Total yearly cost	$ 185,891	$ 196,220	$ 206,550	$ 166,881	$ 166,881	$ 184,484

Note: These costs are obtained from Attachment B in https://www.census.gov/documents/Guidelines_for_RDC _Development_and_Operations_FY2020.pdf. We take the top-range estimate for all administrator-related expenses.

Most operating costs are fixed rather than proportional to actual data use. As a consequence, consortium size is an important determinant of the cost per potential user: adding participating

[23] See Attachment B in https://www.census.gov/documents/Guidelines_for_RDC_Development_and_Operations_FY2020.pdf.

[24] The Executive Director of the Boston FSRDC reported a figure of approximately $187,500 yearly costs. See: https://deepblue.lib.umich.edu/ACDEB_response_FSRDCDirectorsFinal2021.pdf.

[25] This amount, spread over three years, does not include the in-kind contribution of the physical space by the host institution. However, the Bureau will supply much of the technology and equipment for the facility, including the router to connect to the FSRDC Virtual Private Network.

institutions expands the potential user base without a proportional increase in the costs of operating the center.[26] Because FSRDCs also serve researchers outside economics, we allocate costs to economics in proportion to project use. We manually classify 771 active and completed FSRDC projects by discipline, of which 597 are related to economics, and therefore attribute approximately 77% of FSRDC costs to economics. We make two additional assumptions when modeling costs. First, we assume that the cost reported for Year 5 will be repeated in each subsequent year. Second, we assume the hosting consortium avails itself of the maximum grant support from the NSF to establish the FSRDC, amounting to a total of $300,000 to help cover start-up costs.[27]

G.2 Monetary Benchmarks for the Value of a Publication

Assigning a monetary value to an additional scientific publication necessarily requires strong assumptions. We consider three approaches that provide benchmarks for the magnitude of this value, rather than selecting a single estimate. The first uses the relationship between publications and tenure outcomes to infer the value universities place on an additional publication. The second uses estimated salary returns to publications to approximate their private value to researchers. The third uses an existing estimate of the social value of scientific publications from another research setting.

Method 1: Revealed Value to the University. Our first method leverages the fact that an additional publication will likely increase the probability that an assistant professor is awarded tenure during their first employment spell. The basic idea is to use the salary increase from the promotion to associate professor to estimate the revealed monetary value of an additional top publication for the university. For this method, we rely on the estimate from Sarsons (2015) that an additional publication is associated with a 5.7% increase in the likelihood that a professor's first tenure spell is successful. We then leverage the transparency of the University of California system, which publicly releases salary information for all its employees, allowing us to pinpoint the salary premium associated with obtaining tenure at a higher-status campus relative to the likely counterfactual of having to move to a lower-ranked institution.

Specifically, each academic salary at the University of California consists of a scale component (UC-wide) and an off-scale component (employee-specific). The salary scale is based on the professor's rank (i.e., assistant professor, associate professor, or full professor) and the number of years in the position.[28] In contrast, the off-scale component varies systematically across UC campuses: for instance, internal documents suggest the off-scale amounts paid to UC Berkeley faculty leads to 18% higher salaries on average than UC Irvine faculty.[29] We construct two hypothetical earnings timelines, each with a length of 35 years (the typical career of an economist), inspired by the data in Heckman and Moktan (2020).

[26] This consideration is net of eventual congestion costs that will become binding when a facility is used at full capacity. However, this issue never came up in any of our interviews, thus suggesting that FSRDCs are not currently used at their full capacity.

[27] See for instance: https://www.nsf.gov/awardsearch/showAward?AWD_ID=1955355&HistoricalAwards=false.

[28] Economics faculty follow the salary scale for ladder-rank faculty in Business, Economics, and Engineering. See: https://www.ucop.edu/academic-personnel-programs/oct-2023-acad-salary-scales.pdf.

[29] See: https://academicsenate.ucdavis.edu/sites/archive/08_ucd_uc_salaries.pdf.

In the first timeline, an assistant professor is promoted to associate professor after a 7-year tenure, then to full professor after another 7-year tenure. In this base case, we derive the professor's annual salaries from the salary scale. In the second timeline, the assistant professor is denied tenure after the first 7-year term and moves to a lower-tier institution to advance immediately to a tenured position. After a successful second 7-year spell, they are promoted to full professor. To adjust for the downward mobility in this timeline, we apply the off-scale wage differential between UC Berkeley and UC Irvine as a conservative earnings penalty. We calculate the present value of the annual salary difference between the two timelines to be $472,205 using a 5% discount rate. Note that this estimate does not include the option value of more consulting opportunities or book deals. In conjunction with the estimate from Sarsons (2015), this implies the present value of a publication to be $26,916. If we allow for a 2-year lag before the researcher is promoted to associate professor in the second timeline, the value of a paper increases to $32,043.

Method 2: Private Value for the Researcher. Our second method is more direct and just evaluates the researchers' salary raise associated with the marginal top publication. However, this approach relies entirely on the validity of the estimated relationship between publication output and annual salary. For this exercise, we use the estimate from Sauer (1988), which finds that one additional publication in a top-20 journal increases a professor's annual salary by at least $851 in 1982 dollars, equivalent to $2,255 in 2019 dollars. This corresponds closely to our measure of top publications, which considers any publication in top journals. To convert this measure into the lifetime value of a publication, we once more assume a 35-year career length and a 5% discount rate. Since the average economist in our sample is 13 years into their career, this implies the present value of a top publication is $31,167.

We note that the estimate from Sauer (1988) is consistent with more recent estimates from other related fields. For instance, Swidler and Goldreyer (1998) finds that a publication in a top finance journal increases a finance professor's salary by $1,952 in 1993 dollars, which is equivalent to $3,453 in 2019 dollars. Mittal et al. (2008) estimate that a Tier 1 marketing journal publication increases a marketing professor's salary by $2,176 in 2001 dollars, which is equivalent to $3,143 in 2019 dollars. Finally, Li et al. (2023) estimate that a top US management journal publication increases a strategy professor's salary by between $2,200 and $3,800. Taken together, this shows that our preferred estimate is quite conservative.

Method 3: Using Estimates of Social Value from Other Disciplines. The third and final method is adopting an existing estimate of social value from other disciplines. Unfortunately, there are few estimates currently available because estimating social value requires strong assumptions. One recent exception is Morretta et al. (2022), who estimate the marginal social value of publications that use imagery from the Italian Space Agency's Earth Observation (EO) satellite constellation. The study estimates the marginal cost of producing an EO publication by dividing annual researcher salaries (adjusted for the share of time dedicated to research) by annual productivity, then adding the publishing costs borne by publishers. The study then factors in the social value of a citation, which is calculated by augmenting the marginal cost of a publication by the share of time dedicated to reading research. Using this method, the study estimates that 1,235 publications from 1998-2018 produced $36,837,104 in societal value, or $29,828 per publication.

Table G1: Alternative Monetary Benchmarks for a Top Publication

Method	Preferred Estimate
1. Present value of the increased probability of tenure	$26,916
2. Present value of the salary increases	$31,167
3. Benchmark from another social science field (EO)	$29,828

G.3 Cost-Benefit Analysis of Opening an FSRDC Location

While our three methods produce comparable estimates of the dollar value of publication (Table G1), we are hesitant to select any one of them. Instead, we calculate the value that an additional top publication would need to have for the discounted research benefits generated by an FSRDC to equal its discounted costs. This approach allows the break-even threshold to be evaluated under different assumptions about the value of scientific output.

$$\sum_{t=1}^{20} \frac{Price \times Number\ Publications_t}{1.05^{t-1}} = \sum_{t=1}^{20} \frac{0.77[(Total\ Costs_t) - NSF\ Grant_t]}{1.05^{t-1}} \tag{2}$$

Because universities typically apply for an FSRDC as a consortium, we evaluate the decision to open an FSRDC at the consortium level, assuming that costs are largely fixed. We calculate the number of additional publications due to the FSRDC as follows:

$$Number\ Publications_t = 1.035 + (24 \cdot Consortium\ Effect) \cdot Number\ Partners \tag{3}$$

The first term captures additional top publications at the host institution. We calculate it by multiplying the median number of AEA-affiliated economists at FSRDC host institutions in our sample (45) by the estimated treatment effect for host institutions (0.023). The second term captures additional publications at consortium partners. We multiply the median number of economists at partner institutions (24) by the assumed treatment effect for consortium partners and by the number of partner institutions. We then solve for the break-even value of an additional top publication under different combinations of consortium size and partner effects. We discount all costs and benefits using a 5% discount rate.

The results of this analysis are presented in Figure G2. To interpret this table, consider the cell (0.015, 4). This cell can be read as implying that the dollar value of a top publication would have to be at least $47,094 for a consortium to justify opening an FSRDC location, assuming the consortium consists of one hosting institution and four partners and the treatment effect for the partners is 65% as large as the effect for hosting institutions. This exceeds the upper limit of our estimated dollar publication value, though it may still fall within an acceptable range for some institutions. If the same consortium could increase FSRDC adoption among non-hosting partners, the dollar value of a paper would fall within the range of our estimated dollar value of a publication. In practice, the extent of the effect for any consortium member will depend on its geographical distance from the hosting institution (Appendix Figure H5). Alternatively, if the same consortium could include four additional members, then the dollar value of a paper would only have to be $29,772. This is within the range of our estimated dollar value of a publication.

Figure G2: Required Dollar Value of a Publication for Break-even of FSRDC Funding

Consortium effect	Number of consortium partners: 1	2	3	4	5	6	7	8	9
0.005	$100,916	$91,418	$83,554	$76,936	$71,289	$66,415	$62,164	$58,425	$55,110
0.01	$91,418	$76,936	$66,415	$58,425	$52,151	$47,094	$42,931	$39,444	$36,481
0.015	$83,554	$66,415	$55,110	$47,094	$41,114	$36,481	$32,787	$29,772	$27,265
0.02	$76,936	$58,425	$47,094	$39,444	$33,932	$29,772	$26,521	$23,909	$21,766
0.025	$71,289	$52,151	$41,114	$33,932	$28,887	$25,147	$22,265	$19,976	$18,113

Note: These costs are obtained from Equation 2 at increasing values of the treatment effect size for consortium partners (from 0.005 to 0.025) and the number of consortium partners (from 1 to 9). Each cell reports how much a publication would need to be valued to justify opening an FSRDC with that effect size and number of consortium members. See text for details.

While our analysis rests on various assumptions, there are several takeaways from this exercise. First, while increasing access to confidential microdata can result in higher research productivity, the decision to open an FSRDC is not unconditional for university administrators. If the consortium cannot guarantee sufficient engagement with the FSRDC, then the required dollar value might be prohibitively large to justify the FSRDC. On the upside, the analysis in Figure G2 would look much more favorable if universities managed to increase their user base, which we assumed to be fixed. When considering institutions with a larger number of applied scholars and treatment effects reflecting the dynamically increasing impact of FSRDCs documented in Figure 3, even individual universities could justify opening an FSRDC.[30] Second, this exercise highlights multiple potential strategies to maximize the value of an FSRDC. Either the consortium can prioritize efficiency, increasing the number of papers that employ FSRDC data from its current roster of faculty (perhaps by removing barriers to adoption, see Section A), or the consortium can prioritize scale, increasing the potential user base while holding engagement constant.

G.4 Cost-Effectiveness of Opening an FSRDC Location

We next compare the research output associated with FSRDC expenditures with estimates of the effects of other forms of research funding. We draw on studies that estimate the relationship between research funding and publication output and normalize their estimates to $100,000 of funding. Table G2 summarizes these benchmarks.

Table G2: Estimates of Papers per Dollar of Funding from Other Academic Studies

Academic Study	Sattari et al. (2022)	Wahls (2018)	Payne and Siow (2003)	Whalley and Hicks (2014)	Rosenbloom et al. (2015)	Arora and Gambardella (2005)
Field	Biomedicine	Biomedicine	All	All	Chemistry	Economics
Papers per 100k dollars of funding	0.8	0.53-0.87	1	0.83	0.6-1.9	0.48

The first study is Sattari et al. (2022), which examines the extent to which biomedical research was

[30] Note also that our count of potential users does not include doctoral students, whose usual lack of publications during the PhD program prevents from being included in our sample. FSRDCs have an important educational value and enable several PhD dissertations every year. In particular, access during graduate training may shape subsequent research topics, data choices, and collaborations. To the extent these effects are positive, our calculations understate the total research benefits of FSRDC access.

stimulated by National Institutes of Health (NIH) science funding across 72 university campuses in the United States. Using an event-study design, the study finds that an additional $100,000 in funding produces 0.8 additional papers. A related study by Wahls (2018) finds that each $1 million in NIH funding to prestigious institutions produced 5.3 papers, whereas $1 million in NIH funding to less prestigious institutions produced 8.7 papers. A third study by Payne and Siow (2003) examines how federal research funding affects the total quantity of research produced at a university. Using alumni representation on the congressional appropriations committees to instrument for the share of federal R&D funding, the study finds that an additional $1 million in funding results in 10 more articles. Similarly, the fourth study from Whalley and Hicks (2014) estimates how university spending affects total research output. Since universities are constrained in their spending by the market value of their endowments, the study uses stock market shocks to instrument spending. The authors find that an additional $1.1 million in expenditures produces about nine additional papers. In the context of chemical research, Rosenbloom et al. (2015) find that every additional million in Federal R&D funding produces 6-7 more papers, while their IV estimates (which use Federal R&D funding for mathematics and physics as an instrument) imply a larger increase of 19 articles.

Finally, in the setting closest to our own, Arora and Gambardella (2005) look at NSF grants' impact on US-based economists' research output.[31] However, we must first perform some imputations to facilitate a direct comparison. Firstly, their measure of research output is not publications but rather quality-adjusted publication units, with one publication in the highest-ranking journal corresponding to 100 publication units and any publication not among the top 50 journals corresponding to 1 publication unit. We consider 14.63 units equivalent to one top publication.[32] Secondly, the study only reports separate elasticities by academic tenure, so we weight them by the shares of NSF grants received by researchers of different seniority in their sample to reconstruct the aggregate elasticity. With this method, we calculate that adding $100,000 to an NSF grant would result in 7 additional quality-adjusted "publication units" per year, equivalent to half a top publication.

Figure G3: Annual Top Publications per $100,000 of Funding in FSRDCs

Consortium effect \ Number of consortium partners	1	2	3	4	5	6	7	8	9
0.005	0.90	0.99	1.09	1.18	1.27	1.37	1.46	1.55	1.65
0.01	0.99	1.18	1.37	1.55	1.74	1.93	2.11	2.30	2.49
0.015	1.09	1.37	1.65	1.93	2.21	2.49	2.77	3.05	3.33
0.02	1.18	1.55	1.93	2.30	2.67	3.05	3.42	3.79	4.17
0.025	1.27	1.74	2.21	2.67	3.14	3.61	4.07	4.54	5.01

Note: These costs are obtained from Equation 4 at increasing values of the treatment effect size for consortium partners (from 0.005 to 0.025) and the number of consortium partners (from 1 to 9). Each cell reports the additional number of publications per $100,000 of funding in an FSRDC with that effect size and number of consortium members. See text for details.

Then, we calculate the number of additional publications per $100,000 of funding invested in

[31] This is a particularly salient comparison, given NSF's role in seeding the establishment of FSRDCs.

[32] To recall, we defined as a top publication the highest-ranked general purpose and top field journals. We consider a top publication to be worth 14.6, based on this being the minimum number of units for any of the top 20 journals in Arora and Gambardella (2005).

opening and operating an FSRDC. The formula for the number of top publications is identical to before, which means we keep $Consortium\ Effect$ and $Number\ Partners$ free and estimate one elasticity per cell. This time, however, we divide the number of publications by the steady state cost of operating an FSRDC (i.e., the cost from the fifth year onwards). This is the maximum yearly cost in our schedule, since NSF grant support will already have subsided. We then normalize the figure to \$100,000 of funding. Hence, the cost-effectiveness of an FSRDC is calculated as follows:

$$Papers\ per\ \$100,000 = [1.035 + (24 \cdot Consortium\ Effect) \cdot Number\ Partners] \cdot \frac{100,000}{166,881 * 0.77} \quad (4)$$

Figure G3 compares these benchmarks with the implied number of additional top publications per \$100,000 invested in an FSRDC. Across much of the parameter space, the implied publication effect of FSRDC funding is at least as large as the estimates reported in the studies summarized in Table G2. The closest comparison is Arora and Gambardella (2005), who studied NSF grants to economists. Our conversion of their estimates implies approximately 0.48 additional top publications per \$100,000 of funding. The corresponding FSRDC estimate varies with consortium size and researcher response but exceeds this benchmark under the central parameter values considered in Figure G3. These comparisons should be interpreted with caution because the underlying studies differ in their empirical settings and outcome measures. Nonetheless, they indicate that the estimated research output associated with FSRDC expenditures is comparable to, and under many parameter values larger than, existing estimates for direct research funding.

G.5 Potential New FSRDC Consortia

The preceding exercises evaluate the costs and research benefits of FSRDC access for a hypothetical consortium. We next use the geographic structure of the existing FSRDC network to identify locations where an additional consortium could potentially serve a substantial number of empirical economists. This exercise is intended to illustrate how our estimates can inform future expansion rather than to identify an optimal set of new FSRDC locations.

We use existing FSRDC consortia to define the geographic area that a new center could plausibly serve. We first exclude universities without FSRDC access that are located within the median distance between existing FSRDC hosts and their consortium partners. These institutions could plausibly have joined an existing consortium but did not, suggesting limited revealed demand for access. Among the remaining universities, we rank potential hosts by the number of empirical economists at neighboring universities that could join as consortium partners. To avoid double-counting, each university is assigned to only one prospective consortium. Table G3 reports the ten prospective consortia with the largest number of empirical economists. Applying the treatment effect from our main specification, the largest prospective consortium would generate approximately 0.9 additional top publications per year; applying the early cohort (1994–2001) estimate ($\hat{\beta} = 0.055$, from Appendix Table H6), which captures the steady state effect after roughly 20 years, yields approximately 2 additional top publications per year.

Several locations identified in this exercise subsequently obtained FSRDC access after our sample ended. The Wasatch Front FSRDC opened at the University of Utah in 2020 with Brigham Young University and Utah State University as consortium members. The University of Florida opened an FSRDC in 2022 with the University of Central Florida as a consortium member, and Iowa

State University opened an FSRDC in 2025. The subsequent expansion of the network to several locations identified by our procedure provides additional validation of the geographic exercise. Taken together, these exercises suggest that the research benefits associated with FSRDC access can be large relative to operating costs, particularly when centers serve multiple institutions and generate substantial researcher use. At the same time, these calculations are necessarily approximate and do not incorporate all benefits or costs associated with confidential data access.

Table G3: Potential New FSRDC Consortia

				Expected Top Publications per Year	
Rank	Host University	N Universities	N Empiricists	Main Specification	Early Cohort
1	Dartmouth College	7	39	0.905	2.145
2	Michigan State University	4	24	0.557	1.320
3	Miami University	8	24	0.557	1.320
4	University of Florida	2	16	0.371	0.880
5	University of Tennessee at Chattanooga	4	16	0.371	0.880
6	Brigham Young University	2	13	0.302	0.715
7	Iowa State University	2	13	0.302	0.715
8	Kansas State University	3	12	0.278	0.660
9	University of Alabama	3	12	0.278	0.660
10	University of Houston	2	11	0.255	0.605

Note: This table reports the ten potential FSRDC consortia with the largest number of empiricist researchers. We exclude universities without FSRDC access that are located within the median distance between existing FSRDC hosts and their consortium partners, since these institutions could plausibly have joined an existing consortium. For each remaining potential host, we define a service radius of 112 miles, corresponding to the 75th percentile of the observed distance between existing hosts and consortium partners, and assign universities within this radius as potential consortium members. Each university is assigned to at most one consortium. The final two columns report the expected increase in top publications per year, calculated as $\hat{\beta} \times N_{\text{empiricists}}$, using the treatment effect from our main specification ($\hat{\beta} = 0.023$) and the estimate for the early FSRDC cohort, 1994–2001 ($\hat{\beta} = 0.055$), respectively.

Appendix References

Abowd, J. J., J. Haltiwanger, and J. Lane (2004): "Integrated longitudinal employer-employee data for the United States," *American Economic Review*, 94, 224–229.

Abowd, J. M. and I. M. Schmutte (2019): "An economic analysis of privacy protection and statistical accuracy as social choices," *American Economic Review*, 109, 171–202.

Angrist, J., P. Azoulay, G. Ellison, R. Hill, and S. F. Lu (2020): "Inside job or deep impact? Extramural citations and the influence of economic scholarship," *Journal of Economic Literature*, 58, 3–52.

Arora, A. and A. Gambardella (2005): "The impact of NSF support for basic research in economics," *Annales d'Economie et de Statistique*, 91–117.

Atrostic, B. (2007): "The Center for Economic Studies 1982-2007: A brief history," CES working paper.

Azoulay, P., J. S. Graff Zivin, D. Li, and B. N. Sampat (2019): "Public R&D investments and private-sector patenting: Evidence from NIH funding rules," *The Review of Economic Studies*, 86, 117–152.

Baumann, A. and K. Wohlrabe (2020): "Where have all the working papers gone? Evidence from four major economics working paper series," *Scientometrics*, 124, 2433–2441.

Bloom, N., E. Brynjolfsson, L. Foster, R. Jarmin, M. Patnaik, I. Saporta-Eksten, and J. Van Reenen (2019): "What drives differences in management practices?" *American Economic Review*, 109, 1648–1683.

Bowen, C. M. (2024): "Government data of the people, by the people, for the people: Navigating citizen privacy concerns," *Journal of Economic Perspectives*, 38, 181–200.

Callaway, B. and P. H. Sant'Anna (2021): "Difference-in-differences with multiple time periods," *Journal of Econometrics*, 225, 200–230.

Card, D., R. Chetty, M. S. Feldstein, and E. Saez (2010): "Expanding access to administrative data for research in the United States," in *Ten Years and Beyond: Economists Answer NSF's Call for Long-Term Research Agendas*, American Economic Association.

Card, D., S. DellaVigna, P. Funk, and N. Iriberri (2022): "Gender differences in peer recognition by economists," *Econometrica*, 90, 1937–1971.

CES (2017): "Center for Economic Studies and Research Data Centers Research Report: 2016," Available on the U.S. Census Bureau's website.

Chow, M. C., T. C. Fort, C. Goetz, N. Goldschlag, J. Lawrence, E. R. Perlman, M. Stinson, and T. K. White (2021): "Redesigning the Longitudinal Business Database," NBER Working Paper w28839.

Currie, J., H. Kleven, and E. Zwiers (2020): "Technology and big data are changing economics: Mining text to track methods," *AEA Papers and Proceedings*, 110, 42–48.

Einav, L. and J. Levin (2014): "Economics in the age of big data," *Science*, 346, 1243089.

Foster, L., R. Jarmin, and L. Riggs (2009): "Resolving the tension between access and confidentiality: Past experience and future plans at the US Census Bureau," *Statistical Journal of the IAOS*, 26, 113–122.

Ghanem, D., P. H. Sant'Anna, and K. Wüthrich (2025): "Selection and parallel trends," *arXiv preprint arXiv:2203.09001*.

Goroff, D., J. Polonetsky, and O. Tene (2018): "Privacy protective research: Facilitating ethically responsible access to administrative data," *The ANNALS of the American Academy of Political and Social Science*, 675, 46–66.

Greenstone, M., R. Hornbeck, and E. Moretti (2010): "Identifying agglomeration spillovers: Evidence from winners and losers of large plant openings," *Journal of Political Economy*, 118, 536–598.

Heckman, J. J. and S. Moktan (2020): "Publishing and promotion in economics: The tyranny of the top five," *Journal of Economic Literature*, 58, 419–470.

Hsieh, C.-T. and P. J. Klenow (2009): "Misallocation and manufacturing TFP in China and India," *The Quarterly Journal of Economics*, 124, 1403–1448.

Jarmin, R. S. and J. Miranda (2002): "The longitudinal business database," CES working paper.

Kalaitzidakis, P., T. P. Mamuneas, and T. Stengos (2003): "Rankings of academic journals and institutions in economics," *Journal of the European Economic Association*, 1, 1346–1366.

Lane, J. (2007): "Optimizing the use of microdata: An overview of the issues," *Journal of Official Statistics*, 23, 299–317.

——— (2021): *Democratizing Our Data: A Manifesto*, MIT Press.

Li, C., J. Ahn, J. Bu, and K. E. Meyer (2023): "The value of publishing in JIBS," *Journal of International Business Studies*, 54, 1688–1699.

Lusher, L. R., W. Yang, and S. E. Carrell (2021): "Congestion on the information superhighway: Does economics have a working papers problem?" NBER Working Paper w29153.

McGuckin, R. H. (1995): "Establishment microdata for economic research and policy analysis: Looking beyond the aggregates," *Journal of Business & Economic Statistics*, 13, 121–126.

McGuckin, R. H., R. H. McGukin, and A. P. Reznek (1993): "The statistics corner: Research with economic microdata: The Census Bureau's Center for Economic Studies," *Business Economics*, 52–58.

Mittal, V., L. Feick, and F. Murshed (2008): "Publish and prosper: The financial impact of publishing by marketing faculty," *Marketing Science*, 27, 430–442.

Moed, H. F., M. Aisati, and A. Plume (2013): "Studying scientific migration in Scopus," *Scientometrics*, 94, 929–942.

Moretti, E. (2021): "The effect of high-tech clusters on the productivity of top inventors," *American Economic Review*, 111, 3328–3375.

Morretta, V., D. Vurchio, and S. Carrazza (2022): "The socio-economic value of scientific publications: The case of Earth Observation satellites," *Technological Forecasting and Social Change*, 180, 121730.

Nagaraj, A., F. Stipanicic, and M. Tranchero (2025): "Restricted-Access Data in Economics: Adoption, Diffusion, and Impact of US Census Bureau's Microdata," *Harvard Data Science Review*.

National Science Foundation (2015): "Restricted-Access Research Data Centers (RDCs)," Program Solicitation NSF 15-586, National Science Foundation.

Önder, A. S. and S. Schweitzer (2017): "Catching up or falling behind? Promising changes and persistent patterns across cohorts of economics PhDs in German-speaking countries from 1991 to 2008," *Scientometrics*, 110, 1297–1331.

Payne, A. and A. Siow (2003): "Does federal research funding increase university research output?" *Advances in Economic Analysis Policy*, 3, 1018.

Rambachan, A. and J. Roth (2023): "A more credible approach to parallel trends," *Review of Economic Studies*, 90, 2555–2591.

Rosenbloom, J. L., D. K. Ginther, T. Juhl, , and J. A. Heppert (2015): "The effects of research development funding on scientific productivity: Academic chemistry, 1990-2009," *PloS one*, 10, e0138176.

Sarsons, H. (2015): "Gender Differences in Recognition for Group Work," *Working Paper*.

Sarsons, H., K. Gërxhani, E. Reuben, and A. Schram (2021): "Gender differences in recognition for group work," *Journal of Political Economy*, 129, 101–147.

Sattari, R., J. Bae, E. Berkes, and B. A. Weinberg (2022): "The ripple effects of funding on researchers and output," *Science Advances*, 8.

Sauer, R. D. (1988): "Estimates of the returns to quality and coauthorship in economic academia," *Journal of Political Economy*, 96.

Swidler, S. and E. Goldreyer (1998): "The value of a finance journal publication," *The Journal of Finance*, 53, 351–363.

U.S. Bureau of the Census (1998): "Establishing New Research Data Centers (RDCs)," *Federal Register*, 63, 3309–3310.

U.S. Census Bureau (2019): "Guidelines for the Development and Operation of a Federal Statistical Research Data Center (RDC) or RDC Branch," Tech. rep., U.S. Census Bureau, october 16, 2019.

Wahls, W. P. (2018): "High cost of bias: Diminishing marginal returns on NIH grant funding to institutions," *BioRxiv*, 367847.

Whalley, A. and J. Hicks (2014): "Spending wisely? How resources affect knowledge production in universities." *Economic Inquiry*, 52, 35–55.

Yin, Y., Y. Dong, K. Wang, D. Wang, and B. F. Jones (2022): "Public use and public funding of science," *Nature Human Behaviour*, 6, 1344–1350.

H Appendix Tables and Figures

Table H1: Effect of FSRDC Access on Projects Approved and Working Papers

	FSRDC Project Approval (0/1)		CES Working Paper (0/1)	
	(1)	(2)	(3)	(4)
Post-FSRDC	0.00647***	0.00639***	0.00342**	0.00237*
	(0.00136)	(0.00134)	(0.00129)	(0.00113)
Researcher FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
University FE	Yes	No	Yes	No
University Tier × Year FE	No	Yes	No	Yes
N	155615	155622	155615	155622

Note: This table presents estimates from linear probability models evaluating the impact of FSRDC access on additional measures of confidential data adoption. Columns (1) and (2) report the results from linear probability models, where the dependent variable is an indicator of whether the researcher has any new project approved to be carried out in an FSRDC. Columns (3) and (4) report the results from linear probability models, where the dependent variable is an indicator of whether the researcher has any CES working papers published. Post-FSRDC equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. All models include individual fixed effects. Columns (1) and (3) include year fixed effects and university dummies, while columns (2) and (4) include year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively. See details in Appendix C.5.

Table H2: Alternative Specifications for FSRDC Use and Diffusion

Panel A: Extensive Margin Estimates

	Any FSRDC Paper		Any Citing FSRDC	
	(1)	(2)	(3)	(4)
Post-FSRDC	0.00486***	0.00487***	0.0136***	0.00815*
	(0.00132)	(0.00138)	(0.00323)	(0.00320)
Researcher FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
University FE	Yes	No	Yes	No
University Tier × Year FE	No	Yes	No	Yes
N	155,615	155,622	155,615	155,622

Panel B: PPML Estimates

	FSRDC Use		FSRDC Cites	
	(1)	(2)	(3)	(4)
Post-FSRDC	0.788***	0.712***	0.218**	0.134*
	(0.23469)	(0.18430)	(0.07935)	(0.06145)
Researcher FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
University FE	Yes	No	Yes	No
University Tier × Year FE	No	Yes	No	Yes
N	5,604	4,944	44,939	44,386

Note: Panel A reports linear probability models for the extensive margin of FSRDC use and diffusion. The dependent variables indicate whether researcher j publishes at least one paper using confidential FSRDC data or at least one paper citing FSRDC-based research in year t, respectively. Panel B reports Poisson pseudo-maximum likelihood (PPML) estimates using the corresponding publication counts. The sample size is smaller because the model excludes researcher groups with no identifying variation in outcomes. *Post-FSRDC* equals one in all years after a researcher has been affiliated with a university with an operating FSRDC. All models include researcher fixed effects. In each panel, odd-numbered columns include year and university fixed effects, while even-numbered columns include university-tier × year fixed effects. Standard errors are clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H3: Effect of FSRDC Access on Policy Impact By Country of the Policy Source

	Policy-Relevant Publications (1)	U.S. Policy-Relevant Publications (2)	Non-U.S. Policy-Relevant Publications (3)
Post-FSRDC	0.0362**	0.0242**	0.0304**
	(0.01139)	(0.00834)	(0.01089)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
N	155622	155622	155622
Mean of DV	0.252	0.164	0.224

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on the relevance of economic research to US and Non-US policymaking. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles with at least one cite in a policy document. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles with at least one cite in a US-based policy document. Column (3) reports the result from an OLS model, where the dependent variable is the count of articles with at least one cite in a non-US-based policy document. Post-FSRDC equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively

Table H4: Effect of FSRDC Access on the Policy Orientation of Empirical Researchers

	Mentions of Policy (1)	Policy JELs (all) (2)	Policy JELs (subset) (3)
Post-FSRDC	-0.00125	0.000421	0.00141
	(0.00650)	(0.00176)	(0.00149)
Researcher FE	Yes	Yes	Yes
University Tier $\times$ Year FE	Yes	Yes	Yes
N	155622	155622	155622

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of policy orientation of empirical researchers. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that mention the words "policy" or "policies" in the title or abstract. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that report JEL codes including the word "policy". Column (3) reports the result from an OLS model, where the dependent variable is the count of articles that report JEL codes most associated with government, labor, and public policies. Post-FSRDC equals one in all years after a researcher has been affiliated with a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively

Table H5: Effect of FSRDC Access on Research Output Using the Callaway-Sant'Anna Doubly-Robust DID Estimator.

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
ATT	0.00474*	0.0236***	0.0195	1.743*	0.0454*
	(0.002)	(0.006)	(0.016)	(0.854)	(0.021)
Observations	150,567	150,567	150,567	150,567	150,567

Note: This table presents ATT estimates from models evaluating the impact of FSRDC access on measures of confidential data adoption and research output using the doubly-robust difference-in-difference estimator developed by Callaway and Sant'Anna (2021). Column (1) reports the result from a model where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from a model where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from a model where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan, 2020). Column (4) reports the result from a model where the dependent variable is the count of articles weighted by the number of citations received up to five years following their publication. Column (5) reports the result from a model where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC (ATT) equals one in all years after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects, year fixed effects, and university fixed effects. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H6: Heterogeneity in FSRDC Effects by Adoption Cohort

Panel A: Top Publications

	(1) Early (1994–2001)	(2) Middle (2002–2010)	(3) Late (2011–2019)
Post-FSRDC	0.0547**	0.0251*	-0.00937
	(0.01903)	(0.01091)	(0.01250)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
Mean of DV	0.114	0.117	0.117
N	127,884	136,027	137,407

Panel B: Citation-Weighted Publications

	(1) Early (1994–2001)	(2) Middle (2002–2010)	(3) Late (2011–2019)
Post-FSRDC	2.850*	1.612*	0.108
	(1.13148)	(0.80598)	(0.50983)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
Mean of DV	4.348	4.551	4.398
N	127,884	136,027	137,407

Panel C: Policy-Relevant Publications

	(1) Early (1994–2001)	(2) Middle (2002–2010)	(3) Late (2011–2019)
Post-FSRDC	0.0578*	0.0558**	0.00581
	(0.02695)	(0.02010)	(0.01454)
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
Mean of DV	0.234	0.239	0.239
N	127,884	136,027	137,407

Note: This table presents estimates from separate OLS models evaluating the impact of FSRDC access on research output by FSRDC adoption cohort. Panel A reports the results where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Panel B reports the results where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Panel C reports the results where the dependent variable is the count of articles cited in at least one policy document. Columns (1), (2), and (3) respectively restrict the treated group to researchers at universities that gained FSRDC access in 1994–2001, 2002–2010, and 2011–2019; in each column, the comparison group consists of researchers at universities that never gain FSRDC access during the sample period. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H7: Comparing Empiricists with Theorists (Within-University Variation)

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC × Empiricist	0.00852***	0.0251***	0.0445***	2.294***	0.0549***
	(0.00178)	(0.00479)	(0.00914)	(0.53370)	(0.01320)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University × Year FE	Yes	Yes	Yes	Yes	Yes
N	232145	232145	232145	232145	232145

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output using theorists at the same university as the control group. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated to a research institution with an operating FSRDC. Empiricist: 0/1 = 1 for researchers whose first five publications are mostly empirical in nature, as assessed by our machine-learning based paper classifier. All models include individual fixed effects and year fixed effects interacted with university dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H8: Comparing Empiricists in Differentially Impacted Fields (Within-University Variation)

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC × Impacted Field	0.0107***	0.0309***	0.0356*	2.773***	0.0165
	(0.00204)	(0.00675)	(0.01511)	(0.81346)	(0.01748)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University × Year FE	Yes	Yes	Yes	Yes	Yes
N	152988	152988	152988	152988	152988

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output for empiricists in fields impacted by FSRDC data, using empiricists in less impacted fields but at the same university as the control group. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated to a research institution with an operating FSRDC. Impacted Field: 0/1 = 1 if the majority of the researcher's work is classified under environmental economics, international economics, labor economics, applied microeconomics, or public economics, as assessed by our machine-learning based paper classifier. All models include individual fixed effects and year fixed effects interacted with university dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H9: Robustness to Controlling for Field of Research Fixed Effects

	FSRDC Use (1)	FSRDC Cites (2)	Top Pubs (3)	Cite-Weighted (4)	Policy Pubs (5)
Post-FSRDC	0.00599***	0.0110**	0.0230**	1.191**	0.0395***
	(0.00142)	(0.00377)	(0.00821)	(0.42076)	(0.01123)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
Field × Year FE	Yes	Yes	Yes	Yes	Yes
N	154,168	154,168	154,168	154,168	154,168

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output, controlling for field-specific time trends. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects, year fixed effects interacted with university tier dummies, and field × year fixed effects. Field is defined as one of 15 research areas based on the first letter of the researcher's modal JEL code. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H10: Effects of FSRDC Access on Financial and Research Assistant Acknowledgments

	Grant Count	Log (Grant + 1)	RA Count	Log (RA + 1)
	(1)	(2)	(3)	(4)
Post-FSRDC	0.0169	0.00950	0.00527	0.00355
	(0.01094)	(0.00508)	(0.00302)	(0.00204)
Researcher FE	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes
N	155,622	155,622	155,622	155,622

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of research resource support. Column (1) reports the result from an OLS model where the dependent variable is the number of grants acknowledged by researcher i in year t. Column (2) uses the log of one plus the number of grants acknowledged. Column (3) reports the result from an OLS model where the dependent variable is the number of research assistants acknowledged by researcher i in year t. Column (4) uses the log of one plus the number of research assistants acknowledged. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated with a research institution with an operating FSRDC. All models include researcher fixed effects and year fixed effects interacted with university-tier dummies. Standard errors are in parentheses, clustered at the university level. *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively.

Table H11: Comparing Empiricists to Other Members of the Same Consortium (Within-Consortium Variation)

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC	0.00605**	0.0167*	0.0252**	1.306*	0.0222
	(0.00203)	(0.00711)	(0.00908)	(0.61967)	(0.01838)
Researcher FE	Yes	Yes	Yes	Yes	Yes
Consortium × Year FE	Yes	Yes	Yes	Yes	Yes
N	63096	63096	63096	63096	63096

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output, using empiricists in non-hosting institutions but in the same consortium as the control group. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with consortium dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H12: University-Level Effects of FSRDC Access

Panel A: All Researchers

	(1) FSRDC Use	(2) FSRDC Cites	(3) Top Pubs	(4) Cite-Weighted	(5) Policy Pubs
Post-FSRDC	0.215*	1.0223**	2.204*	86.099	3.972**
	(0.08481)	(0.33241)	(1.00187)	(50.61781)	(1.44126)
University FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	8,685	8,685	8,685	8,685	8,685

Panel B: Empiricists

	(1) FSRDC Use	(2) FSRDC Cites	(3) Top Pubs	(4) Cite-Weighted	(5) Policy Pubs
Post-FSRDC	0.221**	0.964***	1.846**	75.144*	3.380**
	(0.08140)	(0.28861)	(0.67505)	(36.77793)	(1.12500)
University FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	8,685	8,685	8,685	8,685	8,685

Panel C: Theorists

	(1) FSRDC Use	(2) FSRDC Cites	(3) Top Pubs	(4) Cite-Weighted	(5) Policy Pubs
Post-FSRDC	-0.00533	0.0583	0.358	10.955	0.593
	(0.01834)	(0.09646)	(0.59970)	(22.40362)	(0.66873)
University FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	8,685	8,685	8,685	8,685	8,685

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output at the university level. Panel A includes all researchers affiliated with the university, while Panels B and C construct the university-level outcomes using publications by empiricists and theorists, respectively. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan, 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received up to five years following their publication. Column (5) reports the result from an OLS model where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC: 0/1 = 1 after a university has an operating FSRDC. All models include university fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H13: Effect of FSRDC Access Excluding NSF Grant Applicants

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC	0.00589***	0.0104**	0.0219**	1.205**	0.0359**
	(0.00139)	(0.00386)	(0.00811)	(0.41970)	(0.01108)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	155039	155039	155039	155039	155039

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output, excluding researchers who were the applicants of the NSF grant leading to an FSRDC. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received up to five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC equals one in all years after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H14: Effect of FSRDC Access Excluding Authors in Institutions Hosting FSRDCs

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
Within 10 Miles of FSRDC	0.00300**	0.00951**	0.0135*	0.973*	0.0223**
	(0.00091)	(0.00307)	(0.00660)	(0.41020)	(0.00832)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	122695	122695	122695	122695	122695

Note: This table presents estimates from OLS models evaluating the impact of FSRDC proximity on measures of confidential data adoption and research output, excluding researchers directly affiliated with an institution with an operating FSRDC center. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Within 10 miles: 0/1 = 1 after a researcher has been affiliated with a research institution located within 10 miles of an operating FSRDC but not hosting it. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H15: Alternative Definitions to Measure FSRDC Access

	Top Pubs	Cite-weighted	Policy-relevant	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)	(6)
Post-FSRDC (City)	0.0146*	0.885*	0.0232**			
	(0.00637)	(0.39640)	(0.00794)			
Post-FSRDC (Consortium)				0.00653	0.961*	0.00742
				(0.00621)	(0.39377)	(0.00806)
Researcher FE	Yes	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	155622	155622	155622	155622	155622	155622

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of research output using alternative definitions of access. Columns (1) and (4) report the results from OLS models, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan, 2020). Columns (2) and (5) report the results from OLS models, where the dependent variable is the count of articles weighted by the number of citations received up to five years following their publication. Columns (3) and (6) report the results from OLS models, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC (City): 0/1 = 1 after a researcher has been affiliated to a research institution located in a city with an operating FSRDC. Post-FSRDC (Consortium): 0/1 = 1 after a researcher has been affiliated to a research institution belonging to a consortium operating an FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H16: Effect of FSRDC Access on Theorists

	Using FSRDC (1)	Citing FSRDC (2)	Top Pubs (3)	Cite-weighted (4)	Policy-relevant (5)
Post-FSRDC	-0.000489	0.00252	-0.00131	0.563	0.0104
	(0.00111)	(0.00400)	(0.01423)	(0.63503)	(0.00853)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	77538	77538	77538	77538	77538

Note: This table presents estimates from an OLS model evaluating the impact of FSRDC access on measures of confidential data adoption and research output when looking at a sample of theorists only. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC equals one in all years after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H17: Robustness to Researcher Age Controls

	(1) FSRDC Use	(2) FSRDC Cites	(3) Top Pubs	(4) Cite-Weighted	(5) Policy Pubs
Post-FSRDC	0.00565***	0.0102**	0.0209**	1.148**	0.0375**
	(0.00140)	(0.00374)	(0.00793)	(0.40218)	(0.01137)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
Researcher Age FE	Yes	Yes	Yes	Yes	Yes
N	155,622	155,622	155,622	155,622	155,622

Note: This table presents estimates from OLS models evaluating the robustness of the impact of FSRDC access to controls for researcher age. Researcher age is defined as the number of years since a researcher's first publication. Column (1) reports the result from an OLS model, where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated with a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies, as well as researcher age fixed effects. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H18: Effect of FSRDC Access Excluding Authors Treated After Mobility Events

	Using FSRDC	Citing FSRDC	Top Pubs	Cite-weighted	Policy-relevant
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC	0.00593***	0.0122*	0.0277**	1.183**	0.0297*
	(0.00153)	(0.00491)	(0.00922)	(0.42848)	(0.01375)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	146491	146491	146491	146491	146491

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of confidential data adoption and research output excluding researchers who were treated after moving to an institution with an operating FSRDC (i.e., considering only researchers treated by the FSRDC opening). Column (1) reports the result from an OLS model where the dependent variable is the count of articles that directly use confidential data accessible only in an FSRDC. Column (2) reports the result from an OLS model, where the dependent variable is the count of articles that cite a paper using confidential data accessible only in an FSRDC (excluding papers that directly use confidential FSRDC data). Column (3) reports the result from an OLS model, where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Column (4) reports the result from an OLS model, where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Column (5) reports the result from an OLS model, where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC equals one in all years after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H19: Effect of FSRDC Access on University Hiring and Researchers' Composition

	Net Inflow	Net Inflow (Stars)	Net Inflow (Superstars)	Share Empiricists	Difference in Pubs
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC	-0.0926	-0.0415	-0.0442	-0.0196	-0.146
	(0.18602)	(0.04727)	(0.02537)	(0.02009)	(0.19047)
University FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	7917	7917	7917	8323	8323

Note: This table presents estimates from an OLS model at the university level that tests whether universities experience changes in faculty composition after establishing an FSRDC. Column (1) reports the result from an OLS model at the university level, where the dependent variable is the number of researchers that join the faculty, minus the number of researchers that leave the faculty. Column (2) reports the result from an OLS model, where the dependent variable is equal to the number of researchers in the top 10% of cumulative citations up to that year ("stars") that join the faculty, minus the number of stars that leave the faculty. Column (3) reports the result from an OLS model, where the dependent variable is equal to the number of researchers in the top 5% of cumulative citations up to that year ("superstars") that join the faculty, minus the number of superstars that leave the faculty. Column (4) reports the result from an OLS model at the university level, where the dependent variable is the share of empiricists out of all economists affiliated with the university. Column (5) reports the result from an OLS model, where the dependent variable is the rolling difference in average lifetime top publications between new arrivals and incumbent faculty (to test whether new hires are more accomplished than standing faculty). Post-FSRDC equals one in all years after a university has an operating FSRDC. All models include university fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H20: Effect of FSRDC Access on University-Level Researcher Exit Rates

	Total Exits	Empiricists	FSRDC JEL	FSRDC Users
	(1)	(2)	(3)	(4)
Post-FSRDC	-0.000292	0.00350	-0.00182	0.0103
	(0.00436)	(0.00556)	(0.00582)	(0.01131)
University FE	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes
N	8,280	8,280	8,280	8,280

Note: This table presents estimates from OLS models at the university-year level evaluating the impact of FSRDC access on researcher exit rates. Column (1) reports the exit rate for all researchers, column (2) for empiricists, column (3) for researchers in FSRDC-related JEL fields, and column (4) for FSRDC users. A researcher exits university j in year t if they were affiliated with j in year $t-1$ but not in year t, considering consecutive years only. The exit rate equals the number of departures divided by the stock of researchers at the university in the prior year. Post-FSRDC: 0/1 = 1 after university j has an operating FSRDC. All models include university fixed effects and year fixed effects interacted with university-tier dummies. Standard errors are in parentheses, clustered at the university level. *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively.

Table H21: Effect of FSRDC Access on Alternative Measures of Research Direction

	Top 20 JELs (0/1)	Top 40 JELs (0/1)	Top 60 JELs (0/1)	Top 100 JELs (0/1)	Top 200 JELs (0/1)
	(1)	(2)	(3)	(4)	(5)
Post-FSRDC	0.0139*	0.0147*	0.0138	0.0143	0.0138
	(0.00590)	(0.00736)	(0.00749)	(0.00850)	(0.00868)
Researcher FE	Yes	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes	Yes
N	155622	155622	155622	155622	155622
Mean of DV	0.149	0.201	0.228	0.271	0.304

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of research direction. We experiment with increasingly less stringent definitions of what constitutes a JEL code associated with FSRDC papers. Column (1) reports the result from an OLS model, where the dependent variable is an indicator equal to one if the researcher used at least one FSRDC JEL code during the year, which for this model we consider to be any of the top 20 JEL codes that appear with the greatest frequency in FSRDC papers. Column (2) uses the same model, but we consider any of the top 40 JEL codes that appear with the greatest frequency in FSRDC papers. The remaining columns are specified analogously. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated with a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively

Table H22: Effect of FSRDC Access on External Grant Funding

	Funding Sources (1)	Funding Sources (0/1) (2)	New Funding Sources (3)	New Funding Sources (0/1) (4)
Post-FSRDC	0.00384	0.00356	0.00478	0.00373
	(0.00723)	(0.00330)	(0.00532)	(0.00276)
Researcher FE	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes
N	155622	155622	155622	155622

Note: This table presents estimates from OLS models evaluating whether researchers report additional funding sources after gaining access to an FSRDC. Column (1) reports the result from an OLS model, where the dependent variable is the count of funding sources acknowledged in all the publications of a given year. Column (2) reports the result from a linear probability model, where the dependent variable is equal to one if the researcher acknowledges any funding source in a given year. Column (3) reports the result from an OLS model, where the dependent variable is the count of new funding sources acknowledged in all the publications of a given year. Column (4) reports the result from a linear probability model, where the dependent variable is equal to one if the researcher acknowledges any new funding source in a given year. Post-FSRDC equals one in all years after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H23: Effect of FSRDC Access on Patterns of Collaboration

	Mean Number of Co-Authors	Mean Number of Unique Co-Authors	Mean Number of New Co-Authors	Mean Number of Team-Papers
	(1)	(2)	(3)	(4)
Post-FSRDC	0.105	0.101	0.0421	0.0310
	(0.07287)	(0.06535)	(0.05163)	(0.02069)
University FE	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes
N	8323	8323	8323	8323

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on measures of collaboration among researchers at the university level. Column (1) reports the result from an OLS model, where the dependent variable is the mean number of co-authors for faculty members who publish at least one paper that year. Column (2) reports the result from an OLS model, where the dependent variable is the mean number of unique co-authors for faculty members who publish at least one paper that year. Column (3) reports the result from an OLS model, where the dependent variable is the mean number of new co-authors for faculty members who publish at least one paper that year. Column (4) reports the result from an OLS model, where the dependent variable is the mean number of team papers with four or more authors for faculty members who publish at least one paper that year. Post-FSRDC equals one in all years after the university has an operating FSRDC. All models include university fixed effects and year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively

Table H24: Heterogeneity in FSRDC Effects by Baseline Researcher Productivity

	Top Publications		Cite-Weighted Pubs		Policy Publications	
	(1)	(2)	(3)	(4)	(5)	(6)
Post-FSRDC (Low Baseline)	0.0660***	0.0662***	1.968***	1.670**	0.0776***	0.0772***
	(0.00873)	(0.00954)	(0.53170)	(0.56433)	(0.01654)	(0.01777)
Post-FSRDC × Medium	-0.0120	-0.0139	-0.934	-1.092	-0.0657***	-0.0640***
	(0.01336)	(0.01223)	(0.56094)	(0.57575)	(0.01859)	(0.01866)
Post-FSRDC × High	-0.107***	-0.109***	-0.791	-0.961	-0.0569	-0.0545
	(0.02029)	(0.02046)	(0.93119)	(0.89061)	(0.03956)	(0.03834)
Researcher FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No
University FE	Yes	No	Yes	No	Yes	No
University Tier × Year FE	No	Yes	No	Yes	No	Yes
N	146,124	146,130	146,124	146,130	146,124	146,130

Note: This table presents estimates from OLS models evaluating heterogeneity in the impact of FSRDC access by researchers' pre-FSRDC productivity. Researchers are divided into terciles based on their average annual number of publications prior to gaining local FSRDC access, and the sample is restricted to researchers with at least five years of pre-treatment observations. Columns (1) and (2) report the results where the dependent variable is the count of articles published in the most prestigious economics journals (the list of top five and top field journals is from Heckman and Moktan 2020). Columns (3) and (4) report the results where the dependent variable is the count of articles weighted by the number of citations received during the five years following their publication. Columns (5) and (6) report the results where the dependent variable is the count of articles cited in at least one policy document. Post-FSRDC (Low Baseline) reports the effect for researchers in the lowest tercile of pre-FSRDC productivity. Post-FSRDC × Medium and Post-FSRDC × High report the differential effects for researchers in the middle and highest terciles relative to researchers in the lowest tercile. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated to a research institution with an operating FSRDC. All models include individual fixed effects. Columns (1), (3), and (5) include year and university fixed effects, while columns (2), (4), and (6) include year fixed effects interacted with university tier dummies. Standard errors are in parentheses, clustered at the university level. *, **,*** denote significance at 5%, 1% and 0.1% level respectively.

Table H25: Effects of FSRDC Access on New Coauthors by Type

Panel A: Junior and Expert Coauthors

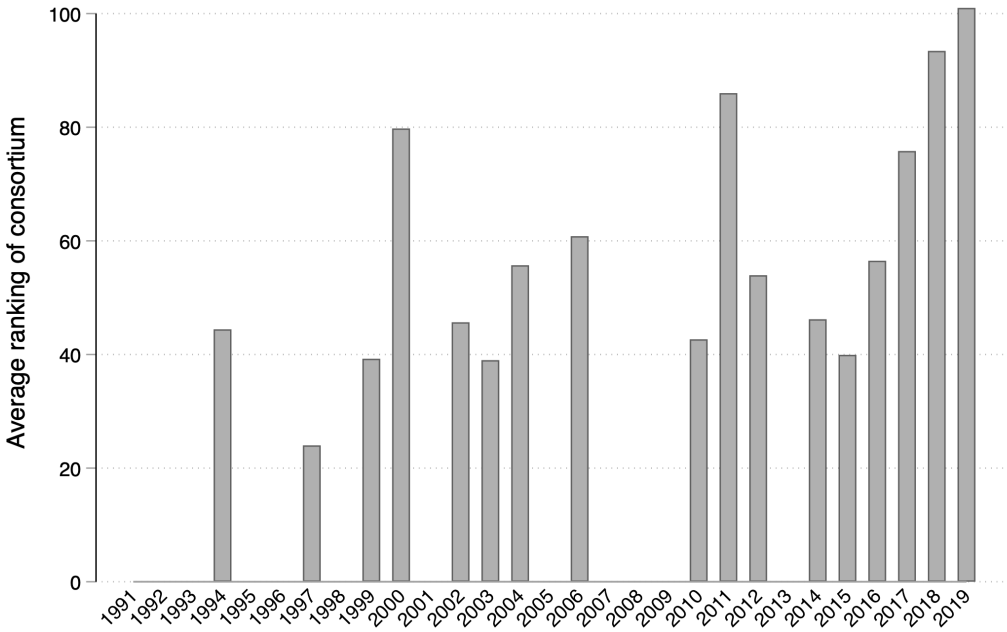
	Junior (5yr)	Junior (10yr)	Census Expert	FSRDC Expert
	(1)	(2)	(3)	(4)
Post-FSRDC	0.0178**	0.0272***	0.0292**	0.00255
	(0.00657)	(0.00797)	(0.00959)	(0.00177)
Mean of the DV	0.073	0.156	0.185	0.008
Researcher FE	Yes	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes	Yes
N	155,622	155,622	155,622	155,622

Panel B: Coauthors by Location

	Local	Distant (No RDC)	Distant (RDC)
	(1)	(2)	(3)
Post-FSRDC	-0.0188*	0.0251**	0.0321***
	(0.00769)	(0.00817)	(0.00432)
Mean of the DV	0.084	0.170	0.028
Researcher FE	Yes	Yes	Yes
University Tier × Year FE	Yes	Yes	Yes
N	155,622	155,622	155,622

Note: This table presents estimates from OLS models evaluating the impact of FSRDC access on the number of new coauthors of different types for researcher i in year t. A coauthor is classified as new if they have not previously coauthored with researcher i. Panel A classifies new coauthors by career stage and expertise. Columns (1) and (2) report the number of new junior coauthors, defined as researchers whose first publication occurred fewer than five or ten years earlier, respectively. Column (3) reports the number of new Census experts, defined as researchers with a prior publication in one of the top 20 RDC-related JEL codes. Column (4) reports the number of new FSRDC experts, defined as researchers with a prior publication using FSRDC data. Panel B classifies new coauthors by location. Column (1) reports the number of new local coauthors, defined as researchers associated with the same nearest RDC as the focal researcher. Columns (2) and (3) report the number of new distant coauthors without and with FSRDC access, respectively. Post-FSRDC: 0/1 = 1 after a researcher has been affiliated with a research institution with an operating FSRDC. All models include researcher fixed effects and year fixed effects interacted with university-tier dummies. Standard errors are in parentheses, clustered at the university level. *, **, *** denote significance at the 5%, 1%, and 0.1% level, respectively.

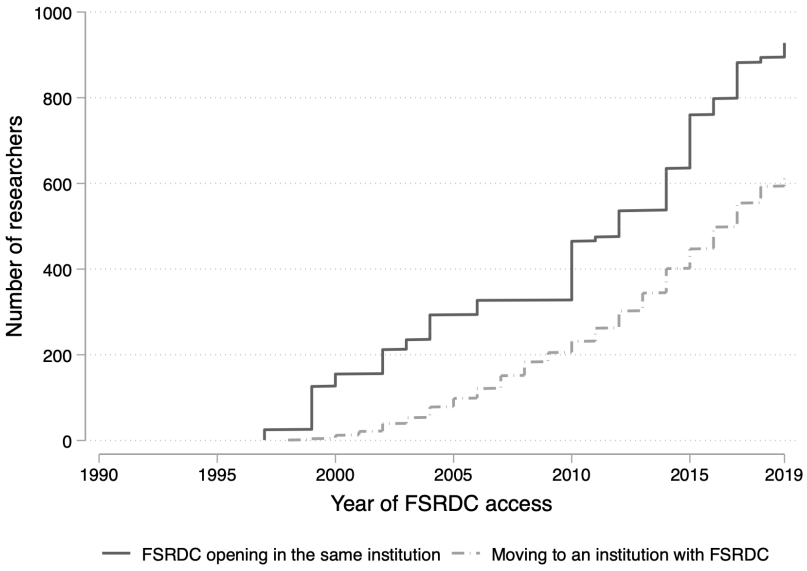
Figure H1: Average Rank of Consortium Members By Time of FSRDC Opening



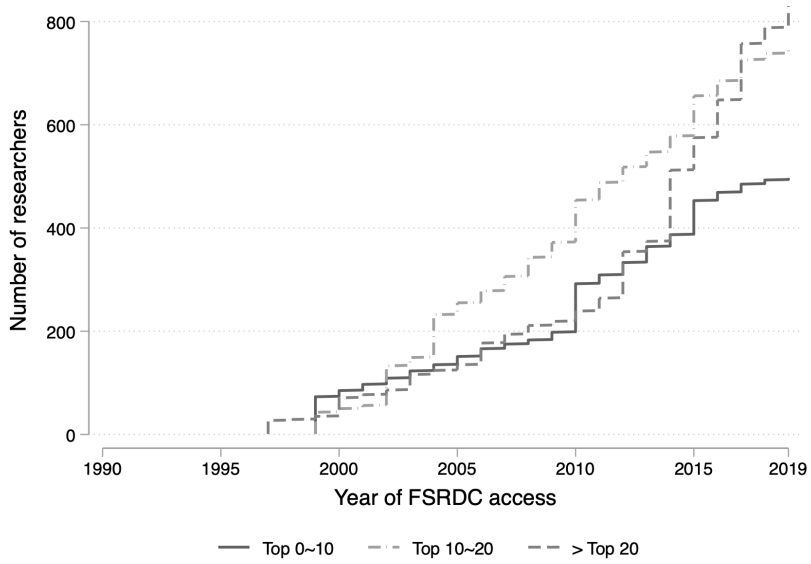
Note: This figure reports the average rank of all members in each FSRDC funding consortium, which is then averaged across all consortia that opened an FSRDC in a given year. When calculating the average rank of each consortium, we include all universities that were a member at any point during the sample period. Information on the ranking of economics departments is taken from Kalaitzidakis et al. (2003). Since Kalaitzidakis et al. (2003) only report information for the top 100 North American institutions, we assign the same residual rank of 101 to all other institutions in our sample. Hence, lower numbers correspond to departments ranked higher. See text for more details.

Figure H2: Cumulative Number of Researchers Gaining FSRDC Access over Time

(i) By Modality of Researchers' Access



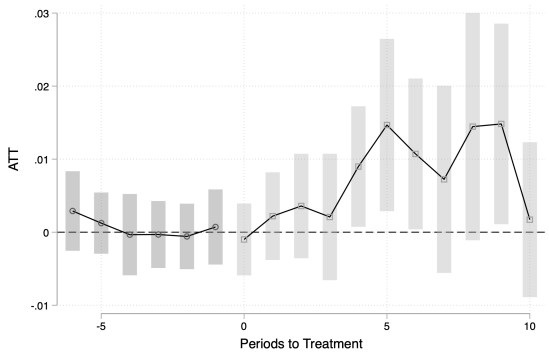
(ii) By Ranking of Researchers' Institution



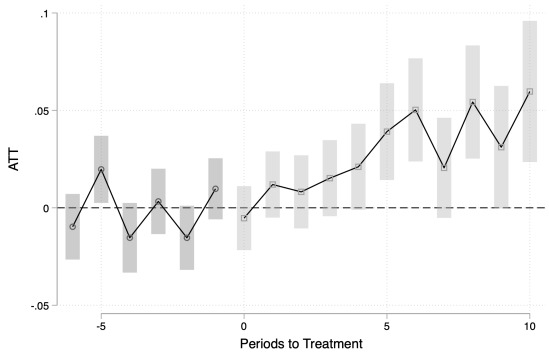
Note: This figure plots the cumulative number of researchers who gain access to an FSRDC through a hosting institution. The cumulative frequencies are split by the modality in which researchers become co-located to an FSRDC (panel (i)) and the ranking of their institution of affiliation (panel (ii)). See text for more details.

Figure H3: Time-Varying Estimates of the Impact of FSRDCs on Research Output Using the Callaway-Sant'Anna Doubly-Robust DID Estimator.

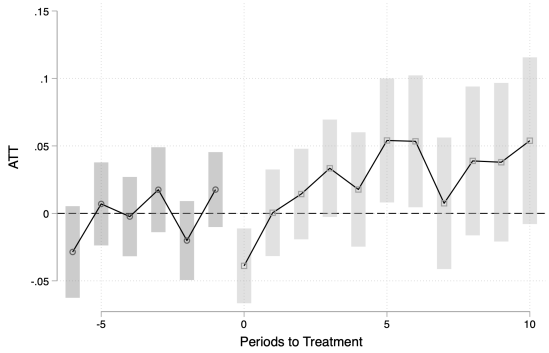
(i) Papers Using FSRDC Data



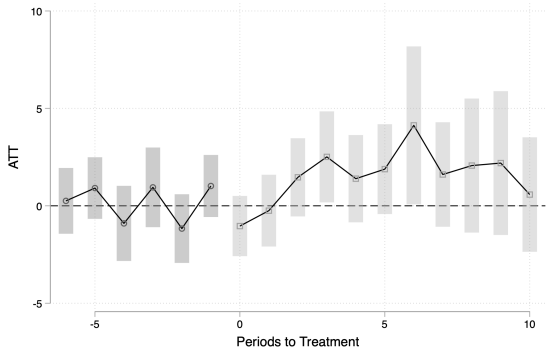
(ii) Papers Citing FSRDC Papers



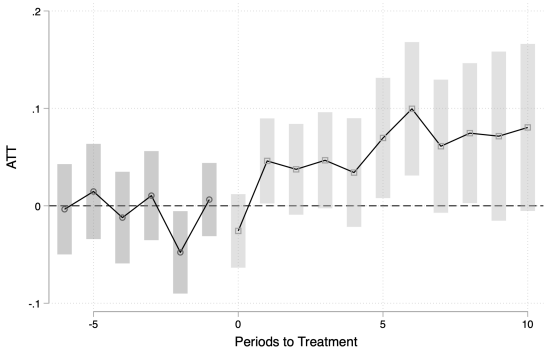
(iii) Top Publications



(iv) Cite-Weighted Publications



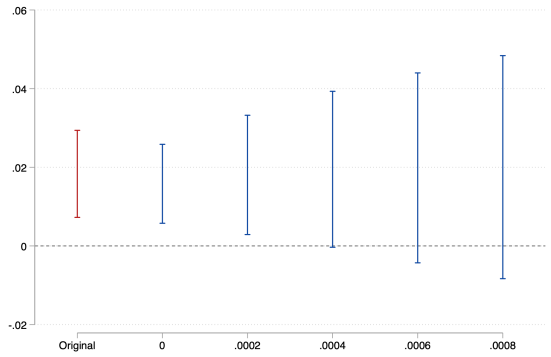
(v) Policy-Relevant Publications



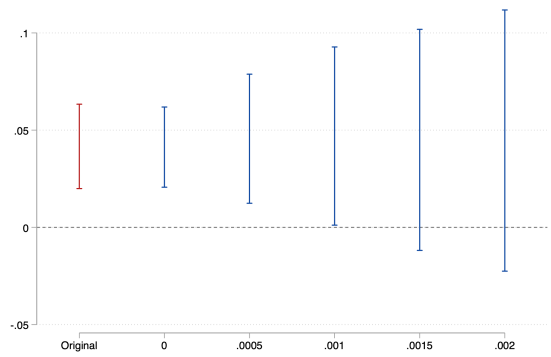
Note: This figure provides visual illustrations of the event study version of the main regressions evaluating the impacts of FSRDC access on measures of research output, estimated using the doubly-robust difference-in-difference estimator developed by Callaway and Sant'Anna (2021). The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). The charts plot values of β for different lags before and after obtaining access to an FSRDC with 95% confidence intervals. Regressions include individual, university, and year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure H4: Sensitivity to Violations of Parallel Trends using the HonestDiD Approach from Rambachan and Roth (2023)

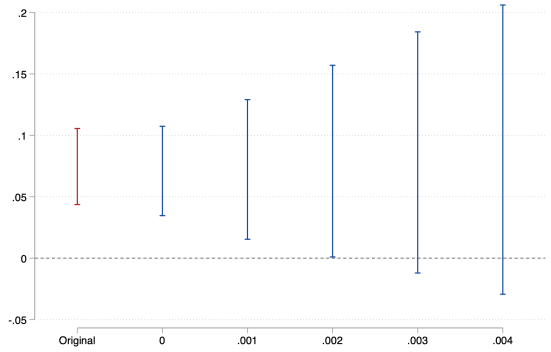
(i) Papers Using FSRDC Data



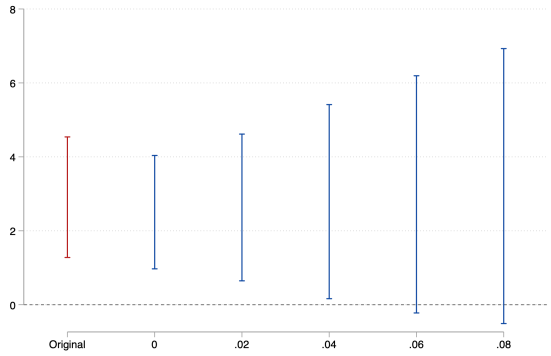
(ii) Papers Citing FSRDC Papers



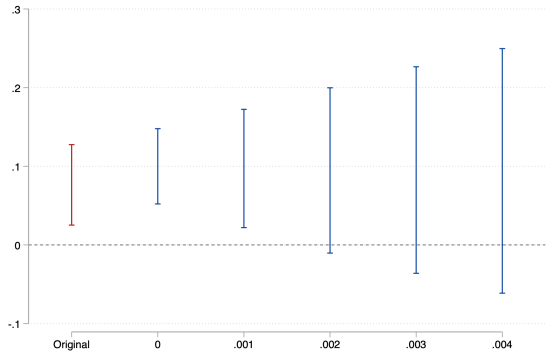
(iii) Top Publications



(iv) Cite-Weighted Publications



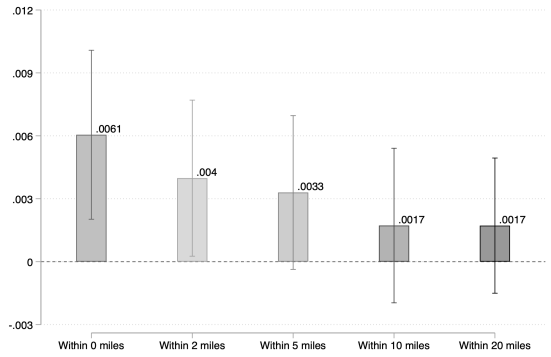
(v) Policy-Relevant Publications



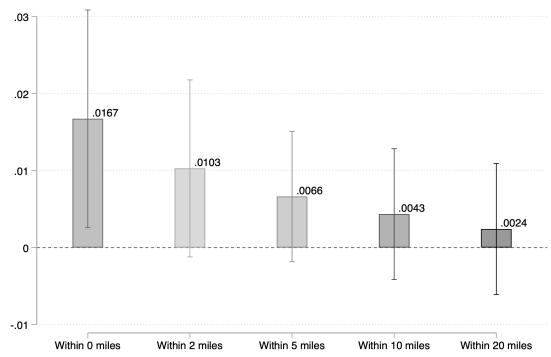
Note: In this figure, we implement the sensitivity analysis from Rambachan and Roth (2023) to test whether our results are robust to violations of parallel trends. In particular, we apply the "smoothness" restriction, which allows for a linear extrapolation of pre-trends to be off by some constant M. For each panel, we report coefficients and 95% confidence intervals for our original specification (in red), for M=0 (which corresponds to a perfectly linear extrapolation of pre-trends), and for progressively larger values of M (which allow for greater deviations from linearity). The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). All models include individual and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure H5: Effect of FSRDC Access for Consortium Members by Distance (Within Consortium Variation)

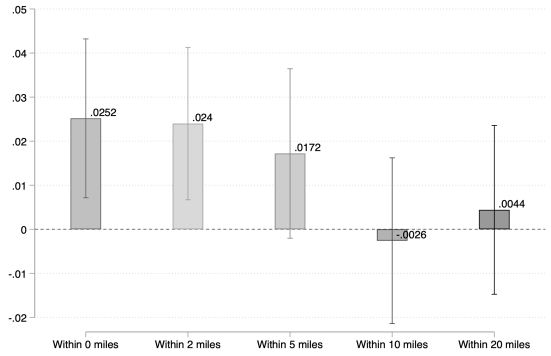
(i) Papers Using FSRDC Data



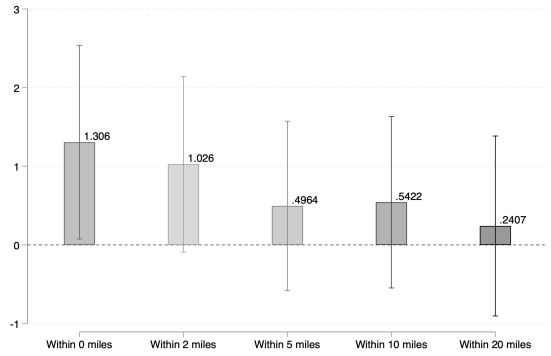
(ii) Papers Citing FSRDC Papers



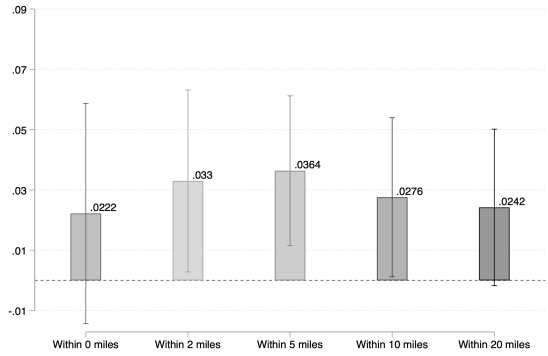
(iii) Top Publications



(iv) Cite-Weighted Publications



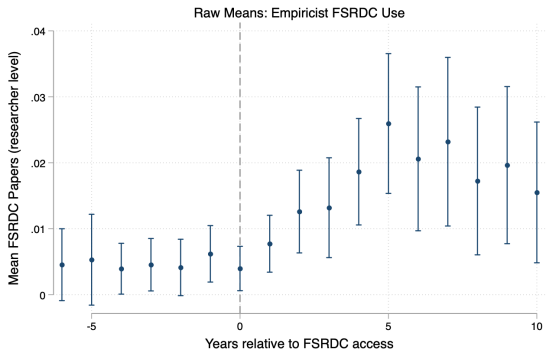
(v) Policy-Relevant Publications



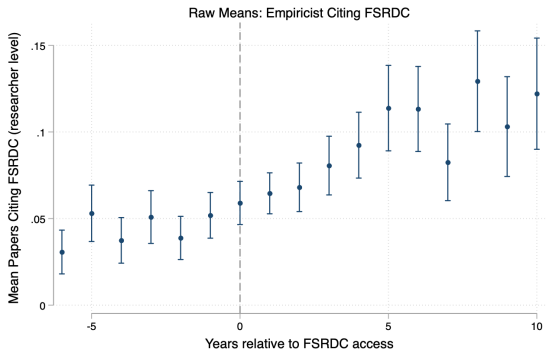
Note: This figure provides visual illustrations of the effect of data access at different distances, using empiricists in non-hosting institutions but in the same consortium as the control group. Distance is the geometric distance between the institution of the researcher and the closest operating FSRDC. The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). The chart plots values of β from different regressions where researchers are considered as treated if they are affiliated to institutions within a progressively larger radius from an FSRDC. All models include university and consortium $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure H6: Raw Means of Research Outcomes for Empiricists Around FSRDC Openings

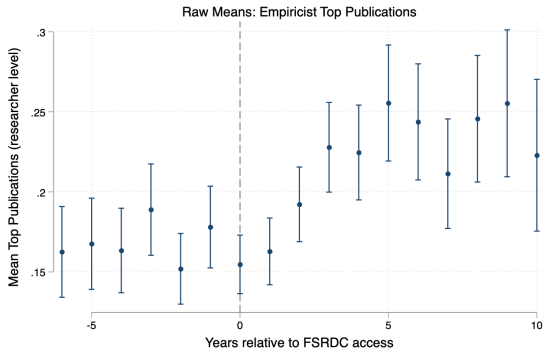
(i) Papers Using FSRDC Data



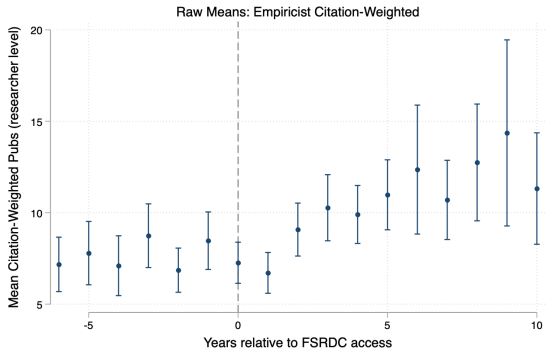
(ii) Papers Citing FSRDC Papers



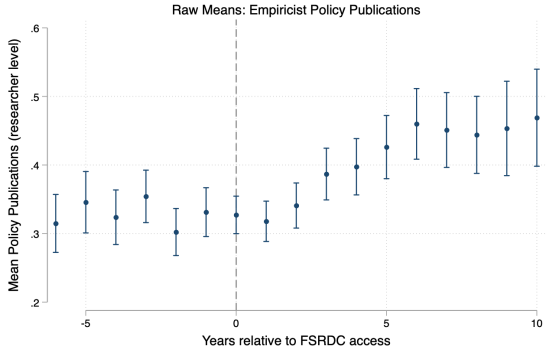
(iii) Top Publications



(iv) Cite-Weighted Publications



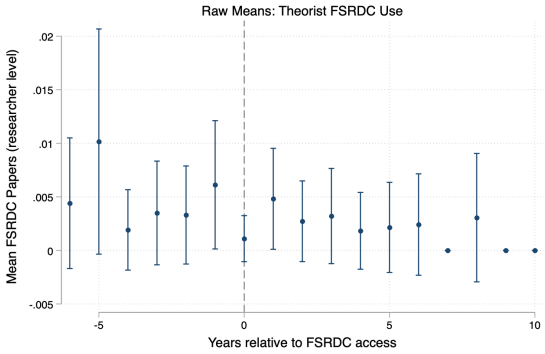
(v) Policy-Relevant Publications



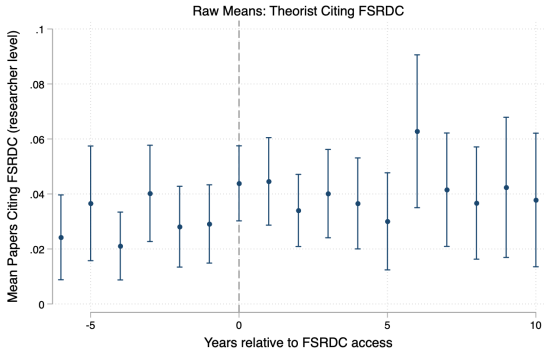
Note: This figure provides visual illustrations of raw means of measures of confidential data adoption and research output for empiricist researchers at universities that gain FSRDC access, centered around the year of FSRDC opening. The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). Event time zero corresponds to the year of FSRDC opening, indicated by the vertical dashed line. Empiricists are researchers whose first five publications are mostly empirical in nature, as assessed by our machine-learning based paper classifier. See text for more details.

Figure H7: Raw Means of Research Outcomes for Theorists Around FSRDC Openings

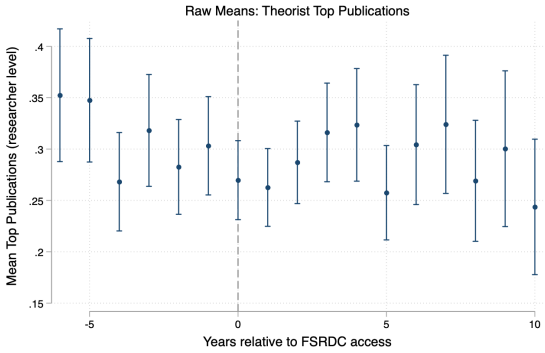
(i) Papers Using FSRDC Data



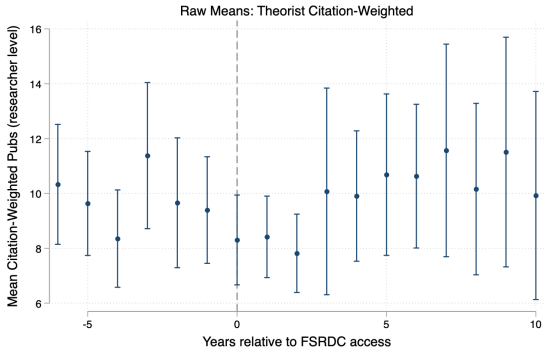
(ii) Papers Citing FSRDC Papers



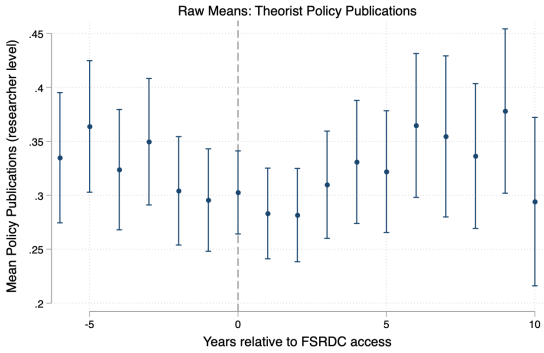
(iii) Top Publications



(iv) Cite-Weighted Publications



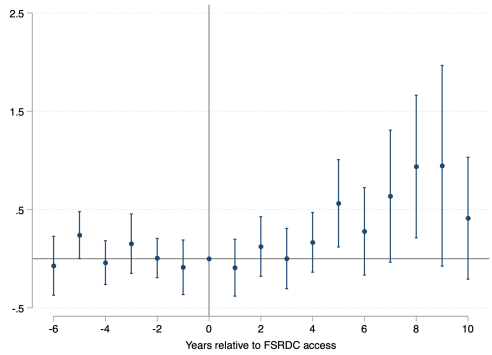
(v) Policy-Relevant Publications



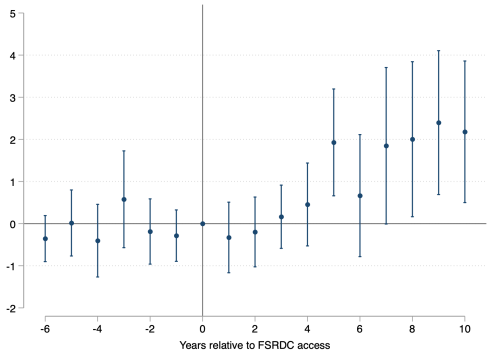
Note: This figure provides visual illustrations of raw means of measures of confidential data adoption and research output for theorist researchers at universities that gain FSRDC access, centered around the year of FSRDC opening. The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). Event time zero corresponds to the year of FSRDC opening, indicated by the vertical dashed line. Theorists are researchers whose first five publications are mostly non-empirical in nature, as assessed by our machine-learning based paper classifier. See text for more details.

Figure H8: University-Level Time-Varying Estimates of the Impact of FSRDCs

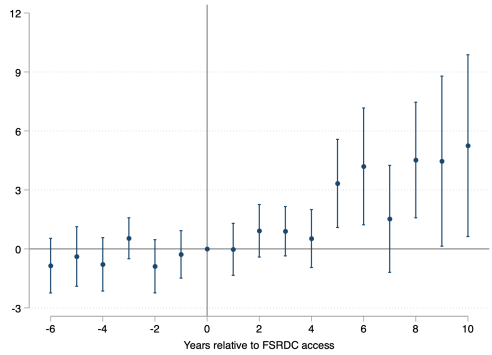
(i) Papers Using FSRDC Data



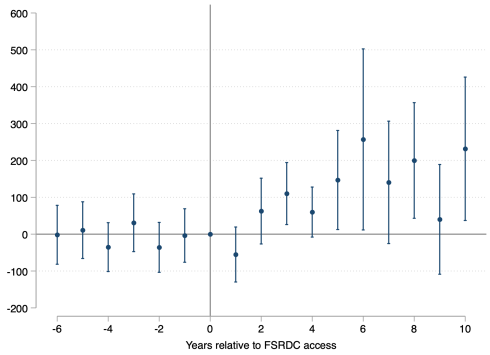
(ii) Papers Citing FSRDC Papers



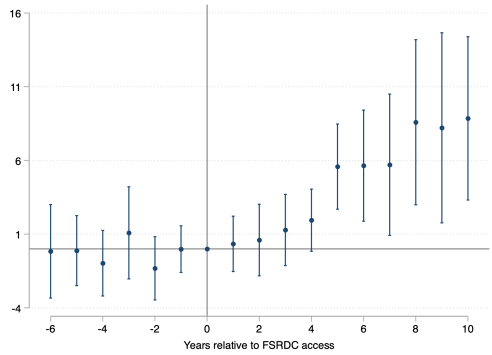
(iii) Top Publications



(iv) Cite-Weighted Publications



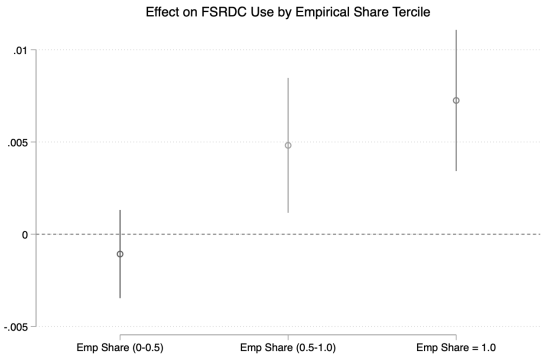
(v) Policy-Relevant Publications



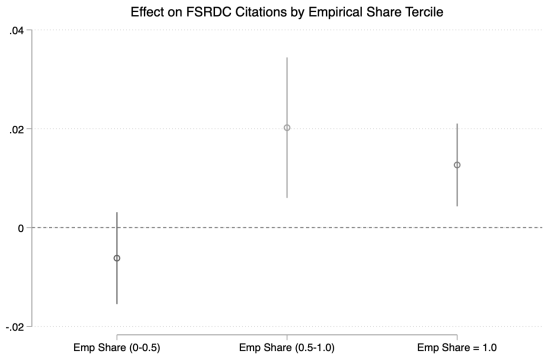
Note: This figure provides visual illustrations of the event study version of the regression evaluating the impact of FSRDC access on measures of research output at the university level. The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). All models include university and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure H9: Effect of FSRDC Access by Share of Empirical Work

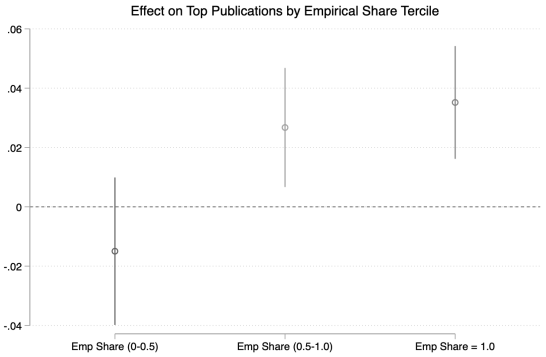
(i) Papers Using FSRDC Data



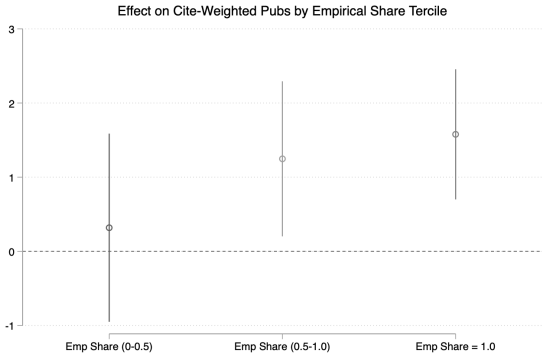
(ii) Papers Citing FSRDC Papers



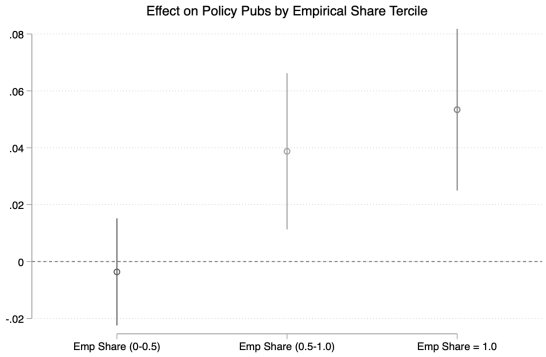
(iii) Top Publications



(iv) Cite-Weighted Publications



(v) Policy-Relevant Publications

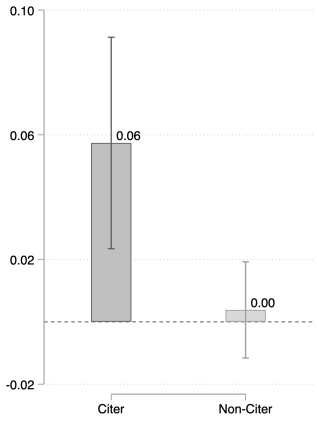


Note: This figure provides visual illustrations of the effect of data access for researchers with different methodological orientations. Empirical share is the proportion of a researcher's first five publications that are empirical. The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). Each chart is estimated from a single interaction-based regression, where each researcher is classified into one of three mutually exclusive categories. All models include individual and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

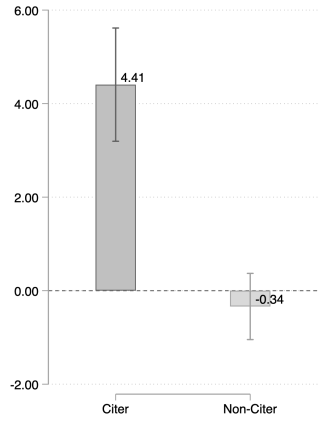
Figure H10: Heterogeneous Effect of FSRDC Access

Panel A: by Researchers Citing FSRDC Research

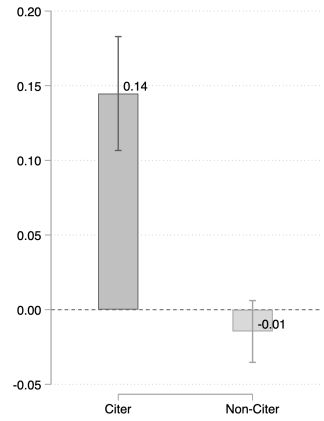
(i) Top Publications



(ii) Cite-Weighted Publications

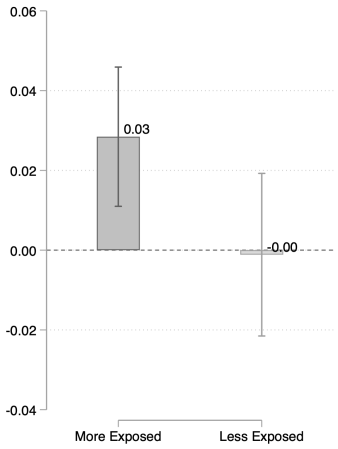


(iii) Policy-Relevant Publications

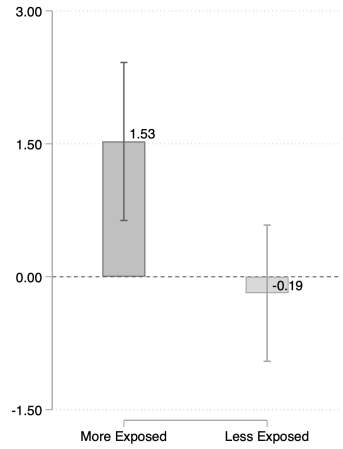


Panel B: by Researchers Exposed to FSRDC Research

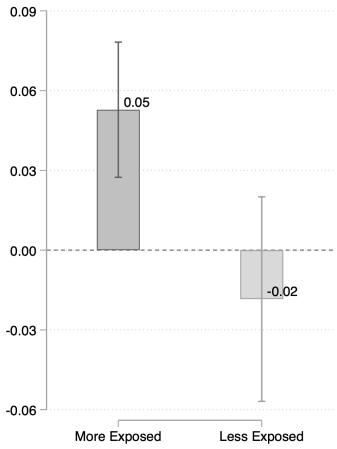
(i) Top Publications



(ii) Cite-Weighted Publications



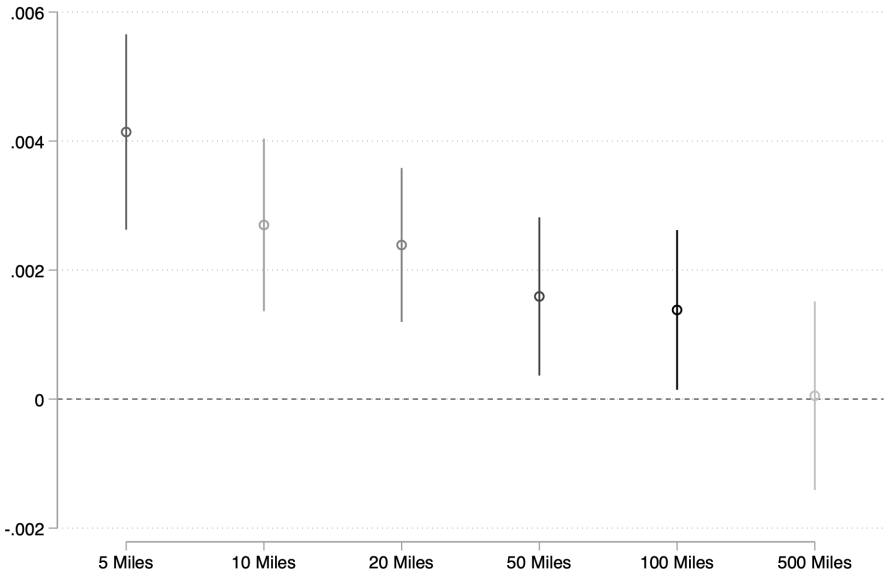
(iii) Policy-Relevant Publications



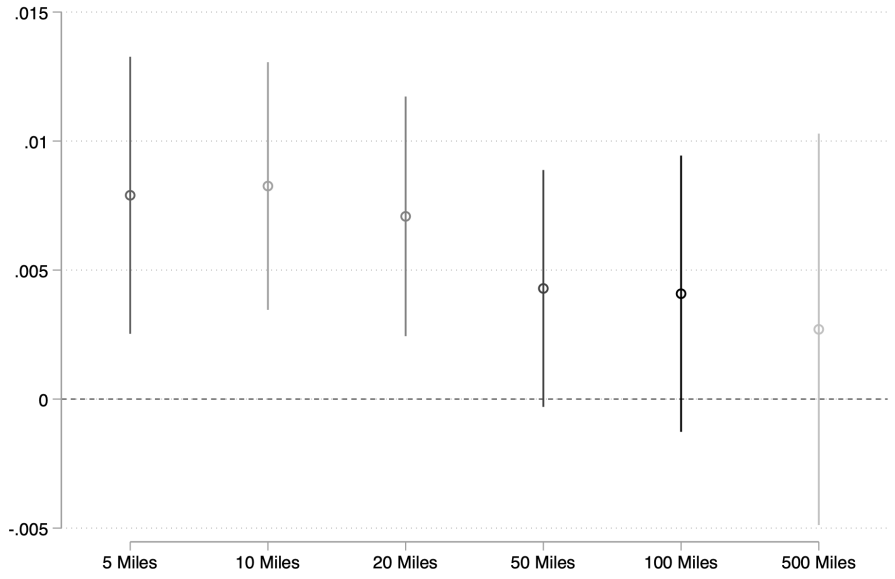
Note: This figure provides visual illustrations of the impacts of FSRDC access on measures of research output for different types of empirical researchers. Each chart is estimated from a single interaction-based regression and excludes economists who have used FSRDC data directly. Columns (i), (ii), and (iii) of Panel A show the effects of being affiliated with an institution with an FSRDC separately for researchers who cited a paper using FSRDC data versus those who have not. Columns (i), (ii), and (iii) of Panel B show the effects separately for researchers who had an above-median number of lifetime colleagues who directly used FSRDC data versus those who did not. The main dependent variables are the number of top publications (columns (i) of Panels A and B), the citation-weighted number of publications (columns (ii) of Panels A and B), and the number of policy-relevant publications (columns (iii) of Panels A and B). Regressions include researcher and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure H11: Effect of FSRDC Access on Confidential Data Diffusion by Distance

(i) Papers Using FSRDC Data



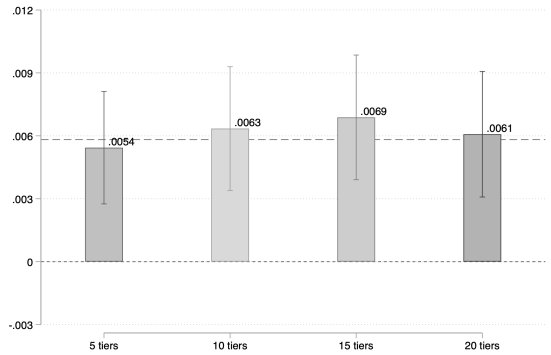
(ii) Papers Citing FSRDC Papers



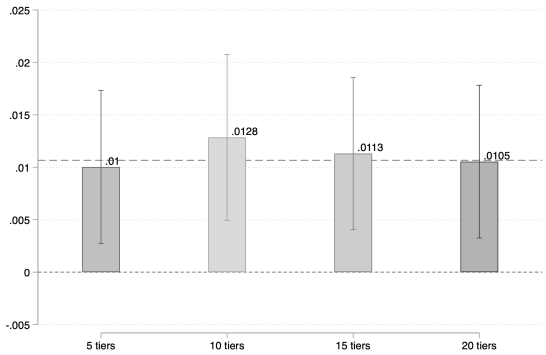
Note: This figure provides visual illustrations of the effect of data access on confidential data diffusion at different distances. Distance is the geometric distance between the institution of the researcher and the closest operating FSRDC. The main dependent variables are the number of papers written using FSRDC data (panel (i)) and the number of papers that cite FSRDC papers (panel (ii)). The chart plots values of β from different regressions where researchers are considered as treated if they are affiliated to institutions within a progressively larger radius from an FSRDC. Regressions include individual and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.

Figure H12: Sensitivity to Alternative Definitions of University Ranking Tiers Fixed Effects

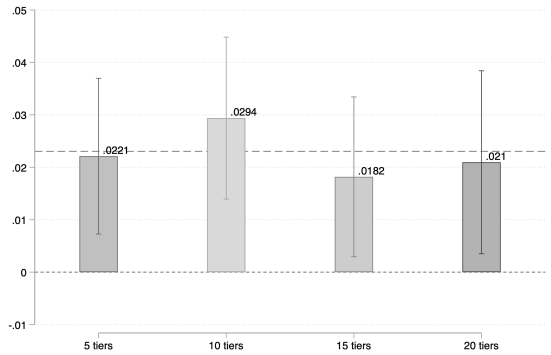
(i) Papers Using FSRDC Data



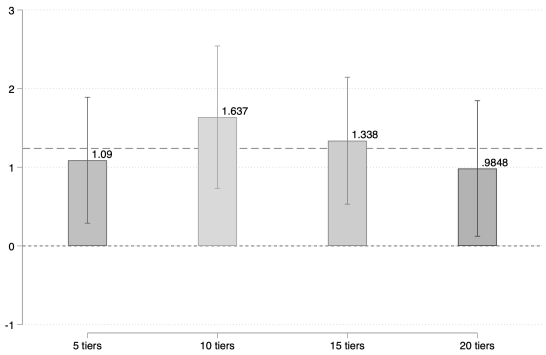
(ii) Papers Citing FSRDC Papers



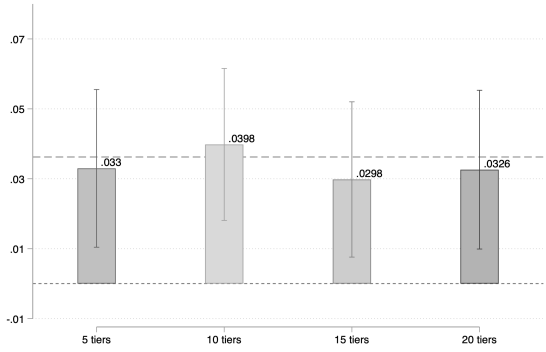
(iii) Top Publications



(iv) Cite-Weighted Publications



(v) Policy-Relevant Publications



Note: This figure provides visual illustrations of the effect of data access when we alter the definition of the university tiers used in our fixed effects structure. Unlike our main specification, where we use tiers of equal size (see B.2 for more details), we show robustness to alternative arbitrary categorizations. We construct the tiers by dividing the 100 ranked universities into equal groups, beginning with five ranking tiers and incrementing by 5 until there are 20 tiers, while keeping researchers affiliated with foreign universities or U.S.-based research institutions not covered by the ranking in two residual tiers. The chart plots values of β from different regressions where we redefine the university tier $\times$ year FEs. We also include the main estimate as a dashed horizontal line. The main dependent variables are the number of papers written using FSRDC data (panel (i)), the number of papers that cite FSRDC papers (panel (ii)), the number of top publications (Panel (iii)), the citation-weighted number of publications (Panel (iv)), and the number of policy-relevant publications (Panel (v)). All models include individual and university tier $\times$ year fixed effects. Standard errors are clustered at the university level. See text for more details.